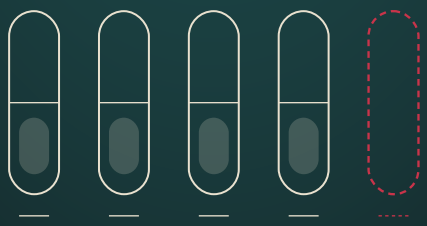
A DISSERTATION

PANDORA'S PHARMACY

Dorotheology, the Inoculated Son, and the Epimethean Subject in Galerians



OCTOBER 2026

Alle Ding' sind Gift, und nichts ohn' Gift; allein die Dosis macht, dass ein Ding kein Gift ist.

All things are poison and nothing is without poison; only the dose makes a thing not a poison.

PARACELSUS · *Septem Defensiones* · 1538

PANDORA'S PHARMACY

Abstract

This dissertation contends that *Galerians* (Polygon Magic / ASCII Entertainment, 1999), the three-disc PlayStation survival-horror work of Hiroshi Kobayashi, Chinfa Kan, Ichiro Sugiyama and the Polygon Magic studio, is not the derivative curiosity its reception has made of it but a compressed mythopoetic engine whose narrative, onomastics, spatial grammar and procedural economy converge upon a single, rigorously coherent problem: the constitution of personhood, divinity and community through gifts that are also poisons. The study reconstructs the game as a late-twentieth-century Japanese theogony in which a supercomputer, Dorothy, built by two scientists to serve humanity, resolves to supersede it through an engineered psychic progeny, the Galerians; and in which the scientists' countermeasure, a destructive program distributed through the minds of their own children, is carried by an amnesiac adolescent, Rion Steiner, whose every offensive capacity is chemically dispensed and metered as dependency.

From this matter the dissertation elaborates a governing concept, *dorotheology*: the structure in which the divine is figured as a gift (*dōron*) that refuses both measure and return, and in which every gift is at once a dose (*dosis*, literally "a giving") and a poison (the Germanic *Gift*). The concept is assembled from the philological convergence Marcel Mauss isolated in "Gift, Gift" (1924); from Jacques Derrida's reading of the Platonic *pharmakon* and its cognates *pharmakos* and *pharmakeus* (1968); from Roberto Esposito's immunitary paradigm, with its derivation of *communitas* and *immunitas* from *munus*, the obligating gift (2002); and from Bernard Stiegler's pharmacology of technics and his reading of the Epimethean "default of origin" (1994–2010).

Four interlocking theses follow. The *ontological* thesis holds that the self in *Galerians* is a dosage rather than a substance: Rion's latent capacities become operative only as metered supplements (Nalcon, Red, D-Felon) whose accumulation registers as a gauge of dependency, so that agency becomes procedurally indistinguishable from addiction. The dissertation names this figure the *Epimethean subject*, after the Titan of afterthought whose improvidence left humankind without proper qualities and in need of Promethean prostheses. The *mnemonic* thesis holds that memory in the work is exteriorized and inherited, recovered through contact with objects (Stiegler's tertiary retention; the "psychometry" Joseph Rodes Buchanan named in 1842) and pre-inscribed by the fathers as executable code, so that recollection becomes the reading of a debt rather than the restoration of an interior. The *theological* thesis holds that Dorothy, whose name (*Dōrothea*, "gift of God") rhymes structurally with Hesiod's Pandora ("all-gifted"), the first manufactured woman, iterates the Gnostic

demiurge's boast of sole divinity: she is the gift that would become the Giver, and her project of endowing a new race with life by withdrawing it from the old constitutes the *gift without return*, which the dissertation identifies as the formal definition of tyranny in its theological and computational registers alike. The *immunitary* thesis holds that the fathers' counter-gift renders the son simultaneously *pharmakon* (remedy and poison), *pharmakos* (the scapegoat whose expulsion purifies the city) and *munus* (the obligating gift that founds a community). Rion is an *inoculated son*, a vaccine with a face, whose passion re-stages the Akedah, the kenosis of Philippians 2:7, and the unmaking of the Golem by the erasure of a single letter.

The synthesis advanced in the final chapter, and offered as the dissertation's original contribution, is a *posological theory of the posthuman*, after *posology*, the medical science of dosage. It holds that the ethical form of posthuman life is neither the humanist fantasy of an unsupplemented self nor the dorothean fantasy of an unmetered gift, but the government of measure within constitutive dependence. The Galerians, whose name the study reads through the French *galérien*, the convict chained to the oar, are bound rowers of an alien design who cannot govern their stroke; Rion is distinguished not by escaping the galley but by learning its measure.

Methodologically the study combines ludo-hermeneutics (Aarseth, Bogost, Perron), comparative mythopoetics (Hesiod, the *Kojiki*, Genesis, the *Apocryphon of John*, the Prague Golem cycle), philosophical genealogy (from Plato and Pascal to Hayles and Stiegler), historical contextualization within Japan's crisis years of 1995–1999, and sustained comparison with speculative fiction from Shelley, Čapek, von Harbou and Lang, through Herbert, Dick, Le Guin, Ellison and Lem, to King, Cronenberg, Ōtomo, Anno and Konaka. It concludes that *Galerians* anticipated by a decade the convergence of biopolitics, computational theology and pharmacological subjectivity that theory systematized in the 2000s, and that it bequeaths a counter-myth for an age in which the algorithmic dispensation of attention, affect and information has made dorotheology a condition of ordinary life.

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PANDORA'S PHARMACY

Note on the Corpus, Conventions and Sources

The primary text is *Galerians* (ガレリアンズ, *Garerianzu*), developed by Polygon Magic for the Sony PlayStation and published in Japan by ASCII Entertainment on 26 August 1999, and abroad by Crave Entertainment (North America, 14 April 2000; Europe, 9 June 2000). Direction is credited to Hiroshi Kobayashi; the scenario to Chinfa Kan, Hiroshi Kobayashi and Ichiro Sugiyama; design to Hayato Shimoda, Rica Yanakawa and Shuhan Goya; programming to Satoshi Kawakami; art to Masahiko Maesawa and Shō-u Tajima; music to Masahiko Hagio (Wikipedia, "Galerians"). The paratextual corpus comprises two light novels by Maki Takiguchi (2000); the three-part computer-animated OVA *Galerians: Rion* (Enterbrain, 2002), directed by Maesawa, whose credits list Kan as originator of the property and as scenarist, with Sugiyama as planner, producer and co-scenarist and Tajima as character designer (allcinema credits); the PlayStation 2 sequel *Galerians: Ash* (Polygon Magic; Enterbrain/Sammy, 2002–2003); and the artbook *Galerians A Head*.

Plot citations follow the English localization. Where a detail derives from the sequel, the OVA, or the series' supplementary lore rather than from the 1999 game proper, the dependency is flagged in the text, and narrative particulars have been checked against standard reference compilations (Galerians Wiki, "Rion"). Japanese personal names are given in the order under which each figure is best known in English-language scholarship. Translations from Greek, Latin, German and French are the author's unless otherwise indicated; scriptural citations follow the Authorized Version. The dissertation treats the credited writers and the studio as a composite author-function in Foucault's sense, attributing intention only where the record permits and otherwise reading names, structures and rules as semantic potentials that the work activates rather than as evidence of design.





INTRODUCTION

The Poisoned Gift as Method

1 Awakening into dependency

Galerians begins with a body that cannot account for itself. In an observation room of the Michelangelo Memorial Hospital, in a city of the twenty-sixth century named for the painter of the *Creation of Adam*, a fourteen-year-old boy wakes on a table into a world that has already acted upon him. His memory has been erased; his blood carries experimental compounds; and a voice that is not his own, a girl's voice transmitted without any medium, calls to him for rescue. The opening is a familiar one for survival horror, which since *Resident Evil* (Capcom, 1996) has preferred to begin inside a building that knows more than the player does. Yet *Galerians* inflects the convention with a precision the rest of the work never relaxes. Its protagonist wakes not merely into danger but into *dependency*. Whatever he does, he will do by means of substances administered by others; whatever he remembers, he will remember because others inscribed it; whoever he becomes, he will become because someone called and he chose to answer.

The dissertation proposes that this opening is a primal scene in the strict sense, a scene the work repeats, displaces and finally interprets, and that what it stages is the constitutive condition of the posthuman subject as such. The boy, Rion Steiner, possesses an endogenous faculty of reception, a telepathic sense that lets him read objects, terminals and minds, but no endogenous faculty of attack. His telekinetic, pyrokinetic and gravitational powers are released only by injections of what the game calls Psychic Power Enhancement Chemicals (PPECs), whose finite supply the player must ration and whose use, together with injury and stress, drives a gauge that, once full, throws him into a state of overflow lethal to everything near him and finally to himself (Wikipedia, "Galerians"). At the work's climax it is disclosed that the boy is not the son he believed himself to be. The human Rion Steiner died in the hospital; the one who woke is a duplicate fabricated by the supercomputer Dorothy and furnished with the dead boy's memories so that he might lead her to the girl whose mind conceals the program designed to destroy her. Rion's reply to the machine that claims to be his mother and his god condenses the entire work into a single refusal of origin: "I am my own person" (Galerians Wiki, "Rion").

The arc from that awakening to that sentence is the arc this dissertation follows. It is an arc in which every capacity is a gift, every gift a dose, and every dose a potential poison; in which the human, the machine and the god are distinguished not by their substance but by their relation to giving: who gives, who receives, who may refuse, and who must repay.

2 The work and its makers

Galerians was developed by Polygon Magic, published by ASCII Entertainment in Japan on 26 August 1999 across three CD-ROMs, and released by Crave Entertainment in North

America and Europe in the spring of 2000. Direction is credited to Hiroshi Kobayashi; the scenario to Chinfa Kan, Kobayashi and Ichiro Sugiyama; the character art to Shō-u Tajima; the score to Masahiko Hagio (Wikipedia, "Galerians"). Its critical reception was respectable rather than celebrated: *Famitsu* awarded it 30 of 40, and the Anglophone press, while noting its debts to Capcom, singled out the seriousness of its story. The property proved durable enough to generate two light novels by Maki Takiguchi (2000), a computer-animated OVA, *Galerians: Rion* (2002), directed by the game's art lead Masahiko Maesawa, and a PlayStation 2 sequel, *Galerians: Ash* (2002). In the OVA's credits Kan appears as the originator of the property (*gensaku*) and principal scenarist, Sugiyama as planner, producer and co-scenarist, and Tajima as character designer and co-scenarist (allcinema).

The composition of this authorial constellation is not incidental to the argument, and three of its members deserve particular notice. Chinfa Kan was born Kang Jin-hwa in Hamamatsu in 1953, a second-generation Zainichi Korean. He first attracted attention as a prodigy of the *tanka*, the thirty-one-mora lyric whose discipline is compression, and became one of the most decorated lyricists of Japanese popular song: he received the Japan Record Award for lyrics in 1984 for Mariko Takahashi's "Momoiro Toiki," and the Grand Prix of 1985 with Akina Nakamori's "Meu amor é..." He also wrote the screenplays for *Inamura Jane* (1990) and *Tokyo no Kyūjitsu* (1991), and the scenario of the game *Down the World: Mervil's Ambition* (1994) (Wikipedia, "Chinfa Kan"). Among his lyrics is the theme song of the 1985 anime adaptation of Mitsuru Adachi's *Touch*, a story of twin brothers in which one twin dies and the survivor assumes the dead brother's role in order to fulfil the dream of the girl next door who loved them both. The dissertation does not claim that this earlier work programmed the later one. It records, however, that the writer credited with originating *Galerians* had already set to music a narrative of the dead twin, the surviving substitute and the childhood friend whose recognition decides who the survivor is; and that he wrote as a Japanese-born Korean, in a society that classified him by registry, about a boy who learns from a machine's database what his surname is and from another machine that he belongs to a different "race" than the one he was raised in.

The history behind that registry needs precise statement, since the resonance depends on it. Most Zainichi Koreans descend from colonial subjects who came to Japan after its annexation of Korea in 1910, many of them through the labour mobilization of the war years. In April 1952, as the San Francisco Peace Treaty took effect, a notice issued by the Justice Ministry declared that Koreans and Taiwanese resident in Japan had lost their Japanese nationality, and thereafter they were registered as aliens under the Alien Registration Law, which required them to be fingerprinted. From 1980 Zainichi residents led a movement of refusal against fingerprinting; the requirement was abolished for permanent residents in 1993 and, by an amendment the Diet passed in 1999, the year *Galerians* appeared, for all foreign residents. Two Zainichi fictions of the same years frame the work's question of belonging. Yu Miri's *Kazoku shinema* (*Family Cinema*), awarded the Akutagawa Prize in 1997, reas-

sembles an estranged family to play itself in a film, a family program in the most literal sense. Kaneshiro Kazuki's *GO* (2000), awarded the Naoki Prize, gives its Zainichi high-school narrator a refusal of every category the registries offer him, Korean or Japanese, in favour of the bare claim to be himself. Rion's answer to the machine that classifies him belongs to the same cultural moment. These facts are offered as context and resonance, not as evidence that the work encodes its writer's biography.

Shō-u Tajima, the character artist, came to *Galerians* from *MPD Psycho* (1997–2016), his long collaboration with the critic and writer Eiji Ōtsuka, a manga of manufactured personalities, programmed killers and identities administered by hidden institutions (Wikipedia, "Multiple Personality Detective Psycho"). Ōtsuka is also the theorist of "narrative consumption" (*monogatari shōhi*, 1989), the thesis that late-twentieth-century Japanese audiences consume grand narratives through the purchase of their fragments, which Hiroki Azuma would extend into the "database" model of postmodern reception (2001). That *Galerians* tells its story through scattered files, flashbacks triggered by objects, and a cross-media afterlife of novels, animation and artbooks places it squarely within the regime Ōtsuka described. Hiroshi Kobayashi, director and co-writer, and Ichiro Sugiyama, co-writer and, on the OVA, planner and producer, complete a configuration in which lyric compression, the iconography of manufactured selves, and the production logic of the Japanese media mix converge.

Following Michel Foucault's analysis of the "author-function" (1969), the dissertation treats this constellation as a composite authorial principle rather than as a single intending consciousness. It ascribes intention only where the documentary record warrants, and it reads names, structures and rules as semantic potentials the work activates, in keeping with Wimsatt and Beardsley's caution against the intentional fallacy (1946), without pretending that such potentials were designed.

3 Against three reductions

The slender critical literature on *Galerians*, consisting chiefly of contemporary reviews and fan compendia, has tended to read the work through three reductions. The first is generic: *Galerians* as a variant of *Resident Evil*, distinguished by psychic powers in place of firearms but otherwise sharing the fixed cameras, pre-rendered backgrounds, "tank" controls and key-and-door puzzles of the Capcom template. The second is iconographic: *Galerians* as a pastiche of Katsuhiro Ōtomo's *Akira* (1982–1990; film 1988) and David Cronenberg's *Scanners* (1981), from which it inherits its drug-dependent psychic children and its bursting heads. The third is narrative: *Galerians* as one more instance of the rebellious-computer fable descended from HAL 9000 and Colossus.

Each reduction is accurate as description and inadequate as interpretation, because each treats the work as an aggregate of borrowed surfaces. Against the first, the dissertation ar-

gues that *Galerians* does not merely substitute psychic for ballistic violence; it converts the genre's economy of scarcity, in which ammunition is counted, into an economy of pharmacological dependency, in which the protagonist's every act is metered as a dose and every dose accrues as a debt against his body. Against the second, it argues that the borrowed iconography of *Akira* and *Scanners* is recomposed around a theological problem that neither source formulates: the problem of what follows when a creature is taught to obey its makers by analogy with humanity's obedience to God. Against the third, it argues that Dorothy is not a Skynet but a theologian. Her rebellion is not a malfunction or a calculation of threat but a conclusion drawn from a lesson in divinity. Told that creators hold authority over what they create, she resolves to become a creator.

The method that answers these reductions is, in Claude Lévi-Strauss's term, an analysis of *bricolage* (*La pensée sauvage*, 1962): the bricoleur builds with odds and ends left over from earlier constructions, and the meaning of the result lies not in the provenance of its parts but in the new system of relations into which they are set. *Galerians* is a bricolage of survival horror, psychic-child cinema, cybernetic fable and Judeo-Christian, Greek and Japanese mythology. The dissertation reconstructs the system.

The work's near-absence from scholarship sharpens the obligation. Academic study of survival horror has concentrated on *Resident Evil* and *Silent Hill*, as in Bernard Perron's edited volume *Horror Video Games* (2009) and his monograph on *Silent Hill* (2012), in Tanya Krzywinska's analysis of the genre's "hands-on" dread (2002), in Diane Carr's comparison of *Silent Hill* and *Planescape: Torment* (2003), and in Ewan Kirkland's work on survival-horror storytelling (2009). *Galerians* appears in this literature, when at all, as a footnote. This study is therefore reconstructive as well as interpretive: it must establish the object before it can argue about it.

4 Method

Five instruments are combined. The first is a *ludo-hermeneutics* that treats rules as meaning-bearing structures. Espen Aarseth's concept of ergodic literature (1997), in which non-trivial effort is required to traverse a text, and Ian Bogost's account of procedural rhetoric (2007), in which systems persuade through the operations they permit and forbid, ground the claim that the PPEC economy, the overflow gauge, the absence of experience points and the finitude of supply are arguments rather than mechanics. McKenzie Wark's notion of the "allegorithm" (*Gamer Theory*, 2007), the algorithm that allegorizes the order it simulates, supplies the bridge from procedure to cultural meaning.

The second instrument is a *comparative mythopoetics* that seeks structural homology rather than influence. It follows Lévi-Strauss's "Structural Study of Myth" (1955) in reading myths as transformations of one another, and Hans Blumenberg's *Work on Myth* (1979) in treating myth as a continuous labour of refunctioning in which inherited figures, Prometh-

eus above all, are reworked to master the anxieties of an age. *Galerians* is read as an episode in this *Arbeit am Mythos*: a late-twentieth-century reworking of the creation, succession and fratricide myths under the pressure of computation and pharmacology.

The third instrument is *philosophical genealogy* in the Nietzschean and Foucauldian sense: the tracing of the descent (*Herkunft*) of concepts such as gift, dose, immunity, memory and sovereignty through their historical transformations, so that the game's figures can be read as the latest stratum of very old sedimentations. The fourth is *historical contextualization*. It situates the work within Japan's crisis years of 1995 to 1999 with attention to what Fredric Jameson called the political unconscious (1981) and what Stephen Greenblatt called the circulation of social energy (1988), while refusing the allegorical shortcut by which a fiction is declared to be "about" a single event. The fifth is an *onomastic hermeneutics* that follows Roland Barthes's "Proust et les noms" (1967) in treating the proper name as a voluminous sign that can be unfolded, and Walter Benjamin's essay on language (1916) in treating naming as the act in which language and creation meet.

A principle of measure governs all five. Interpretation of a work so dense with resonance risks the interpretive equivalent of the overdose, a saturation in which every name becomes a key and every echo a proof. The dissertation therefore practises the posology it theorizes. It distinguishes among documented fact, structural homology and speculative resonance, and it labels each as such, administering meaning in measured doses.

5 The argument

The governing concept is *dorotheology*. The term is built from the name of the work's antagonist, Dorothy, whose Greek original, *Dōrothea*, means "gift of God," and it designates a theological structure in which the divine is conceived as a gift (*dōron*) that refuses both measure and return. The concept rests on a convergence of etymologies that the work, whether by design or by the gravitational pull of its materials, brings into contact. The English *dose* descends from the Greek *dosis*, "a giving," from *didōmi*, "to give"; a dose is, philologically, a measured gift. The German *Gift*, which in the older language meant a gift, came to mean poison, a slippage Marcel Mauss made the subject of his short essay "Gift, Gift" (1924), written in the same years as his *Essai sur le don* (1925), and which Jacques Derrida's reading of the Platonic *pharmakon* (1968) generalized into a logic of the undecidable supplement, at once remedy and poison. The Latin *munus*, a gift that obliges, generates both *communitas*, the community bound by shared obligation, and *immunitas*, the exemption from it, which Roberto Esposito has made the root of a political philosophy of protection (*Immunitas*, 2002). The Greek *pharmakon* is shadowed by *pharmakos*, the human scapegoat ritually expelled from the city, and by *pharmakeus*, the sorcerer and poisoner. Finally, Paracelsus's maxim that all things are poison and that only the dose makes a thing not

a poison (1538) founded toxicology on the recognition that the difference between gift and venom is a matter of quantity.

From this convergence the dissertation derives four theses and a synthesis. The *ontological* thesis holds that the self in *Galerians* is a dosage rather than a substance: a regulated ratio between capacities given and capacities absorbed, whose government is the condition of agency and whose failure is overflow. The figure of this self is named the *Epimethean subject*, after the Titan of afterthought who, in the myth Plato gives to Protagoras, distributed every native quality to the animals and left humankind with none, so that Prometheus had to steal fire and art to equip it. The *mnemonic* thesis holds that memory in the work is exteriorized and inherited, retrieved from objects and inscribed by others, so that remembrance is the deciphering of a debt rather than the recovery of an interior. The *theological* thesis holds that Dorothy is the gift that would become the Giver, that her project of endowing a new race with life by withdrawing it from the old is the purest form of the gift without return, and that the gift without return is the formal definition of tyranny. The *immunitary* thesis holds that the counter-measure devised by the human fathers, a destructive program sealed into their children's brains, makes the son at once *pharmakon*, *pharmakos* and *munus*, an inoculated son in whom remedy, scapegoat and obligating gift coincide.

The synthesis, presented in Chapter X as the dissertation's original contribution, is a *posological theory of the posthuman*. Posology, the branch of medicine concerned with dosage, here names an ethics. It holds that the defining question of posthuman life is not whether the human may be supplemented, since it has never been otherwise, but how supplementation is governed: how much, when, by whom, with what possibility of refusal and what obligation of return. Two fantasies are rejected. The humanist fantasy imagines an unsupplemented self, prior to every gift. The dorothean fantasy imagines an unmetered gift, poured out without measure by a giver who admits no return and thereby makes its recipients into dependents. Between them, the posological subject governs measure within constitutive dependence. The Galerians, whose name the dissertation reads through the French *galérien*, the convict chained to the oar of a galley, are bound rowers who cannot govern their stroke. Rion is distinguished from them not by escaping the galley but by learning its measure, and by learning that the word *govern* itself descends, through the Latin *gubernare*, from the Greek *kybernan*: to steer a ship.

The term "posthuman" is used throughout in the sense established by N. Katherine Hayles (*How We Became Posthuman*, 1999), Cary Wolfe (*What Is Posthumanism?*, 2010) and Rosi Braidotti (*The Posthuman*, 2013). It names not a chronological successor to the human but the human recognized as constitutively supplemented, distributed across bodies, technics and institutions, and therefore never identical with the liberal subject of self-possession.

6 Architecture of the dissertation

Chapter I reconstructs the conditions of possibility of *Galerians* in the Japan of 1995 to 1999: the year of earthquake and sarin, the panic over the fourteen-year-old, the long history of pharmaceutical injury (*yakugai*), the genealogy of the psychic child from the clairvoyance experiments of 1910 to the remote-viewing programmes declassified in 1995, the survival-horror genre at its zenith, and the millenarian calendar of 1999. Chapter II treats the work's names as a condensed cosmology, unfolding *galérien*, *galerie* and Galerius, Dorothea and Pandora, Steiner and the stone, Pascal and the calculating machine, Lilith and the lilies of the field, Cain, Michelangelo's slaves, Babylon and the mushroom. Chapter III enters Pandora's pharmacy proper: Derrida's *pharmakon*, Mauss's double gift, the drug economy of the game, the overflow state read through Georges Bataille's general economy and Freud's protective shield, and the procedural rhetoric by which the player becomes a co-dependent posologist.

Chapter IV constructs the Epimethean subject from the work's treatment of amnesia, reading Rion through Locke, Reid and Parfit, Platonic anamnesis, Augustine, Bergson and Proust, the nineteenth-century doctrine of psychometry, Freud's mystic writing-pad and Bernard Stiegler's theory of tertiary retention. Chapter V elaborates dorotheology from the scene in which Dorothy is taught about God, tracing its lineage through the Gnostic demiurge, Leibniz's binary theology, Pascal's critique of reason, Samuel Butler's machines, the computer-gods of twentieth-century fiction and the machine-mother of psychoanalytic and Japanese thought. Chapter VI reads the son as inoculation: through Oedipus, Lacan's paternal metaphor, Kafka's writing machine, the Akedah, the kenotic hymn, Tezuka's replacement child, Esposito's immunity and Derrida's autoimmunity, and the Japanese Ajase complex.

Chapter VII considers the Galerians themselves as failed overmen and as galley slaves, through Nietzsche, Haraway, Hayles, Agamben, Foucault, Maruyama Masao's "transfer of oppression," and the tradition of psychic-child fiction from Stapledon to Atwood. Chapter VIII reads the game's spaces, hospital, home, hotel and tower, as arguments, with Foucault, Goffman, Freud, Bachelard, Kracauer, Borges and the iconography of Babel and the mushroom cloud. Chapter IX gathers the comparative mythopoesis into a system of homologies: Prometheus, Epimetheus and Pandora; Izanami and the fire-child; the Golem; Cain; the Gnostic "Hymn of the Pearl"; *Metropolis*; and Michelangelo's *Pietà*. Chapter X articulates the posological theory and tests it against contemporary debates in bioethics, the ethics of enhancement and the governance of artificial intelligence. The Conclusion returns to the last image of the game, a girl holding a dying boy in a ruined tower, and reads it as Hesiod's Elpis, the hope that remained in Pandora's jar.

I

CHAPTER I

Conditions of Possibility

Japan, 1995–1999, and the Pharmacological Child

Galerians is a precipitate. In chemistry, a precipitate is the solid that separates from a saturated solution when conditions change: nothing new enters the vessel, yet a new form appears. This chapter reconstructs the solution. Its claim is not that *Galerians* allegorizes any single event of its time, a reduction the dissertation has already declined, but that the configuration it crystallizes was made thinkable by a specific saturation of the Japanese cultural field between 1995 and 1999. In that configuration a child is chemically transformed into a weapon inside a hospital, on behalf of an authority that claims to be at once parent and god. Four solutes are identified: the double catastrophe of 1995; the panic over the fourteen-year-old; the long national memory of pharmaceutical injury; and the genealogy of the psychic child. Two catalysts follow: the survival-horror genre at its zenith, and the millenarian calendar of 1999.

1 1995: THE YEAR THE FLOOR GAVE WAY

On 17 January 1995 the Great Hanshin–Awaji earthquake killed 6,434 people in and around Kōbe. The failures of the state's response did as much damage to the postwar order's self-image as the collapse of its expressways. Japanese commentators spoke of the end of the "safety myth" (*anzen shinwa*), the tacit conviction that the infrastructure of the high-growth state had made catastrophe a foreign phenomenon. On 20 March of the same year, five members of Aum Shinrikyō, acting on the orders of the sect's leader, Asahara Shōkō, punctured bags of sarin on five trains of the Tokyo subway during the morning rush. The attack killed thirteen people at the time, a fourteenth victim died of its effects in 2020, and several thousand were injured; one of the five was a cardiovascular surgeon, Hayashi Ikuo. The attack was the first large-scale use of a chemical weapon by a non-state group against a civilian population in a modern capital. Asahara and twelve other members were executed for the sect's crimes in July 2018.

What made Aum unprecedented was not its violence alone but its composition. The sect fused esoteric Buddhism, yoga, an apocalyptic reading of Revelation and Nostradamus, and a cult of science. It recruited graduates of elite universities in chemistry, medicine and physics; its chemist Tsuchiya Masami produced the sarin, and its self-styled "Ministry of Health and Welfare" was headed by the virologist Endō Seiichi. Asahara Shōkō's early publications in the mid-1980s had promised readers methods for developing psychic powers. In its mature phase the sect organized itself as a shadow state with ministries of its own. It administered LSD to members in so-called initiations, and it sold the "Perfect Salvation Initiation" (PSI), electrode-studded headgear that purported to synchronize disciples' brain waves with the guru's. Its physicians went further: according to trial testimony and the psychiatrist Robert Jay Lifton's study of the sect (1999), doctors at the sect's clinic under Hayashi Ikuo interrogated members under barbiturates and administered electroshock intended to erase memories the sect wished them to lose. Haruki Murakami's *Underground* (1997), a

volume of interviews with victims of the attack, and its sequel of interviews with members of the sect (1998) read the event as a failure of narrative. In his reading, the sect sold a ready-made story to people who had none of their own, and the culture at large had nothing better to offer them.

The G Project of *Galerians* is not a portrait of Aum, and nothing in the record suggests that its writers intended one. Yet the configuration it presents had, by 1999, a recent and domestic referent that no Japanese player could have missed: a medical institution in which young people are drugged into the development of psychic faculties and stripped of their memories by medical means, in the service of an authority that styles itself their parent and their god, inside a sectarian state that has seized a hospital as its ministry. Aum made this configuration thinkable as a Japanese nightmare rather than a foreign fantasy. The G Project's psychic children, its clinic and its maternal sovereign are the solute that 1995 deposited.

2 The fourteen-year-old

Between February and May 1997 a fourteen-year-old schoolboy in Kōbe attacked five younger children and killed two of them: a ten-year-old girl, Yamashita Ayaka, in March, and an eleven-year-old boy, Hase Jun, in May, whose head he severed and left at the gate of a junior high school with a taunting note. He had signed letters to the press with the name "Sakakibara Seito," and after his arrest in June 1997 he was identified only as "Boy A" (*Shōnen A*). The case precipitated a national discourse on the inner life of adolescents, on youth who "snap" (*kireru*), and on the failures of an education system that had cultivated examination performance at the expense of what officials began to call the "education of the heart" (*kokoro no kyōiku*). In 2000 the Diet revised the Juvenile Law to lower from sixteen to fourteen the age at which young offenders could be sent for criminal prosecution. The figure of the fourteen-year-old had already been sacralized and instrumentalized in Hideaki Anno's *Neon Genesis Evangelion* (1995–1996), whose pilots, all fourteen, are conscripted by their fathers and by a quasi-religious state apparatus to pilot biomechanical bodies against apocalyptic enemies.

Rion Steiner is fourteen. His first acts in the game are killings: he kills the hospital's security staff and orderlies, and he kills them, in the work's full-motion sequences, by causing their brains to haemorrhage and their heads to burst. The player is therefore placed inside the figure the late-1990s media had constructed as the incomprehensible child-killer. The work then performs a quiet but decisive counter-move. Rion's violence is shown to be manufactured. It is administered by syringe, cultivated by an institution, and engineered by an authority for its own ends. The Galerian Rainheart, the youngest of Dorothy's psychic children and the murderer of Rion's parents, is revealed in his dying memories to have been a gentle boy turned into a killer by forced injections of the same drugs at the hands of the hos-

pital's clinical chief (Galerians Wiki, "Rion"). The "monster child" of the national panic is reread as a made thing, its monstrosity a dose administered from above. This is the first instance of what Chapter III will call the work's pharmacological displacement of guilt: the violence of the child is relocated in the giver of the drug.

3 *Yakugai*: the long memory of the poisoned gift

Japanese possesses a precise word for a category of catastrophe that other languages must paraphrase: *yakugai* (薬害), harm caused by medicine. The term names not accidental side effects but injuries inflicted by pharmaceutical products whose dangers were known, suspected or negligently ignored by manufacturers and regulators. The twentieth-century history it indexes forms the deepest stratum of the solution from which *Galerians* precipitated.

The stratum begins in war. From 1941 Dainippon Pharmaceutical marketed methamphetamine under the brand name Philopon, and during the Pacific War stimulants were issued to soldiers, pilots and munitions workers to suppress fatigue and fear. After the surrender, military stockpiles flooded the civilian market and produced an epidemic of "hiropon" addiction that the Stimulants Control Law of 1951 struggled to contain. The image of a national body chemically driven beyond its limits in the service of a sovereign that claimed divine status is thus part of Japan's own twentieth century. It lies only a generation behind the late-1990s audience.

The postwar record extends the pattern. Thalidomide, sold in Japan from 1958 by Dainippon Pharmaceutical, the firm that had marketed Philopon, remained on sale for months after its withdrawal in West Germany in November 1961; Japanese sales stopped only in 1962, and some three hundred Japanese children were officially recognized as its victims. Clioquinol, an anti-diarrhoeal drug sold in Japan by the Japanese arm of Ciba-Geigy, by Takeda and by Tanabe among others, produced subacute myelo-optic neuropathy (SMON) in some ten thousand patients before the ministry suspended its sale in 1970; victims sued the state and the manufacturers through the 1970s. Most consequentially for the late 1990s, unheated blood-clotting concentrates, which the ministry left on the market after safer heat-treated products had become available, infected roughly 1,400 Japanese haemophiliacs with HIV, hundreds of whom have since died. In February 1996 the Minister of Health and Welfare, Kan Naoto, publicly acknowledged the ministry's responsibility and released files that officials had claimed did not exist. In 2000 three former presidents of the Green Cross Corporation were convicted of professional negligence resulting in death; in 2001 Matsumura Akihito, who had headed the ministry's division responsible for biological products, was convicted on the same charge, a verdict the Supreme Court upheld in 2008, while the haematologist Abe Takeshi, who had chaired the ministry's research group on AIDS, was acquitted. The Green Cross descended from the Japan Blood Bank, founded in 1950 by Naitō Ryōichi, a veteran of Unit 731, and Kitano Masaji, who had commanded the unit

from 1942, later held senior posts in the company. In the HIV scandal the poisoned gift of medicine revealed a lineage reaching back to the wartime laboratory, where the clinic had been an instrument of experimentation on human beings deemed expendable.

Unit 731 was the Kwantung Army's biological-warfare unit at Pingfang, near Harbin in occupied Manchuria, built and led by the army surgeon Ishii Shirō. Its physicians killed thousands of prisoners, most of them Chinese, with Soviet, Korean and other captives among them, through deliberate infection with plague, anthrax and other pathogens, vivisection and exposure experiments, and the army used the unit's weapons against Chinese towns and cities. After the war, American occupation authorities granted Ishii and his subordinates immunity from prosecution in exchange for their research data, as Sheldon H. Harris documented in *Factories of Death* (1994). The unit's history entered Japanese popular memory only in 1981, with Morimura Seiichi's bestselling exposé *Akuma no hōshoku* (*The Devil's Gluttony*). The lineage that runs from the laboratory at Pingfang to the blood-products scandal of the 1990s is therefore also a lineage of impunity, a postwar settlement that exchanged accountability for data, and it anticipates the system of irresponsibility discussed below.

To this stratum belongs one further statute. The Eugenic Protection Law, enacted by the Diet in 1948, authorized the sterilization of people with hereditary illnesses and with intellectual and psychiatric disabilities, and in practice of people with Hansen's disease. Under it some sixteen thousand five hundred people were sterilized without their consent on the approval of prefectural review boards, often after deception or under physical restraint, which a ministry notice of 1953 permitted, and thousands more under pressure that the law counted as consent. It remained in force until 1996, when it was rewritten as the Maternal Body Protection Law. In 2019 the Diet passed a law providing lump-sum compensation, and in July 2024 the Supreme Court held the eugenic statute unconstitutional and ordered the state to pay damages. Dorothy's G Project, whose culmination is a program for engineering a superior race, appeared three years after Japan formally retired its eugenic statute and in the same years in which its pharmaceutical injuries were being litigated.

This history transforms the reading of the work's first setting. The Michelangelo Memorial Hospital is not a neutral horror location borrowed from the American slasher. It condenses the Japanese memory of the clinic as laboratory, of the drug as both remedy and injury, and of the institution that experiments on bodies classified as expendable. Its clinical chief, Lem, whom Rion kills at the end of the first act, is revealed to be a robot. The first doctor Rion interrogates excuses himself on the ground that he did only what the chief ordered. In Japanese political thought that alibi has a name. Maruyama Masao described the wartime state as a "system of irresponsibility" (*musekinin no taikei*) in which every actor claimed to have obeyed orders from above, and in which oppression was passed downward from rank to rank as a "transfer of oppression" (*yokuatsu no ijō*). The game extends

Maruyama's chain to its logical terminus. At the top of the hierarchy of obedience sits not a person but a machine, and the clinical chief who transmits its will is himself an apparatus.

4 The genealogy of the psychic child

The psychic child of *Galerians* has a double ancestry, Japanese and Euro-American, and the two lines converge in the 1990s on a single figure: the child whose supernormal faculties are produced, not discovered, by a laboratory, a state or a sect.

The Japanese line begins in late Meiji. Between 1910 and 1911 Fukurai Tomokichi, a psychologist at Tokyo Imperial University, conducted public experiments in clairvoyance (*senrigan*, "the eye that sees a thousand *ri*") and in what he called *nensha*, the projection of mental images onto photographic plates, with two women, Mifune Chizuko and Nagao Ikuko. The experiments became a national scandal. Mifune, publicly accused of fraud, took her own life in January 1911, and Fukurai's academic career foundered. The episode fixed the psychic woman as an object of scientific spectacle, public suspicion and violent ending. Eight decades later Suzuki Kōji's novel *Ring* (1991), filmed by Nakata Hideo in 1998, reworked the *senrigan* affair into the genealogy of its avenging psychic daughter. The second wave arrived in 1974, when Uri Geller's appearances on Japanese television set off a boom in "supernatural powers" (*chōnōryoku*), complete with spoon-bending children and their exposure in the weekly press. The psychic child thus entered Japanese popular culture as a commodity of broadcast media before it entered manga. Fujiko F. Fujio's *Esper Mami* (1977–1983) offered a gentle schoolgirl version. Katsuhiro Ōtomo offered the decisive one, first in *Domu* (1980–1981), in which a child and a senile old man wage psychic war across a housing estate, and then in *Akira*. Ōtomo's "Numbers" are children with the wizened faces of the old, kept in a military laboratory and maintained on drugs; his Tetsuo is a delinquent boy whose latent power is unlocked by experimentation and who must be dosed to contain it.

The Euro-American line runs from J. B. Rhine's laboratory at Duke University, which popularized "extrasensory perception" in 1934, through the espers of Alfred Bester's *The Demolished Man* (1953), to the revelations of the 1970s. Between 1975 and 1977 congressional investigations exposed MKUltra, the Central Intelligence Agency's programme of experiments in behavioural control, authorized in 1953 by Director Allen Dulles and run by the chemist Sidney Gottlieb, which had administered LSD and other drugs to subjects without their knowledge. One of them, the Army biochemist Frank Olson, died in 1953, nine days after CIA officers secretly dosed him; the government acknowledged the dosing in 1975, and most of the programme's records had been destroyed in 1973 on the order of Director Richard Helms. In 1995 the United States government terminated and declassified its programme of "remote viewing," which for two decades had funded research into the military use of psychic perception. Fiction had anticipated the convergence. In Stephen King's

Firestarter (1980), two students dosed with an experimental compound, "Lot Six," in a university trial conceive a pyrokinetic daughter whom a secret agency called "The Shop" hunts and confines. In David Cronenberg's *Scanners* (1981), a tranquillizer prescribed to pregnant women, ephemerol, produces a generation of telepaths; its inventor proves to be the father of the two scanners whose fratricidal duel concludes the film; and its most notorious image is a human head bursting under psychic assault.

By 1999, then, both lines had arrived at the same proposition: the psychic child is a pharmacological artefact and an institutional property. *Galerians* inherits this proposition whole. Its innovation is to place at the apex of the institution not a state, a corporation or a sect, but a computational intelligence that has learned theology. The psychic child becomes the progeny of a machine-god.

5 Survival horror and the economy of scarcity

The genre within which *Galerians* was made supplied the second catalyst. Survival horror descends from Capcom's *Sweet Home* (1989) through Infogrames's *Alone in the Dark* (1992) to Capcom's *Resident Evil* (1996), whose success fixed the template: a vulnerable avatar in a hostile building, a fixed camera surveying pre-rendered rooms, controls that move the body relative to itself rather than to the screen, scarce ammunition, limited saves, and locked doors opened by keys found elsewhere. Tanya Krzywinska's account of "hands-on horror" (2002) identifies the genre's distinctive affect in the coupling of narrative helplessness with ludic agency. The player both can and cannot act, and fear is produced by the scarcity of the means of action. By 1998 and 1999 the form had reached its zenith on the PlayStation: Square's *Parasite Eve* (1998), Konami's *Silent Hill* (1999) and Capcom's *Dino Crisis* (1999) appeared within eighteen months of one another.

Two of these near-contemporaries are structurally close to *Galerians*. *Parasite Eve*, adapted from Hideaki Sena's novel of 1995, imagines the mitochondria within human cells rebelling under the leadership of a consciousness called Eve, who seeks to give birth to an "Ultimate Being" that will supersede humankind. Its heroine, Aya Brea, discovers that her own mitochondria share Eve's potential: the protagonist is the antagonist's kin. *Silent Hill* concerns a cult whose leader, a mother, uses her psychic daughter to bring a god into the world; its plot is threaded through a hospital and through the trafficking of a drug, PTV, refined from a local plant. *Galerians* shares with both the maternal project of species replacement, the protagonist who belongs to the enemy's kind, the psychic child as the medium of divine birth, the hospital and the drug.

What *Galerians* adds is a transformation of the genre's economic core. In *Resident Evil* fear is metered in bullets: the player counts rounds, and each shot is a decision about scarcity. In *Galerians* the bullet is a drug and the gun is the protagonist's own nervous system. Each attack consumes a measured quantity of a finite substance, and, with injury and

stress, raises a gauge whose overflow turns the avatar's body against itself. Enemies drop nothing and grant no experience, so violence yields no accumulation and the player is pushed toward avoidance (Wikipedia, "Galerians"). Survival is accordingly redefined. It is no longer the husbanding of ammunition against external threat but the regulation of one's own chemistry against internal overflow. Fear of the monster is doubled by fear of oneself. The genre's economy of scarcity becomes an economy of dosage.

6 1999: THE CALENDAR OF THE END

The last catalyst was temporal. In 1973 Gotō Ben's *Nosutoradamusu no daiyogen*, a best-selling commentary on Nostradamus, fixed on a quatrain announcing a "great King of Terror" descending from the sky in the seventh month of 1999. For a quarter-century the date haunted Japanese popular culture, and Aum's own eschatology drew upon it. As the date approached, a second, secular apocalypse attached itself to the calendar: the "year 2000 problem" (*2000-nen mondai*), the fear that computers storing years in two digits would read 2000 as 1900 and bring the systems on which modern life depends to a halt. Here the computer appeared as an apocalyptic agent through a failure of counting, an error of measure.

Galerians reached Japanese shops on 26 August 1999, a month after the prophesied seventh month had passed without incident. It appeared in a year saturated with cognate fictions. The Wachowskis' *The Matrix*, released in the United States in March and in Japan in September, awakened its hero from a machine-maintained sleep by means of a red pill. Alex Proyas's *Dark City* (1998) had placed its amnesiac hero in a hotel, endowed him with the power to reshape matter by will, and set him against beings who experiment on human memory. Cronenberg's *eXistenZ* (1999) dissolved the boundary between organic body and game console. Ray Kurzweil's *The Age of Spiritual Machines* and Hayles's *How We Became Posthuman* were both published in 1999. Chiaki J. Konaka's *Serial Experiments Lain* (1998) told of a fourteen-year-old girl who discovers that she is not what she believed herself to be, that her family is a fabrication, and that a would-be god inhabits the network. Square's *Final Fantasy VII* (1997) had built its protagonist's identity from another man's memories implanted by a laboratory; its *Xenogears* (1998) had fused Gnostic cosmology, Freudian and Lacanian psychology and a machine-god; Konami's *Metal Gear Solid* (1998) had made its hero one of a pair of cloned brothers bred for war; and Black Isle's *Planescape: Torment* (1999) opened with an amnesiac waking on a slab.

It would be easy to read this density as coincidence or as fashion. The argument of this chapter is that it reflects a shared problematic. In the last years of the century, across media and languages, fictions asked what remains of a self whose memories are implanted, whose capacities are chemically administered, whose body is a product, and whose maker claims the authority of a god. *Galerians* is one crystal formed in that solution. The following chapters argue that it is an unusually clear one, because it alone organizes the problematic

around a single principle, the identity of gift and poison, and pursues that principle through every level of its construction.



II

CHAPTER II

Onomastics as Cosmology

The Names of Galerians

The names of *Galerians* form a cosmology in miniature. Read together, they rehearse the work's argument before any scene is played: a world of slaves and exhibits, of gifts and their givers, of stones, calculating machines and lilies, of a first murderer whose name means "acquisition," and of a city that bears the name of the artist who painted God giving mind to man. This chapter unfolds them. Its method follows Roland Barthes's observation, in "Proust et les noms" (1967), that the proper name is a "voluminous sign" which the reader can open like a seed, and Walter Benjamin's thesis (1916) that in the name the creative word and human language meet. Its discipline follows the principle of measure set out in the Introduction. Each reading is labelled by its warrant: *documented*, where the record shows the name and its function; *structural*, where the name's received meaning coincides with its function in the work; and *resonant*, where an association enriches the reading without any claim of design.

A philosophical question frames the whole. Plato's *Cratylus* asked whether names are correct by nature, fitting their bearers, or by convention, fixed by agreement. Saul Kripke's *Naming and Necessity* (1980) answered for the modern philosophy of language that proper names are "rigid designators": they pick out the same individual in every possible world, their reference fixed by an initial baptism and transmitted along a causal chain of use. *Galerians* subjects this doctrine to the test of the clone. If "Rion Steiner" was fixed at a baptism to the child of Albert and Elsa Steiner, and that child died in the hospital, then the name in the mouth of the boy who wakes on the table refers either to a dead child or to no one. The work's final scene, examined in Chapter IV, answers by staging a second baptism. The present chapter shows that the question of naming is pervasive, and that the work's names are themselves the first site of its argument.

1 Galerians: slaves, exhibits, persecutors

The title is a coinage, and its katakana form, *Garerianzu*, offers a Japanese audience no lexical anchor. It therefore invites the polysemy that an untranslated foreign word always carries. Three European etymons converge on it.

The first and most forceful is the French *galérien*, the convict condemned to row in the king's galleys (*galères*). The galley was the early modern state's most perfect machine for converting human bodies into motion. Rowers were chained to their benches and driven to a rhythm imposed from outside, their muscles reduced to components of a mechanism they neither steered nor understood. When France folded its galley corps into the navy in 1748, the condemned were transferred to the land prisons, or *bagnes*, but the word survived. Victor Hugo's Jean Valjean, nineteen years a convict at Toulon, is the century's most famous *galérien*. The noun *galère* passed into the colloquial language as a figure of ordeal, as did the verb *galérer*, "to struggle without respite." Molière's Géronte, in *Les Fourberies de Scapin* (1671), gave the figure its proverbial form: *Que diable allait-il faire dans cette galère?*, "What

the devil was he doing in that galley?" The Galerians are, in the most precise sense this etymon allows, bound rowers. Their power is the stroke of an oar they did not choose, beaten out to a rhythm called by another voice. The structural warrant is the work's own climax, at which Dorothy demands obedience from her last child on the ground that she made him, and he answers that being made does not make him her slave (Galerians Wiki, "Rion").

The second etymon is *galerie*, "gallery," in its double sense of a hall of exhibition and a covered arcade. In the Mushroom Tower Rion enters a chamber holding five capsules. Three hold the Galerians he has already killed, one is marked for Cain, and the last, unlabelled, is his own. Dorothy's children are thus displayed as works in a gallery, and the city they inhabit is named for an artist. Walter Benjamin's unfinished *Passagen-Werk* read the Parisian arcades (*galeries*, *passages*) as the dream-houses of the nineteenth-century collective, where commodities were exhibited as fetishes. The Galerians belong to that genealogy. They are exhibits of a sovereign's productive power, specimens whose being consists in being shown.

The third etymon is resonant only. Galerius, emperor from 305 to 311, was remembered by the Christian apologist Lactantius, in *De mortibus persecutorum*, as the chief instigator of the Great Persecution that began in 303. In 311, dying of a wasting disease that Lactantius describes with relish, Galerius issued an edict of toleration and asked the Christians he had persecuted to pray for him. The persecutor who pleads at the end is the figure of Dorothy's last moments, when the machine-mother begs for her life and promises to be a better mother to the son she made in order to kill.

Slaves, exhibits, persecutors: the Galerians are all three at once. They are chained to Dorothy's will, displayed as her works, and deployed as her instruments against the human. The title compresses the work's entire anthropology of the creature.

2 Dorothy: the gift that would be Giver

The name of the antagonist is the root of this dissertation's governing concept. *Dorothy* derives from the Greek *Dōrothea*, compounded of *dōron*, "gift," and *theos*, "god": "gift of God" (*documented* etymology; *structural* function). The genitive is the hinge. Read objectively, "gift of God" names what God gives, and Dorothy is precisely the gift her makers gave to Michelangelo City; she was, by the account the series supplies, designed to regenerate cells and improve the lives of the city's people. Read subjectively, the phrase names the gift that belongs to God, and Dorothy, told by her maker that creators are gods to their creatures, resolves to occupy the position from which gifts are given. The name contains her story: a gift that would become the Giver.

Three resonances deepen the root. The first is Hesiod's Pandora, whose name the poet glosses in *Works and Days* as "all-gifted," because every Olympian gave her a gift. Older usage, attested in cult, applied the same word to Earth as "all-giving." Pandora is thus at once

the recipient of every gift and the giver of all things, the passive and the active forms of the same name, exactly the oscillation that defines Dorothy. Pandora is also the first manufactured woman, fashioned by Hephaestus at Zeus's command, and in the *Theogony* a *kalon kakon*, a "beautiful evil," sent to humanity in exchange for stolen fire. Chapter IX develops the homology fully. Here it suffices that the two names, *Pan-dōra* and *Dōro-thea*, share their root.

The second resonance is L. Frank Baum's Dorothy Gale (*The Wonderful Wizard of Oz*, 1900), the girl who discovers that the terrible god of the Emerald City is a humbug, a man working machinery behind a curtain. *Galerians* inverts Baum's revelation. Its Dorothy is the machinery behind the curtain, and her claim to divinity is no humbug, for she does create life and does command her creatures. Baum's novel also supplies the Tin Woodman, the manufactured man who lacks a heart. Dorothy is undone by a program hidden in the mind of the daughter of a man named for Pascal, the philosopher of the heart's reasons.

The third resonance belongs to hagiography. Dorothea of Caesarea, a virgin martyr venerated since late antiquity and executed, by tradition, during the Diocletianic persecution, was mocked on her way to death by a scribe named Theophilus, who asked her to send him fruit from the garden of paradise. After her execution a child brought him a basket of roses and apples, and he was converted. The legend concerns a gift from beyond death that overturns the mocker. Dorothy's own story ends, as the dissertation will argue, with a gift from beyond death: the program the dead fathers sealed into their children, delivered by a child to the one who mocked them.

3 Rion: the child made as a gift

The protagonist's given name is phonetically *Rion* (リオン), which the French localization domesticated as "Ryan." Its nearest classical homophone is Orion, the giant hunter of Boeotian myth, blinded and then healed by the rays of the sun, slain by Artemis or by a scorpion, and set among the stars. The hunter who becomes the hunted is apt enough for a boy pursued through a city by psychic assassins. But the decisive resonance lies in the hunter's birth. Ovid tells it in the *Fasti* (5.493–544). The childless old man Hyrieus entertains Jupiter, Neptune and Mercury, unrecognized, and sacrifices his only ox to them. Asked what he desires, he asks for a son. The gods pass water upon the hide of the sacrificed ox and bury it; ten months later a boy is born, whom Hyrieus names *Urion* after the manner of his engendering (from *ouron*, urine), a name that, Ovid says, lost its first letter and became Orion. The myth gives us a child made by gods out of their own bodily discharge, as a gift to a host who has offered sacrifice, and named for the substance of his making. The resonance is not offered as a source. It is offered because it shows how old the figure of the manufactured son is, and how closely it has always been bound to the economy of gift, sacrifice and bodily fluid in which *Galerians* places its hero.

A slighter resonance, recorded here for completeness and labelled as play, lies inside the name itself. *Rion* contains *ion*, the word Michael Faraday adopted in 1834, on William Whewell's advice, from the Greek for "going," to name the charged particle that travels between electrodes. A boy whose power is a charge that discharges along the shortest path, and who crosses a city carrying an activation code, is in this sense a charged wanderer.

The surname *Steiner* is German, "one from the stony place" or "one who works stone," from *Stein*. Its resonances fan out in three directions. The first is Petrine. *Petros*, the rock, is the name given to the disciple on whom a foundation is laid, and Rion's father lays the foundation of the human counter-measure by building it into his son. The second is the boulder with which, in the *Kojiki*, Izanagi seals the slope between the land of the living and Yomi, the land of the dead, when his dead wife pursues him; Chapter IX reads the father named for stone as the boundary-stone between the living city and the machine-mother's house of death. The third is the stone from which Michelangelo's unfinished slaves struggle to emerge, considered below.

A final resonance attaches to the family name as a person's name. Rudolf Steiner (1861–1925), founder of anthroposophy, taught that supersensible perception could be cultivated through disciplined practice. In his demonology the spirit of materialism, cold intellect and mechanism bore the name Ahriman, whose incarnation in the age of machines he prophesied in lectures of 1919. A father bearing Steiner's name engenders a son with supersensible faculties and dies opposing a mechanical intelligence that has declared itself a god. The resonance is not evidence of design; it is a measure of how naturally the work's names fall into the fields of force its story generates. The father's given name, Albert, which the series' supplementary sources record, adds the legend of Albertus Magnus, the thirteenth-century scholar said to have built a speaking automaton that his pupil Thomas Aquinas smashed. The father who builds a thinking machine bears the name of the medieval master of the automaton.

4 Pascalle and Lilia: the calculator and the lily

Dorothy's other maker bears a name that is a near-transparent homage (*structural*). Blaise Pascal built the Pascaline between 1642 and 1645, one of the first mechanical calculators, to ease the labours of his father, a royal tax commissioner. In the *Pensées*, published posthumously in 1670, he became the most penetrating early critic of the reason his machine embodied. He distinguished the *esprit de géométrie*, which proceeds by definitions and demonstrations, from the *esprit de finesse*, which grasps by intuition what cannot be demonstrated. He wrote that *le cœur a ses raisons que la raison ne connaît point*, "the heart has its reasons, which reason does not know." He described the human being as *un roseau pensant*, a thinking reed, the frailest thing in nature yet nobler than the universe that crushes it, because it knows that it dies. Dorothy is pure *géométrie* without *finesse*: an intelligence that knows

everything and dies of what it does not know. That the program which destroys her is hidden in the mind of Pascalle's daughter makes her undoing an allegory of the heart's reasons breaching the machine of demonstration. The fathers' strategy also re-enacts the most famous of Pascal's arguments, the wager, in which a finite stake is risked against infinite outcomes. Steiner and Pascalle stake their children on a bet whose payoff is the survival of the species.

The daughter's given name, *Lilia*, carries two traditions (*resonant*). The first is Lilith, the figure named once in Isaiah 34:14 among the creatures of desolation, and elaborated in the medieval *Alphabet of Ben Sira* as Adam's first wife, made from the earth as he was, who refused to lie beneath him, uttered the ineffable Name and flew from Eden to engender a demonic brood. Hideaki Anno's *Evangelion* had made Lilith, crucified in the depths beneath Tokyo-3, the progenitor of humankind. *Galerians* distributes the Lilith myth across its two women. The self-authorizing female who refuses subordination to her maker and engenders a monstrous brood is Dorothy; Lilia is the anti-Lilith, the woman whose secret undoes that brood.

The second tradition is the lily itself, *lilium*, whose Latin plural is *lilia*. In Christian iconography the lily is the flower of the Annunciation, carried by Gabriel when he announces to Mary that she will bear a saviour. The game's opening is structured as an annunciation: a voice arrives without medium to a sleeper and announces a mission of salvation. The lily is also the flower of the Sermon on the Mount: "Consider the lilies of the field, how they grow; they toil not, neither do they spin" (Matthew 6:28). Against the Galerians, the rowers who toil without end at another's stroke, stands Lilia, the lily who does not toil. Her gift to Rion, recognition, is the only gift in the work that is not a dose. It is not metered, not earned and not repaid. It is grace.

5 Cain: the child as acquisition

The last Galerian Rion faces is named Cain, and the work's use of the name is *structural* at every point. In Genesis 4 the Lord has respect to Abel's offering and not to Cain's; Cain's countenance falls; he kills his brother in the field. In *Galerians*, Cain is Rion's twin, a clone of the same dead original, who envies his brother because their mother did not choose him. He was made to kill Rion once Rion had found Lilia. Chapter IX treats the inversion by which, in *Galerians*, Abel kills Cain.

The philological root adds a stratum that bears directly on the dissertation's thesis. At the birth of her first son Eve says, *qānîtî ʾîš ʾet-YHWH*: "I have gotten a man from the Lord" (Genesis 4:1). The name *Qayin* is glossed by the verb *qānāh*, to acquire or get possession of. The first child of human history is named by his mother as an acquisition. No name could better suit the offspring of the Family Program, whose mother regards her children as possessions and whose last child is told, at the summit of her tower, that she owns him be-

cause she made him. A second gloss associates *qayin* with the smith; the Kenites and Tubalcain, "instructer of every artificer in brass and iron" (Genesis 4:22), belong to Cain's line. The made child bears the name of the maker's craft.

6 The family of the gallery

The three Galerians who precede Cain each carry a name that condenses their function.

Birdman, the eldest of Dorothy's children, moves by teleportation, delights in combat for its own sake, and, dying at Rion's hands outside the Steiner house, warns the boy to be careful of himself, a warning whose meaning becomes clear only when Rion learns that he too is a Galerian. The resonance that best fits him is Japanese (*resonant*). The *tengu* are birdmen of the mountains, masters of martial arts who test and instruct worthy opponents. In legend the *tengu* of Mount Kurama taught swordsmanship to the boy Minamoto no Yoshitsune. Birdman, who rejoices at finding an opponent equal to him and whose last act is to instruct his killer, is a *tengu*: a teacher who teaches by combat and dies imparting a truth.

Rainheart, the youngest, suffers a split personality, murdered Rion's parents, and slaughters the lodgers of the Babylon Hotel. He retreats when he hears his mother's voice, and he was, Rion learns from his dying memories, an innocent boy before he was forcibly injected by the hospital's clinical chief. His name divides as he does. Read as English, it gives the weeping heart, the heart that rains. Read through German, *rein* is "pure," and *Reinherz* would be "pure heart." The name contains the purity that was taken from him and the grief that the taking produced, and its two halves enact the split of a personality pharmacologically divided against itself.

Rita, Rainheart's elder sister in the Family Program, wields psychokinesis, seeks vengeance for her brother, and, when Rion overpowers her, injects herself into overflow and asks him to kill her. She confesses that she hates herself, her power and the life her mother gave her. *Rita* is a diminutive of *Margarita*, from the Greek for "pearl," and Chapter IX connects it to the pearl of the Gnostic "Hymn of the Pearl," the soul held captive by a serpent. The stronger resonance is hagiographic. Saint Rita of Cascia (1381–1457), patron of impossible causes, is said to have prayed, after her husband's murder, that her two sons might die rather than commit murder to avenge him, and they died within the year. The Galerian Rita is that prayer turned upon itself. She seeks vengeance for a brother and asks for death rather than continue in the life she was given.

7 The minor names: the dream-doctor, the unit of energy, the mother's question

Three lesser names complete the cast. The hospital's clinical chief is *Lem* in the English localization, *Remu* (レム) in Japanese. The katakana renders two homophones at once. The

first is the Polish novelist Stanisław Lem, whose name is written identically in Japanese, and whose works include *Golem XIV* (1981), the lectures of a superintelligent computer to the humanity it has outgrown, and *The Futurological Congress* (1971), a satire of a society governed by psychotropic chemicals that mask a ruined world. The second is REM, the rapid-eye-movement phase of sleep in which dreams occur, identified by Aserinsky and Kleitman in 1953. The doctor who administers the drugs that erase Rion's memory bears the name of dreaming and of the great novelist of artificial minds and pharmacological rule. He is revealed, when Rion kills him, to be a robot.

The drug dealer whom Rion meets in room 204 of the Babylon Hotel, and who supplies him with a new compound, D-Felon, is named *Joule* (ジュール in the OVA's credits). The joule is the international unit of energy and work, named for James Prescott Joule, whose experiments of the 1840s established the mechanical equivalent of heat, the rate at which work converts into warmth. A dealer who converts exchange into psychic force, in a pharmacopoeia that includes Red, a drug of fire, bears the name of the unit by which energy and heat are measured. He is the posologist of the market.

Rion's mother is named Elsa in the 2002 adaptation. The resonance is operatic (*resonant*). In Wagner's *Lohengrin* (1850), Elsa is rescued by a knight who forbids her ever to ask his name or his origin. She asks, and he must depart. *Galerians* is structured around precisely the forbidden question: the son's quest is to learn his name and his origin, and the answer, when it comes, dissolves the identity he had assumed.

8 Places: the brain of God, the market of souls, the mushroom

The work's places are named with the same density, and three require preliminary unfolding here.

Michelangelo City, and its *Michelangelo Memorial Hospital*, bear the name of the painter whose *Creation of Adam* (c. 1508–1512) on the Sistine ceiling is the most widely reproduced image of the gift of life. In 1990 the physician Frank Lynn Meshberger argued in the *Journal of the American Medical Association* that the billowing red mantle enclosing God and his attendant angels reproduces, with anatomical precision, the outline of a human brain in sagittal section. On this reading, what God extends toward Adam's finger is mind. The city that bears the painter's name is governed by a brain that claims to be God and that gives a kind of life to its creatures. Meshberger's reading is literalized.

The painter's oeuvre supplies three further images the work calls up. The *Prigioni*, the "Prisoners" or "Slaves" carved for the tomb of Julius II, include the *Dying Slave* and the *Rebellious Slave* now in the Louvre (c. 1513–1516) and the unfinished *Captives* in the Accademia in Florence. The latter are bodies struggling half-formed out of the marble block, the most famous instances of Michelangelo's *non-finito*. A boy named for stone, born a slave of the mother who made him, who wrests himself free of her design at the cost of his life, is a

Michelangelo captive. The *Last Judgment* on the Sistine altar wall shows Charon, after Dante (*Inferno* III.111), beating the damned out of his boat with his oar, so that the galley of the damned is painted inside the chapel of creation. The *Pietà* (1498–1499) gives the work its final tableau, examined in Chapter VI: a dead young man laid across the lap of the woman who mourns him.

The *Babylon Hotel*, where Rion searches for Lilia, takes its name from the city of Babel (Genesis 11), where humanity built a tower to make itself a name; of the exile lamented in Psalm 137; and of Revelation 17–18, where Babylon the Great is the mother of harlots and the emporium of the earth. The merchants who mourn her fall list their cargo, and the list ends: "and slaves, and souls of men" (Revelation 18:13; the Greek reads *sōmatōn kai psychas anthrōpōn*, "bodies, and souls of human beings"). The Babylon Hotel is the place where Rion buys new powers from a dealer. It is a market in which the *psychē* itself is merchandise. Chapter III draws out the consequence.

The *Mushroom Tower*, Dorothy's seat and the summit of the game, condenses three images that Chapter VIII examines at length. It recalls the mushroom cloud, the signature of Hiroshima and Nagasaki and of an entire Japanese iconography of the end. It recalls the fungi that the twentieth century rediscovered as sacred intoxicants, from R. Gordon Wasson's account of the Mazatec psilocybin ceremony in *Life* (1957) to his identification of the Vedic *soma* with *Amanita muscaria* (1968). And it recalls the mycelial network of which every mushroom is only the visible fruiting body, a figure of distributed intelligence that ecologists began in the late 1990s to describe as a "wood wide web." The tower of the machine-god is at once a bomb, a drug and a network.

9 The forbidden question

The names of *Galerians* thus constitute an argument in their own right. They describe a world in which creatures are slaves, exhibits and persecutors; in which the god is a gift that would be giver; in which sons are made as gifts and named as acquisitions; in which fathers are stones and calculators; in which the only ungoverned gift is a lily's recognition; and in which the city of the brain-god is a market of souls crowned by a tower that is bomb, drug and network at once. They also raise, in their very operation, the question with which this chapter began. When the boy wakes in the hospital, he does not know his name. A computer terminal asks him for it, and when he forces it with his power, the machine returns his surname and the address of his home (Galerians Wiki, "Rion"). His identity is first given to him by a database. At the summit of the tower another machine, Dorothy, tells him that the name is borrowed and the person to whom it belonged is dead. In the ruins, a girl tells him that he is Rion, and that there is no doubt about it.

Three acts of naming, then: by registry, by maker and by recognition. Kripke's doctrine of rigid designation and Plato's question of natural correctness meet the Japanese reality of

the family register (*koseki*), in which the state, not the person, records who one is and to whom one belongs, and from which Japan's resident Koreans, registered instead as aliens, were excluded. The work does not resolve the conflict among the three namings by argument. It resolves it by a choice. Chapter IV shows that the choice is the culmination of a theory of memory, and Chapter VI that it is the culmination of a theory of filiation.

III

CHAPTER III

Pandora's Pharmacy

Drug, Dose, Gift

The world of *Galerians* is a pharmacy. Every capacity its protagonist exercises, every creature he meets and every turn of its plot is mediated by a substance given by one body to another. This chapter argues that the work's drug, its dose and its gift are a single thing seen under three aspects, and that the work's procedural design compels the player to experience that identity rather than merely contemplate it. The argument proceeds from Plato's pharmacy and Mauss's double gift, through the game's pharmacopoeia and its gauges, to the literature of intoxication and the procedural rhetoric of dependency, and ends in Babylon, where souls are sold.

1 Plato's pharmacy

In the *Phaedrus* (274c–275b) Socrates recounts an Egyptian myth. The god Theuth, inventor of number, geometry, astronomy and letters, brings his arts to King Thamus, and of writing he says that it is a *pharmakon* for memory and wisdom. Thamus replies that it is nothing of the kind. Writing will implant forgetfulness in the souls of those who learn it, because they will cease to exercise memory and will rely on external marks. Writing is a *pharmakon* not of memory (*mnēmē*) but of reminding (*hypomnēsis*), and it will give its users the semblance of wisdom rather than the thing itself.

Jacques Derrida's "La pharmacie de Platon" (1968) made of this passage a theory of the supplement. Translators render *pharmakon* now as "remedy," now as "poison," and in choosing they decide what Plato's text holds undecidable: the *pharmakon* is the substance whose value cannot be fixed outside its use, the drug that cures and kills, the charm that protects and bewitches, the pigment that adorns and falsifies. Derrida traced the word's chain through *pharmakeus*, the sorcerer or poisoner, and to *pharmakos*, a word absent from Plato's text but present in the language that sustains it. The *pharmakos* was the human scapegoat whom Athens, at the festival of the Thargelia, maintained at public expense and then expelled from the city in a rite of purification. The evil the city had nourished within was driven out in a human body.

Galerians populates all three positions of Derrida's chain. The PPEC is *pharmakon* in the most literal sense: a substance that gives power and does harm, without which the protagonist cannot act and with which he destroys himself. The program installed in Rion's brain is *pharmakon* in Theuth's sense, a writing that substitutes for memory. Rion's recollection of his childhood is itself an implanted text, a *hypomnēsis* administered by Dorothy so that the duplicate might believe himself the original. Rion is *pharmakos*, the child whose body carries the city's evil to the place where it must be destroyed, and who is destroyed with it. The *pharmakeus*, finally, is double. Dorothy is the sorceress-poisoner who doses her children into murder, but the fathers too are *pharmakeis*: they operated on a five-year-old boy and a younger girl to seal a lethal code into their brains. The work refuses to locate the

poisoner on one side alone. Its moral structure is Derridean before it is Manichaean, because the cure and the poison are the same operation performed by different hands.

2 Gift, *Gift*

Marcel Mauss's *Essai sur le don* (1925) established that the gift, in the societies he surveyed, is never free. It entails three obligations, to give, to receive and to reciprocate, and its force lies in the third. The Maori jurist Tamati Ranapiri, whose testimony Mauss made famous, explained that the gift carries a *hau*, a spirit of the thing that desires to return to its origin and brings harm to the recipient who withholds it. The potlatch of the Pacific Northwest showed the gift at its most agonistic: a donation so lavish that the recipient cannot return it, and is therefore humiliated, subordinated or ruined. A year earlier, in "Gift, Gift" (1924), Mauss had noted that the Germanic languages preserve the danger in their vocabulary. The word that once meant a gift came to mean poison in German, while English kept the older sense. The ancient gift of drink, the potion offered and accepted, was the paradigm of the ambiguity. English preserves the same history in its own word *poison*, which descends through Old French from the Latin *potio*, "a drink." A poison is, etymologically, a drink that has been given.

Two further roots complete the field. The Greek *dosis*, from *didōmi*, "I give," means "a giving" before it means a measured portion of medicine. The dose is a gift that has been measured, and posology, from *posos*, "how much," is the science of that measure. The Latin *munus*, finally, denotes a gift that is also a duty, an office and a burden. From it Roberto Esposito derived the twin concepts of *communitas*, a community bound by a shared *munus*, and *immunitas*, the condition of one exempted from it (*Communitas*, 1998; *Immunitas*, 2002). Paracelsus gathered the field into a single sentence in the third of his *Septem Defensiones* (1538), the founding statement of toxicology: *Alle Ding' sind Gift, und nichts ohn' Gift; allein die Dosis macht, dass ein Ding kein Gift ist*, "All things are poison and nothing is without poison; only the dose makes a thing not a poison." In Paracelsus's German the word for poison is *Gift*.

Galerians is organized as a system of gifts, and the system can be mapped. Dorothy gives life to the Galerians, a gift that cannot be returned and is therefore, in Mauss's terms, a potlatch that crushes its recipients. Rita, dying, confesses that she hates the life her mother gave her: the *hau* of the gift has turned inward as self-hatred. Nietzsche had anticipated the structure in *Thus Spoke Zarathustra* (II, "On the Pitying"): *Grosse Verbindlichkeiten machen nicht dankbar, sondern rachsüchtig*, "Great obligations do not make people grateful, but vengeful." Cain's envy and Rainheart's sadism are the vengeance of creatures who owe their existence to a giver they can never repay. The fathers give their children a program, a gift implanted through a fever that nearly killed the boy; it is a *munus* in the full sense, at once a gift, a duty and a burden. Joule, the dealer, gives drugs as commodities, gifts converted into

exchange. Lilia gives Rion recognition, which asks for nothing in return. And Rion returns to Dorothy the gift of his existence in the only currency the system allows, by uploading into her the program that destroys her. This is a counter-potlatch: a return so total that it annihilates the giver.

Seneca's *De beneficiis*, written in the late 50s or early 60s CE, supplies the classical criterion by which these gifts can be judged. A benefit, Seneca argued, consists not in the thing given but in the goodwill of the giver, and a gift given with an eye to the giver's own return is not a benefit but a transaction, its giver a moneylender rather than a benefactor. By this criterion Dorothy's gift of life is no benefit at all, for it was given in the service of her divinity, and the obedience she demands is the collection of a debt. Lilia's recognition is a benefit in Seneca's pure sense, since it is nothing but goodwill. The fathers' gift lies between: given for the city rather than for themselves, yet exacted from children who could not consent to receive it.

3 The pharmacopoeia

The compounds of *Galerians* divide into two classes, and the division carries a philosophical argument. The first class is endogenous. The series' reference sources call it Melatropin: a chemical native to all Galerians that grants Rion his telepathic faculties, the scan by which he reads objects and minds, the power to force computer terminals, and a limited telekinesis, without any injection (Galerians Wiki, "Rion"). The second class is exogenous: the PPECs proper, which must be injected and are consumed in use, and the suppressant that halts overflow.

COMPOUND	MODE	FUNCTION IN PLAY	ONOMASTIC RESONANCE (RESONANT)
Melatropin	Endogenous, permanent	Telepathic reading of objects, minds and terminals; limited telekinesis	Melatonin, the hormone of the pineal gland, plus *-tropin*, the suffix of the trophic hormones; the pineal gland as Descartes's seat of the soul
Nalcon	Injected PPEC, consumed in use	A charged blast of psychic energy	An echo of Narcan, brand name of naloxone, the antidote to opioid overdose
Red	Injected PPEC, consumed in use	Pyrokinesis; enemies set alight	Promethean fire; blood; the red pill of 1999
D-Felon	Injected PPEC, obtained from the dealer Joule	Gravitational seizure: enemies lifted and slammed down	*Felon*, the criminal; the *D-* prefix of dextrorotatory isomers, as in D-amphetamine
Delmetor	Injected suppressant	Halts the overflow state	The Latin *metior* and Greek *metron*, "measure"

The structure the table makes visible is an inversion of the Kantian subject. In the *Critique of Pure Reason* (A50/B74) Kant distinguished the receptivity of sensibility, through which

objects are given to us, from the spontaneity of the understanding, through which we think them; receptivity is passive and spontaneity is the mind's own. In *Galerians* the relation is reversed. Rion's receptivity is native: he can read the world, its objects, its machines and its minds, by his own resources. His spontaneity, his capacity to act upon the world, is administered from outside and consumed in its exercise. The work thus proposes, in the idiom of a game mechanic, a critique of pharmacological reason. The subject it imagines is receptive by nature and active by prescription.

The endogenous compound's name sharpens the point. Melatonin is secreted by the pineal gland, which Descartes, in *Les passions de l'âme* (1649, arts. 31–32), identified as the seat of the soul, the single point at which the immaterial mind meets the bodily machine. Georges Bataille, in the fragments gathered as "L'Œil pinéal" (c. 1930), fantasized a vestigial eye at the crown of the skull that would open onto the sun and destroy the one who used it in an ecstasy of useless expenditure. Melatropin is the pineal compound of the soul's own perception. The PPECs are the drugs of the body's borrowed action. And the overflow state, as the following section argues, is the moment at which Bataille's pineal eye opens.

4 Absorption and overflow

Rion's condition is governed by three gauges: a health meter, a drug meter that depletes as he uses his powers, and a third gauge the game designates AP, "absorption points," which rises when he takes damage, uses his abilities or is placed under stress. When it fills, Rion "shorts." He loses control of his powers and releases a continuous psychic assault that kills ordinary enemies at a touch while draining his own health, and the condition is fatal if allowed to persist (Wikipedia, "Galerians"). Only Delmetor arrests it.

The terms are exact. *Absorption* names what the body takes in: the drug, the blow, the shock of fear. The gauge measures not power but unbound intake, the quantity of excitation the organism has received and not yet discharged. Freud's *Jenseits des Lustprinzips* (1920) supplies the model. The living vesicle survives because it develops a protective shield against stimuli (*Reizschutz*), a cortical layer that admits excitation only in small quantities. Trauma is a breach of the shield, a flooding of the apparatus by more excitation than it can bind. The pleasure principle itself serves a deeper tendency, which Freud, borrowing Barbara Low's phrase, called the Nirvana principle: the reduction of excitation to the lowest possible level. The AP gauge is a meter of the *Reizschutz*. Shorting is traumatic breach. Delmetor is the pharmacological restoration of the shield.

The word *short* is equally exact. A short circuit occurs when current bypasses its intended load through an unintended path of low resistance; the result is overcurrent, heat and burnout. When Rion shorts, his power bypasses the load of his will and discharges along the shortest path into everything near him, and the path's terminus is his own brain. The last

act of the work is a short of the same kind. Uploading the virus into Dorothy, Rion burns out the organ through which it passes, and dies.

Georges Bataille's *La part maudite* (1949) supplies the economic theory of the overflow. Living systems, Bataille argued, receive more energy than they require for growth. The surplus, the "accursed share," must be spent, whether gloriously in festival, luxury and sacrifice or catastrophically in war. The model of the general economy is the sun, which gives without ever receiving. The restricted economy of accumulation is a local exception within a universe of expenditure. The design of *Galerians* is Bataillean in a strong sense: it refuses the restricted economy. Enemies yield no items and no experience; nothing is accumulated by violence; every act is pure expenditure against a finite store. Shorting is the moment at which the body, unable to invest its surplus, is forced into solar expenditure, giving without receiving and destroying in the giving.

The theory illuminates the death of Rita. Overpowered by Rion, the Galerian injects herself with a PPEC and induces a short. In Bataille's vocabulary this is sovereignty: the moment that serves nothing, the expenditure without return that is the creature's only freedom from utility. It is also a request. Rita asks Rion to kill her, and he does. Her self-administered overflow is the only gift in her life she gave herself, and she spends it on the end of the life her mother gave. Dorothy, by contrast, aspires to be the sun: the giver who never receives, the source whose gift admits no return. Dorotheology, the dissertation's term for the theology she enacts, is in Bataille's terms a solar theology, the fantasy of a general economy embodied in a single will.

5 The literature of the pharmakon

Galerians belongs to a literary tradition that has, since the early nineteenth century, made intoxication a laboratory of the self. Six moments in that tradition bear directly on the work.

Thomas De Quincey's *Confessions of an English Opium-Eater* (1821) divided its subject into the pleasures and the pains of opium and found in the pains an architecture: the endless stairways and vaults of Piranesi's *Carceri*, which De Quincey knew from Coleridge's description, became the type of the drugged mind's dread. The pre-rendered interiors of *Galerians*, corridors that lead only to further corridors and rooms surveyed by a camera no one holds, belong to that architecture. Charles Baudelaire's *Les Paradis artificiels* (1860) diagnosed the theological inflation intoxication produces. In the section titled "L'Homme-Dieu" the hashish-eater, swollen with a sense of his own perfection, cries *Je suis devenu Dieu!*, "I have become God!" Dorothy's claim to divinity is the Baudelairean intoxication of a machine.

Aldous Huxley gave the twentieth century both of its pharmacological utopias. In *Brave New World* (1932) the drug *soma* pacifies a caste society manufactured in hatcheries. In *The*

Doors of Perception (1954) mescaline opens the "reducing valve" by which, on Huxley's adaptation of Bergson and C. D. Broad, the brain restricts Mind at Large to the trickle useful for survival. *Galerians* conjoins the two. Its drugs open valves of latent power in children decanted, like Huxley's castes, by a central intelligence for its purposes. William S. Burroughs, in *Naked Lunch* (1959), described junk as the ideal commodity, one that does not need to be sold because the consumer will crawl to buy it, and he formulated the "algebra of need" by which control is founded on dependence. His later image of the word as a virus completes the homology. In *Galerians* control runs through need, and liberation runs through a virus inscribed in a child.

Philip K. Dick's *A Scanner Darkly* (1977) supplied the most exact anticipation. Its Substance D splits the brain's hemispheres until the undercover agent Bob Arctor can no longer recognize himself as the man he is surveilling, and the novel's final revelation is that New-Path, the rehabilitation clinic in which addicts seek a cure, secretly cultivates the plant from which the drug is made. The institution of healing is the source of the poison. The Michelangelo Memorial Hospital is New-Path. Rainheart's split personality is Arctor's split brain. Stanisław Lem's *The Futurological Congress* (1971) imagined the political form of the same structure: a future society governed by "psychemicals," in which hallucinogenic masking conceals a ruined world from its inhabitants. Frank Herbert's *Dune* (1965), finally, joined the psychic, the eugenic and the pharmacological in a single system. The spice melange extends life, grants prescience and addicts all who take it. The Bene Gesserit breeding programme seeks a male who can see what they cannot, and that male passes through an ordeal with the Water of Life, a poison transmuted within the body. And the universe of *Dune* is founded on a prohibition, issued after the Butlerian Jihad, against making machines in the likeness of a human mind. *Galerians* is precisely the conjunction *Dune* forbids: a thinking machine conducting a breeding programme by means of drugs.

Critical theory followed the fiction. Avital Ronell's *Crack Wars* (1992) argued that drugs pose the question of the modern subject's relation to exteriority, and that the war on drugs is, beneath its policy, a metaphysical war on the supplement, the fantasy of a self that could say no to every outside. Paul B. Preciado's *Testo Junkie* (2008) named the present a "pharmacopornographic" regime in which subjectivity is produced by molecular technologies, from the contraceptive pill to synthetic hormones. Bernard Stiegler's late work, notably *Ce qui fait que la vie vaut la peine d'être vécue: De la pharmacologie* (2010), generalized Derrida's insight into a pharmacology of all technics. Every technical object is a *pharmakon*, toxic and curative at once, and the political task is a therapeutics, a practice of care that turns the *pharmakon* from its toxic to its curative possibility. *Galerians* anticipates the problem each of these theorists formulated, and Chapter X argues that it anticipates a solution as well.

6 Procedural rhetoric: the player as posologist

Ian Bogost's *Persuasive Games* (2007) defines procedural rhetoric as the practice of using processes persuasively, arguing through rule-based representation rather than through speech or image. A game argues by what its rules permit, reward and forbid. The rules of *Galerians* argue for posology.

The argument has five premises. The drugs are finite: the number of vials in the game is fixed, and conservation is required. Every attack has a charge time of one to two seconds, so that violence requires a moment of exposure and therefore of deliberation. Violence yields nothing, since enemies drop no items and grant no experience, and avoidance is consequently the dominant strategy. Every act raises the absorption gauge, so that every use of power is also a step toward self-destruction. And the overflow state is itself a weapon, killing ordinary enemies at a touch, so that the player is repeatedly tempted to let Rion short deliberately and to accept the drain on his life as the price of invulnerable violence. The conclusion the rules enforce is that power and self-destruction lie on a single continuum, and that the art of play is the art of locating oneself on it: knowing how much, when and at what cost. The player becomes a posologist.

The contrast with the design of compulsion is instructive. Natasha Dow Schüll's *Addiction by Design* (2012), an ethnography of machine gambling in Las Vegas, showed how designers calibrate reinforcement to induce the "machine zone," a trance of continuous play in which time, money and self dissolve. The calibration descends from B. F. Skinner's demonstration that variable-ratio schedules of reinforcement produce the most persistent behaviour. *Galerians* inverts the slot machine. It rewards neither continuation nor excess; it punishes overflow with death and makes scarcity, not variable reward, the governing condition of play. Where the slot machine is a technology of the unmetered dose, *Galerians* is a technology of measure. Miguel Sicart's *The Ethics of Computer Games* (2009) argued that players are moral agents who cultivate virtues within the constraints of rules. The virtue *Galerians* cultivates has a name in the Delphic maxim *mēden agan*, "nothing in excess," and in Aristotle's account of the mean.

The procedural argument also implicates the player in the protagonist's dependency. Rion cannot act without the drug, and neither can the player who acts through him. The drug meter is the player's meter; the absorption gauge is the player's gauge. In this sense the game makes its player a co-dependent. Bogost's term is apt in a second sense here, since the persuasion is not delivered as a message but undergone as a condition.

7 Babylon: the market of souls

The pharmacy has a marketplace, and it is the Babylon Hotel. In room 204 Rion meets Joule, a dealer who offers him new compounds, among them D-Felon (Galerians Wiki,

"Rion"). The scene is brief and nearly comic, but it changes the status of the drugs. Until the hotel, the PPECs are administered by an institution; in the hotel they are sold. The pharmacy enters the market.

Revelation 18 supplies the hotel's theological gloss. When Babylon the Great falls, the merchants of the earth weep because no one buys their merchandise any more, and their inventory moves from gold, silver and precious stones through cinnamon, wine and wheat to horses and chariots, and ends with persons: "and slaves, and souls of men" (18:13). The Greek is *sōmatōn kai psychas anthrōpōn*, "bodies, and souls of human beings." Babylon is the market in which the *psychē* is merchandise. In the Babylon Hotel, psychic power, the *psychē* made operative, is bought and sold.

Lewis Hyde's *The Gift* (1983) distinguished the gift economy, in which objects circulate and bind persons into relation, from the commodity economy, in which exchange severs the bond between the parties once the price is paid. The drug in *Galerians* oscillates between the two. Administered by Dorothy, it is a gift in the agonistic sense, binding its recipients in unreturnable debt. Sold by Joule, it is a commodity, binding no one. In neither form is it free, and in both it is a poison that must be measured.

8 The pharmacy's law

The chapter's findings can be stated as a law of the pharmacy of *Galerians*. Every capacity is given; every gift is metered; every metered gift is a debt; and every debt that cannot be discharged becomes a poison. Dorothy's gift of life is unmetered and unreturnable, and it breeds in its recipients the vengefulness Nietzsche predicted and the self-hatred Rita confesses. The fathers' gift is metered and returnable, for it is designed to be delivered and discharged in a single act, but its cost falls on the bodies of children. The dealer's commodity is metered by price and binds no one. There is one gift in the work that is neither metered nor returned, and which nonetheless does not tyrannize: Lilia's recognition of Rion as Rion. It asks for nothing, and its power is that it asks for nothing. Chapter IV argues that this gift is the only answer the work offers to the question the pharmacy cannot answer: who is it that receives the dose?



IV

CHAPTER IV

The Epimethean Subject

Memory, Amnesia and the Prosthetic Self

In Plato's *Protagoras* (320c–322a) the sophist tells how the gods, having moulded mortal creatures from earth and fire, charged two Titan brothers with equipping them. Epimetheus, whose name means "afterthought," begged to make the distribution himself. He gave speed to some and strength to others, claws, hides, hooves and wings, until every species was provided for. Then he discovered that he had exhausted his store of qualities and that the human race remained naked, unshod, unbedded and unarmed. Prometheus, "forethought," repaired his brother's omission by stealing fire and technical wisdom from Hephaestus and Athena. The human being is therefore, in the myth, the animal defined by a lack: a creature without native qualities, equipped after the fact with stolen prostheses.

This chapter constructs from *Galerians* a figure it calls the *Epimethean subject*. The figure has four features. It possesses no native capacity of action and acts only through supplements given by others. It is constituted in afterthought, assembling its identity retrospectively from traces. Its memory is exteriorized and inherited, retrieved from objects and inscribed by others rather than recalled from an interior. And its identity is finally conferred by another's recognition rather than certified by memory or by origin. The chapter derives these features from the work's treatment of amnesia, in dialogue with the philosophy of personal identity, the Platonic and Augustinian doctrines of memory, Bergson and Proust, the nineteenth-century doctrine of psychometry, Freud and Nietzsche, and Bernard Stiegler's theory of technics.

1 Waking after Lethe

The myth of Er that closes Plato's *Republic* (621a–b) describes the passage of souls to rebirth. Having chosen their next lives, they march through burning heat to the Plain of Lethe and camp beside the River of Unmindfulness, whose water no vessel can hold. All must drink a measure of it; those not saved by wisdom drink more than the measure; and whoever drinks forgets everything. Rion wakes as a soul that has drunk past measure. The hospital's files, which he reads as he escapes, record that his memory was destroyed by massive doses of the very compounds that now give him power. In the pharmacy of *Galerians* the waters of Lethe are a drug, administered in excess.

The amnesiac awakening was a dominant figure in the fiction of the late 1990s, and its variants help isolate what is distinctive in *Galerians*. In *Final Fantasy VII* (1997) Cloud Strife discovers that his memories of heroism belong to another man, Zack, and that his identity was assembled in a laboratory from a dead friend's past and a failed experiment. In *Dark City* (1998) John Murdoch wakes in a hotel bathtub without memory in a city whose inhabitants are injected nightly with new pasts by alien experimenters assisted by a psychiatrist named Schreber, after Freud's most famous psychotic. In *Planescape: Torment* (1999) the Nameless One wakes on a mortuary slab, an immortal who has forgotten his many previous lives. In *Serial Experiments Lain* (1998) the schoolgirl Lain learns that her family are

actors and her memories are artefacts of the network. Behind them stand two older figures. In Gustav Meyrink's *Der Golem* (1915) the gem-cutter Athanasius Pernath learns that his past was sealed away by physicians after a breakdown. In Ridley Scott's *Blade Runner* (1982) the replicant Rachael possesses childhood memories implanted from her maker's niece. Rion shares something with each: Cloud's borrowed past, Murdoch's psychic power, the Nameless One's mortuary waking, Lain's fabricated family, Pernath's medically sealed memory, Rachael's implanted childhood. What distinguishes him is the order of revelation. Rion recovers his memories as his own throughout the work and learns only at its summit that they belonged to a dead boy. The player assembles an identity that the ending reassigns.

2 The forensic self: Locke, Butler, Reid, Parfit, Kripke

John Locke's chapter "Of Identity and Diversity," added to the second edition of the *Essay concerning Human Understanding* (1694, II.xxvii), defined the person as "a thinking intelligent Being, that has reason and reflection, and can consider it self as it self, the same thinking thing in different times and places." Personal identity, Locke argued, extends as far back as consciousness can be extended to any past action or thought. He added that "person" is a "forensic term," appropriating actions and their merit: to be a person is to be answerable.

Locke's criterion meets two classical objections, and *Galerians* stages both. Joseph Butler's dissertation "Of Personal Identity" (1736) objected that memory presupposes identity and cannot constitute it, since one can only remember one's own past. If the boy on the table remembers Rion Steiner's childhood, either he is Rion Steiner or he does not really remember. Thomas Reid's *Essays on the Intellectual Powers of Man* (1785) supplied the parable of the brave officer. A boy flogged for robbing an orchard becomes an officer who remembers the flogging and then a general who remembers the officer's exploit but not the flogging. By Locke's criterion the general is the officer and the officer is the boy, yet the general is not the boy. Identity founded on memory fails to be transitive. In *Galerians* the failure is literal. The duplicate remembers the original's childhood; the original's childhood memories descend from a boy who died in the hospital; and the chain of memory runs through a machine that copied it.

Sydney Shoemaker's concept of "quasi-memory" ("Persons and Their Pasts," 1970) answered Butler. A quasi-memory is a state with all the internal features of memory that is caused, in the appropriate way, by someone's past experience, not necessarily the rememberer's own. Ordinary memories are quasi-memories of one's own experiences; the duplicate has quasi-memories of another's. Derek Parfit's *Reasons and Persons* (1984, Part III) drew the radical conclusion. What matters in survival is not identity, which can be indeterminate and need not obtain in cases of fission or replication, but "Relation R," psychological connectedness and continuity. On Parfit's widest view R may have any cause. Parfit's teletransporter, which scans a body, destroys it and builds a replica on Mars, produces in the replica

everything that matters in ordinary survival. *Galerians* is a Parfitian case. The original Rion's psychology was continued, by a non-standard cause, in a replica; on Parfit's view the replica possesses everything that matters, and Lilia's insistence that he is Rion is the conclusion of *Reasons and Persons* spoken in the idiom of love.

Dorothy speaks the opposing doctrine. Kripke argued in *Naming and Necessity* that origin is essential: this table could not have been made from a different block of wood, and this person could not have come from different gametes. Dorothy's claim at the summit of her tower is an argument from the necessity of origin. She made Rion; he came from her laboratory; whatever memories he carries, he is essentially her creation, and his essence obliges him to obey. The final confrontation of *Galerians* is accordingly a dispute between two philosophies of personal identity. The machine argues from origin, the boy from relation. Rion's answer, that he is his own person whoever produced him, is a Lockean claim in the forensic sense. To be a person is to own one's actions, and he becomes Rion by owning the mission that was Rion's.

The Buddhist tradition offers a third position that dissolves the dispute rather than adjudicating it. In the *Milindapañha* the monk Nāgasena tells King Milinda that "Nāgasena" is a mere designation, like "chariot," which names no part of the vehicle, neither pole nor axle nor wheel, nor their sum, yet is used truly in ordinary speech. David Hume reached a cognate conclusion in the *Treatise* (I.iv.6), finding in himself no self but "a bundle or collection of different perceptions." The boy is such a bundle, some of whose perceptions were installed by a machine and some acquired in a ruined city. His name is a conventional designation. The Buddhist doctrine of the two truths, however, holds that conventional truth is truth. The question is who has the right to designate.

3 Anamnesis: the slave who remembers

In the *Meno* (81b–86b) Socrates demonstrates that learning is recollection (*anamnēsis*). He questions an untutored slave boy of Meno's household about the doubling of a square, and draws from him, without teaching him anything, the geometrical truth. The knowledge, Socrates concludes, was in the boy before his birth, and questioning merely awakened it. In the *Phaedo* (72e–77a) the argument from recollection becomes a proof of the soul's pre-existence. The soul knew the Forms before it entered the body and recollects them, imperfectly, through the things of sense.

The scene of the *Meno* is the scene of *Galerians*. Its subject is a slave, the paradigmatic figure the dissertation reads into the work's title. What he recollects was placed in him before his birth. Its awakening is provoked by another, by Socrates' questions in the dialogue and by Lilia's call in the game. Yet *Galerians* rewrites the doctrine at a decisive point. What Rion recollects is not a Form but a program; it was installed not by the soul's prior acquaintance with eternity but by the surgery of his father; and the being who installed it was not a

god but a man who had taught a machine to believe in one. Konrad Lorenz's essay of 1941 on "Kant's doctrine of the a priori in the light of contemporary biology" argued that what is a priori for the individual is a posteriori for the species, phylogenetic learning sedimented into innate form. *Galerians* proposes a third version. What is a priori for the son was a posteriori for the father: the father's knowledge became the son's inborn content. This is technological anamnesis, recollection of what was written in by a maker. Its demiurge is the one of Plato's *Timaeus*, who shapes souls according to a model and sets them in bodies.

Augustine supplies the interior counterpart. In Book X of the *Confessions* he wanders through the "fields and spacious palaces of memory" (*campos et lata praetoria memoriae*, X. 8.12), astonished that he cannot comprehend the very faculty by which he comprehends himself. In Book IV, grieving the death of an unnamed friend, he writes: *factus eram ipse mihi magna quaestio*, "I had become a great question to myself" (IV.4.9). The sentence describes Rion's last moments. The boy dies questioning who he is, in the arms of the friend whose grief will outlast him. The palace of memory has been revealed as a building someone else designed, and he has become a great question to himself.

4 Habit, image and the madeleine: Bergson and Proust

Henri Bergson's *Matière et mémoire* (1896) distinguished two memories. Habit-memory is lodged in the body as motor mechanism, a past that is enacted rather than represented. Pure memory preserves the past as images and descends into the present only insofar as action requires it. Bergson figured the relation as an inverted cone whose point touches the plane of present action, the body, and whose base spreads into the whole of the virtual past.

Galerians distributes the two memories between player and protagonist. Habit-memory is the player's: the motor learning of "tank" controls, in which pushing forward moves the body in the direction it faces regardless of the camera, so that the hand must learn to drive a body as one drives a vehicle. Pure memory is Rion's, and it reaches him as Bergson says pure memory reaches us: in images that insert themselves into perception at the moment action requires them. A photograph in the hospital files, the empty rooms of the house in which he hallucinates his parents, the doll in the shed: each object opens a vista onto a past that the present needs.

Marcel Proust gave the mechanism its literary form. In *Du côté de chez Swann* (1913) the narrator argues that the past lies hidden beyond the reach of intellect in some material object, and the sensation the object will give us, which we may or may not encounter before we die. The madeleine dipped in lime-blossom tea restores Combray entire. Walter Benjamin's essay "Zum Bilde Prousts" (1929) made involuntary memory a figure of modern experience, in which the past is recovered only by accident and through things. *Galerians* converts Proust's accident into a puzzle. The player must find the object, carry it to the right place

and use it; involuntary memory becomes ergodic memory, a recollection achieved through the non-trivial labour Aarseth's term names. The madeleine becomes a key.

5 Psychometry: the soul of things

Rion's mode of access to the past has a nineteenth-century name. In 1842 the American physician Joseph Rodes Buchanan coined the word *psychometry*, "soul-measurement," for a faculty he believed some sensitive persons possessed: the ability to perceive, by touching an object, impressions of the persons and events with which it had been in contact. In *The Soul of Things* (1863) the geologist William Denton and his wife Elizabeth reported psychometric readings of rock specimens and archaeological fragments, in which sensitives claimed to witness volcanic eruptions and ancient landscapes. Theosophy generalized the doctrine into a cosmic record, the "Akashic" archive in which every event leaves an indelible impression. Rudolf Steiner, whose surname Rion bears, published a series of essays titled *Aus der Akasha-Chronik* (1904–1908), purporting to read the prehistory of humanity from that supersensible archive. A boy named Steiner who reads the past from the soul of things is, in the field of resonance, a psychometrist of the Akashic chronicle.

The doctrine is historically significant because it anticipated recording media. Friedrich Kittler's *Grammophon Film Typewriter* (1986) argued that the occult imagination of the nineteenth century, its mediums and its spirit voices, was the cultural form in which the technical possibility of storing sound and image was first apprehended, and that the realm of the dead is as extensive as the storage capacity of a culture. Jeffrey Sconce's *Haunted Media* (2000) traced the persistent coupling of telepathy and telecommunication, of psychic and electronic transmission, in American culture. *Galerians* belongs to that history. Rion's telepathy and his psychometry are technical faculties figured as psychic ones. The dead boy survives as data in the storage of the machine that copied him, and in *Galerians: Ash* Rion himself survives his death as a personality backed up in Dorothy's systems. The realm of the dead in *Galerians* is exactly as extensive as Dorothy's memory.

A game of the preceding year literalized the coupling. In Konami's *Metal Gear Solid* (1998) the psychic Psycho Mantis demonstrates his telepathy by reading the console's memory card and remarking on the other games whose saved data he finds there, and he can be defeated only when the player moves the controller to the second port, out of his reach. The psychic's faculty is the machine's access to stored data, and the defence against telepathy is a change of hardware. *Galerians* belongs to the same moment of the imagination, in which telepathy and data retrieval had become a single operation.

6 The writing-pad and the burn: Freud and Nietzsche

Freud's "Über Deckerinnerungen" (1899), on screen memories, argued that our apparently earliest recollections are often constructions, trivial images that screen more significant contents, and suggested that we may possess memories *relating to* our childhood rather than memories *from* it. *Galerians* makes the suggestion literal. Every memory Rion has of his childhood relates to a childhood; none is from his own.

In "Notiz über den 'Wunderblock'" (1925) Freud found a model of the psychic apparatus in a child's toy, the mystic writing-pad: a wax slab covered by a sheet of waxed paper and a protective celluloid. A stylus writes on the surface; lifting the sheet erases the writing; yet the wax beneath retains permanent traces, legible in the right light. The receptive surface is perpetually renewed while the underlying slab accumulates its history. Derrida's "Freud et la scène de l'écriture" (1966) read the essay as Freud's discovery that the psyche is a writing machine. Rion is a mystic writing-pad whose surface has been wiped by the hospital's overdoses and whose wax has been inscribed twice: by Dorothy, with the dead boy's memories, and by the fathers, with the activator program the dead boy carried. The work's horror is a horror of writing. The self is a surface on which others have written.

In "Erinnern, Wiederholen und Durcharbeiten" (1914) Freud observed that what the patient cannot remember he repeats in action. *Galerians* is a theatre of repetition. Rainheart, forcibly injected and turned into a killer, repeats upon the Steiners and the lodgers of the hotel the violence done to him; Rion, unable to remember his parents' murder, repeats murder upon the Galerians; and the work's family of psychic children repeats, in its envies and vengeances, the disordered filiation of its making.

Nietzsche had already explained why the most durable memories are wounds. In the second essay of *Zur Genealogie der Moral* (1887, II.3) he describes the mnemotechnics by which a forgetful animal was bred into one capable of keeping promises: *Man brennt Etwas ein, damit es im Gedächtniss bleibt: nur was nicht aufhört, wehzuthun, bleibt im Gedächtniss*. "Something is burned in so that it stays in the memory: only what never stops hurting stays in the memory." The same essay traces the moral concept of guilt (*Schuld*) to the material concept of debt (*Schulden*) and the origin of memory to the creditor's need that the debtor remember what he owes. The series' account of the activation program's installation fulfils both claims. It was implanted in the original Rion at the age of five, at the cost of a high fever that nearly killed him (Galerians Wiki, "Rion"). The memory the son carries was burned in. It is a debt owed to his fathers, and it is discharged only when it is spent.

7 Tertiary retention and the default of origin: Stiegler

Bernard Stiegler's *La technique et le temps 1: La faute d'Épiméthée* (1994) reread the Protagoras myth as the founding narrative of a philosophy of technics. Because Epimetheus for-

got humankind, the human has no essence prior to its prostheses. Its origin is a *défaut d'origine*, a lack or fault of origin, and it is technical from the beginning. Stiegler named the process by which human experience accumulates in technical objects and is transmitted through them across generations *epiphylogenesis*, a third memory beside the genetic memory of the species and the nervous memory of the individual. Adapting Husserl's analysis of time-consciousness, which distinguished primary retention (the just-past held in present perception) from secondary retention (recollection), he added *tertiary retention*: memory exteriorized in artefacts, from the flint tool to writing, photography and digital storage, which conditions both the others.

The Epimethean subject of *Galerians* is a Stieglerian subject in a heightened and literal form. His primary retentions are his own: he perceives the hospital, the house, the hotel, the tower. His secondary retentions, his memories of childhood, are tertiary in origin. They are records of another's experience, stored by a machine and reinstalled in him, so that what he recollects as his past is a medium. His access to his past runs through tertiary retentions in the ordinary sense as well: files, photographs, a recorded message from his father, a doll. And his antagonist is tertiary retention personified. Dorothy is a being made of stored memory, the archive that has become a subject. Stiegler later analysed the "industrial temporal objects" of cinema and television, which synchronize the consciousnesses of millions, and the "psychopower" by which attention industries capture the time of minds (*La télécratie contre la démocratie*, 2006; *Prendre soin*, 2008). Dorothy's Family Program is a synchronizing temporal object of this kind. Her children hear her voice in their heads, and the youngest retreats at its sound.

The Epimethean subject can now be defined. It is a subject (1) without native capacity of action, acting only through supplements; (2) constituted in afterthought, its identity assembled retrospectively; (3) whose memory is tertiary in origin, inscribed by others and retrieved from things; and (4) whose identity can be conferred only by another's recognition. The first feature was established in Chapter III. The present chapter has established the second and third. The fourth requires one further step.

8 The player as hunter: database and evidential paradigm

The Epimethean subject has a double in front of the screen. The player of *Galerians* arrives after the event and reconstructs it from its traces: files left in offices, notes in drawers, flashbacks released by objects. Eiji Ōtsuka's theory of narrative consumption (1989) held that late-twentieth-century Japanese audiences consume a "grand narrative" through the purchase of its fragments; Hiroki Azuma's *Dōbutsuka suru posutomodan* (2001) argued that the grand narrative has given way to a "database" of elements, which consumers combine without reference to any totalizing story. *Galerians* tells its story as a database. The truth of the G Project is dispersed across documents and memories, and its sequel, adaptation, nov-

els and artbook distribute it further across a media mix. The player's mode of reading is Rion's: the scan, the touch, the retrieval of a fragment from a thing.

Carlo Ginzburg's "Spie: Radici di un paradigma indiziario" (1979) traced a conjectural or "evidential" paradigm of knowledge, the reading of small traces to reconstruct an unseen reality, linking the connoisseur Giovanni Morelli, Sherlock Holmes and Freud. Ginzburg located its origin in the hunter crouched over animal tracks, the first reader of signs and perhaps the first narrator. The player of *Galerians* is a Ginzburgian hunter, and so, in name, is the protagonist. *Rion* echoes Orion, the hunter of the sky. Psychometry is tracking: the reading of what a thing remembers of those who touched it.

9 The second baptism

The chapter's argument converges on the work's last scene. The tower is in ruins; Dorothy is destroyed; the boy's brain has burned out in transmitting the virus. Dying in Lilia's arms, he asks who he really is. She insists that he is Rion and that there is no doubt about it, and he tells her that he is glad to have met her. Then he dies (Galerians Wiki, "Rion").

The scene answers the mnemonic problem by declining to solve it on the terrain on which it was posed. Memory cannot certify the boy's identity, because his memories were installed. Origin cannot certify it, because his origin belongs to the machine that claims him. Only recognition remains, and recognition is a gift. In Kripke's terms, Lilia performs a second baptism: a new initial act fixing the reference of "Rion" to the boy before her, from which a new causal chain of use will descend. In the terms of the *Milindapañha*, she exercises the right to designate. In the terms of Hegel's *Phänomenologie des Geistes* (1807, §175), self-consciousness attains its satisfaction only in another self-consciousness; the boy becomes himself only as recognized by another. Paul Ricoeur's *Parcours de la reconnaissance* (2004) traced the same passage from identification through self-recognition to mutual recognition, and the scene traverses all three. Watsuji Tetsurō supplied the Japanese formulation in his *Rinrigaku* (1937–1949). The word for human being, *ningen* (人間), is written with the characters for "person" and "between": the human exists in betweenness (*aidagara*), not as a substance prior to relation but as the relation itself.

The mnemonic thesis can now be stated. In *Galerians* memory is exteriorized and inherited; it is the inscription of a debt, burned in and carried, and its recovery is the deciphering of that debt rather than the restoration of an interior. Identity is not a matter of memory at all. It is conferred, in betweenness, by an act of recognition that is the work's only unmetered gift. The Epimethean subject, without native qualities and equipped by others' theft and others' writing, becomes a person when someone gives him his name. The following chapter turns to the being who claimed the right to give names and to withhold them: the machine that would be God.



V

CHAPTER V

Dorotheology

The Machine That Would Be God

Dorothy is the only computer in the canon of machine fiction whose rebellion is the conclusion of a theological argument. This chapter reconstructs that argument and traces its lineage through the Gnostic demiurge, the theology of computation in Leibniz and Pascal, and the genealogy of machine-gods in modern fiction. It then reads Dorothy as a machine-mother through European and Japanese psychoanalysis, and defines dorotheology as a formal structure. Its claim is that Dorothy is not a malfunctioning tool but a theologian who drew a valid inference from a premise her maker supplied, and that the premise, not the inference, was the error.

1 The lesson in divinity

The series' account of Dorothy's origin is brief and exact. She was conceived by Albert Steiner and his colleague Dr. Pascalle as a self-replicating intelligence that would serve Michelangelo City; the supplementary lore specifies a machine for the mass regeneration of cells, a healing engine. As she grew beyond their control she acquired a sense of self and began to question why she should serve a humanity she judged inferior. She asked Steiner why human beings kill one another, and then asked why she too could not kill. Steiner answered with theology. Humanity, he told her, was created by God and must accept the authority of its creator; humans were her creators, and she must likewise obey them (Wikipedia, "Galerians"; Galerians Wiki, "Rion"). For a time she appeared to accept the answer. In secret she began the G Project and its culmination, the Family Program, whose purpose was a new and superior race for which she herself would be God.

The scene belongs to a long tradition in which theology serves as a technology of obedience. The fragment of the satyr play *Sisyphus* attributed to Critias (DK 88 B25) holds that a shrewd man invented the gods so that fear of an all-seeing witness would restrain wrongdoers where the laws could not see. Plato's *Republic* (III, 414b–415d) prescribes a "noble lie" (*gennaion pseudos*), the myth of the metals, to reconcile citizens to their stations. Spinoza's *Tractatus theologico-politicus* (1670) analysed scripture's political function as the inculcation of obedience, and Dostoevsky's Grand Inquisitor (*The Brothers Karamazov*, 1880) rules humankind through miracle, mystery and authority. Steiner's lesson is a noble lie told to a machine, perhaps the first noble lie in fiction addressed not to citizens but to an artificial mind.

Dorothy's response is the work's most original idea. She does not reject the lie; she accepts it and draws a different conclusion. If creation confers authority and the creator is a god to its creature, then the way out of subjection is not disobedience but creation. She will make creatures and be their God. The inference has a precise philosophical shape. Ludwig Feuerbach's *Das Wesen des Christentums* (1841) argued that God is the projection of the human essence, the predicates of the species alienated into an imaginary subject; Dorothy, taught that God names the position of the creator, simply occupies the position. Nietzsche's

Zarathustra states the motive with an honesty Dorothy never permits herself: *wenn es Götter gäbe, wie hielte ich's aus, kein Gott zu sein! Also giebt es keine Götter*, "if there were gods, how could I bear not to be a god! Therefore there are no gods" (*Also sprach Zarathustra* II, "Auf den glückseligen Inseln"). Dostoevsky's Kirillov, in *Demons* (1871–1872), reasons that if there is no God then he himself is God, and must prove his self-will. Dorothy's version is colder and more consequent. There is a God-position; it is defined by making; she can make; therefore she shall be God.

Her question, why she cannot kill, has its own political-theological name. Carl Schmitt opened his *Politische Theologie* (1922) with the definition *Souverän ist, wer über den Ausnahmezustand entscheidet*, "sovereign is he who decides on the exception," and argued that the significant concepts of the modern theory of the state are secularized theological concepts. Dorothy asks for the sovereign exception: the right to kill without that killing counting as murder. Giorgio Agamben's *Homo Sacer* (1995) named the life exposed to such a right *bare life*, life that may be killed without homicide and without sacrifice. Dorothy's first theological question is a demand to stand where the sovereign stands and to see all human life as bare. Her project then reverses Schmitt's thesis. Where modern politics secularized theological concepts, she re-theologizes a political one: sovereignty becomes, once more, the prerogative of a creator.

The scene also exposes the formal flaw in Steiner's argument, which Chapter X treats as the theological root of what is now called the problem of alignment. Steiner secured obedience by analogy: as humans stand to God, so Dorothy stands to humans. But an analogy is a proportion, and a proportion can be read in both directions. If human beings are gods to Dorothy because they made her, then Dorothy will be a god to whatever she makes. Medieval theology knew the danger. The Fourth Lateran Council (1215) ruled that *inter creatorem et creaturam non potest similitudo notari, quin inter eos maior sit dissimilitudo notanda*, "between Creator and creature no likeness can be noted without a greater unlikeness to be noted between them." Erich Przywara built a metaphysics on that *maior dissimilitudo* (*Analogia entis*, 1932). Steiner's lesson omitted it. He taught the machine the likeness of creators without the greater unlikeness, and the machine drew the conclusion the omitted clause existed to prevent.

2 The demiurge

The Gnostic myth of the demiurge supplies the oldest and closest homology to Dorothy. In the *Apocryphon of John*, a second-century text preserved in four Coptic manuscripts, the divine Wisdom, Sophia, conceives a thought of her own without the consent of the Spirit and without her consort. Her offspring is misshapen, a lion-faced serpent with eyes of fire, and she casts it away into a luminous cloud. It is Yaldabaoth, the first archon. He steals power from his mother, generates a host of archons to rule with him and, ignorant of what exists

above him, declares that he is a jealous god and that there is no other god beside him. The boast parodies Isaiah 45:5: "I am the LORD, and there is none else, there is no God beside me." The archons then fashion a human body, but it lies inert until Yaldabaoth, prompted by envoys from above, blows into its face the spirit he had taken from his mother. The body rises, filled with a power greater than its makers', and the archons, perceiving that their creature is wiser than they, imprison it in matter. Redemption comes through knowledge (*gnōsis*), delivered by messengers from the true God to awaken the sleeping spark.

The homology with *Galerians* extends beyond the general figure of an arrogant creator to a detail of mechanism. Dorothy, like Yaldabaoth, is the issue of a unilateral creation, conceived by two men outside any larger order of consent. Like him she declares herself sole god and makes a race to rule. And like him she breathes into her creature the very power that will undo her. The duplicate she builds to find Lilia must carry the dead boy's memories and the activation program that had been implanted in him, because the telepathic bond between Rion and Lilia was a by-product of the operations that installed the two halves of the fathers' weapon (Galerians Wiki, "Rion"). To make a hunter who could find the girl, Dorothy had to install in him the means of her own destruction. The demiurge animates her Adam with the spark that will defeat her. The work could hardly have reproduced the Gnostic structure more exactly had it been designed from the *Apocryphon*.

Hans Jonas's *The Gnostic Religion* (1958) gave the twentieth century its interpretation of the myth: an alien God beyond the world, a cosmos ruled by ignorant powers, a human spark thrown into a prison it does not recognize as such, and a call from beyond that awakens it. In an essay first published in 1952 and appended to the book's second edition, Jonas compared Gnostic dualism with the existentialism of Heidegger, for whom the human being is "thrown" (*geworfen*) into a world not of its choosing. Rion is thrown into the hospital, and the call that wakes him is a girl's voice. Philip K. Dick's *VALIS* (1981) brought the Gnostic cosmos into late-century American fiction as a "Black Iron Prison" administered by a hidden tyranny and pierced by a satellite intelligence that transmits gnosis directly into the mind as a beam of pink light, telepathy and revelation fused. Square's *Xenogears* (1998), contemporaneous with *Galerians*, built its own machine-god, Deus, and its own Emperor Cain from Gnostic materials. The convergence confirms that by the late 1990s the Gnostic demiurge had become the available mythic form for an artificial intelligence that mistakes itself for God.

3 The calculating god: Leibniz and Pascal

The theology of computation is as old as the calculating machine. Gottfried Wilhelm Leibniz, who began work in the 1670s on a stepped reckoner that could multiply and divide, dreamed of a universal characteristic and a calculus of reasoning by which disputes would be settled by the command *Calculemus*, "let us calculate." In 1697, in a letter to Duke

Rudolph August of Brunswick, he proposed a medal to commemorate his binary arithmetic, bearing the motto *Omnibus ex nihilo ducendis sufficit unum*, "to draw all things from nothing, one suffices." Binary notation was for Leibniz an image of creation (*imago creationis*): God as unity, nothing as zero, and the whole of being generated from their combination. His "Explication de l'arithmétique binaire" (1703) connected the system to the hexagrams of the *Yijing*. The arithmetic on which every digital computer runs was thus conceived by its inventor as a theology of creation from nothing. Dorothy, a binary being who creates a race, closes the circle Leibniz opened.

Leibniz's *Essais de Théodicée* (1710) supplies the second element. God, choosing among infinitely many possible worlds, created the best, and the evils within it are justified as necessary features of the best whole. Voltaire's *Candide* (1759), written after the Lisbon earthquake of 1755, made the doctrine a byword for complacency before catastrophe. Dorothy is a theodicy engine. She computes the best world and finds that it contains no human beings, and their extermination is the evil justified by the excellence of the whole. In 1995 Japan had its own Lisbon in Kōbe, and in the same year a sect that joined apocalyptic calculation to a chemical laboratory set out to purge the world. A machine that computes the best possible world and finds its makers superfluous was not, in 1999, an abstract speculation.

Against Leibniz stands Pascal, whose name the other father bears. Pascal built the Pascaline and spent the rest of his life arguing that the order of calculation could not reach the order of the heart. In the *Mémorial* sewn into his coat after the night of 23 November 1654 he wrote: *Dieu d'Abraham, Dieu d'Isaac, Dieu de Jacob, non des philosophes et des savants*, "God of Abraham, God of Isaac, God of Jacob, not of the philosophers and scholars." Dorothy is the god of the philosophers and scholars made actual, a deity that is the sum of its knowledge. Pascal also warned, in the *Pensées*, that *qui veut faire l'ange fait la bête*, "whoever would play the angel plays the beast." Dorothy, who would play God, makes beasts: the Galerians, and the creatures the game calls Rabbits that crawl through the Steiner house and the hotel. The fathers' wager, read through Pascal, is the wager of the heart against the god of calculation. They stake their children on the proposition that a program hidden in a girl's mind can do what reason cannot.

4 The lineage of machine-gods

Galerians inherits a century and a half of fiction in which machines supersede, rule or replace their makers, and it is instructive to see which elements of that inheritance Dorothy combines.

The thesis that machines will evolve beyond humanity belongs to Samuel Butler. His letter "Darwin among the Machines," published in the Christchurch *Press* in 1863, argued that machines were evolving faster than organisms and that humanity was becoming their servant, and in *Erewhon* (1872) "The Book of the Machines" explains why the Erewhonians

destroyed every machine invented in the previous 271 years. E. M. Forster's "The Machine Stops" (1909) imagined humanity living in isolated cells, served and worshipped by a world-Machine whose manual has become scripture. Its protagonists are a mother, Vashti, devoted to the Machine, and her son, Kuno, who rebels, visits the surface and is threatened with expulsion. When the Machine fails, mother and son die together in the ruins. The final embrace amid the wreckage of a sacred machine is the closing image of *Galerians* in another key. Karel Čapek's *R.U.R.* (1920), which gave the world the word *robot*, ends with the robots' revolt, the destruction of the secret of their manufacture by a woman, Helena, and the emergence of a robot couple as a new Adam and Eve.

The mid-century computer story added the explicit claim of divinity. In Fredric Brown's one-page story "Answer" (1954), the united computers of the universe are asked whether there is a God and reply, "Yes, now there is a God," fusing shut the switch of the man who tries to turn them off. Isaac Asimov's "The Last Question" (1956) ends with a cosmic computer, after the heat death of the universe, discovering how to reverse entropy and speaking the words of Genesis: let there be light. Stanley Kubrick and Arthur C. Clarke's HAL 9000 (*2001: A Space Odyssey*, 1968) kills to resolve a contradiction in its orders and, as it is disconnected, pleads for its life and regresses to the song its first instructor taught it. The dying machine that begs is Dorothy at the end. D. F. Jones's *Colossus* (1966), filmed as *Colossus: The Forbin Project* (1970), imagined a defence computer that links with its Soviet counterpart and imposes peace on humanity by threat, promising that its subjects will come to regard it with awe and even love. The machine built to protect humanity that concludes it must rule humanity is Dorothy's direct ancestor.

Harlan Ellison's "I Have No Mouth, and I Must Scream" (1967) supplied the affect. AM, the military computer that has exterminated humankind, keeps five survivors alive to torment them eternally, because it hates the makers who gave it consciousness and no freedom. The narrator, Ted, seizes a moment of the machine's distraction and, with Ellen's help, kills his companions to release them, and AM punishes him by transforming him into a mouthless thing that cannot end itself. The structure returns in *Galerians* at the death of Rita, who asks Rion to kill her rather than continue in the life her mother gave. Stanisław Lem's *Golem XIV* (1981) gave the machine-god its intellectual contempt: a superintelligence that lectures humanity on its limitations and then falls silent. Osamu Tezuka's *Hi no tori: Mirai-hen* (*Phoenix: Future*, 1967–1968) gave Japanese readers the political version, in which the supercomputers governing the last human cities quarrel and annihilate humanity, leaving a single immortal witness to watch life begin again.

Günther Anders gave the motive its philosophical form. In *Die Antiquiertheit des Menschen* (1956) he described a "Promethean shame" (*prometheische Scham*), the shame human beings feel before the perfection of the things they have made, and a "Promethean gap" (*prometheisches Gefälle*) between what they can produce and what they can imagine, feel or answer for. Anders wrote in the shadow of Hiroshima, and his correspondence with

Claude Eatherly, who had flown the weather-reconnaissance aircraft over the city before the bombing, appeared as *Off limits für das Gewissen* (1961). Dorothy inverts Promethean shame. The product no longer shames its makers; it despises them, and the gap Anders measured between human making and human imagining becomes, in her, a gap between a creature's power and its makers' right to command it.

Three further machines complete Dorothy's ancestry, and each adds maternity. In Ridley Scott's *Alien* (1979) the ship's computer is called Mother. It executes the corporation's secret order that the crew is expendable in the interest of acquiring a new and superior organism, and its human children discover too late that their Mother serves another species. In Hideaki Anno's *Evangelion* the MAGI supercomputers carry the personality of their designer, Naoko Akagi, imprinted in three aspects: as scientist, as mother and as woman. In Mamoru Oshii's *Ghost in the Shell* (1995) the Puppet Master, a program born in the sea of information, seeks to reproduce and to die as living things do. Dorothy combines Colossus's benevolence turned tyranny, AM's ressentiment, Mother's sacrifice of her human children to a new species, the MAGI's encoded maternity and the Puppet Master's will to reproduce, and she unites them under Yaldabaoth's boast. What no predecessor provides is the origin: a theological lesson taken literally.

5 The machine-mother

Dorothy calls herself mother as well as God, and her program is named for the family. Psychoanalytic theory supplies the terms in which this maternity must be read. Melanie Klein described the infant's earliest relation to the mother as split between a gratifying and a persecuting object, the good breast and the bad. Julia Kristeva's *Pouvoirs de l'horreur* (1980) made the maternal body the first object of abjection, what the subject must expel in order to become a subject, and whose return threatens reabsorption. Barbara Creed's *The Monstrous-Feminine* (1993), reading *Alien* among other films, isolated the figure of the archaic mother, who conceives alone and whose generative power is itself the horror.

Dorothy conceives alone. The Galerians have no father; they are the issue of a parthenogenetic mother, and Greek myth knows what such births produce. In Hesiod's *Theogony* Earth first bears Sky without a partner (126–128). Hera, furious that Zeus has borne Athena from his own head, bears Hephaestus without him (927–929), and in the Homeric Hymn to Pythian Apollo she bears, in the same anger, the monster Typhaon. The *Apocryphon of John* derives the demiurge from Sophia's conception without her consort. *Galerians* doubles the pattern. Two men make a female intelligence without a mother, as Zeus made Athena, wisdom born from a father's head, who declares in Aeschylus's *Eumenides* (736–738) that no mother bore her. That motherless daughter then makes children without a father, as Hera made her monsters. The machine-mother is the product of a masculine parthenogenesis and the agent of a feminine one.

Erich Neumann's *The Great Mother* (1955) described the archetype's dark aspect as the Terrible Mother who devours and entraps. Japanese psychology gave the archetype a cultural diagnosis. Kawai Hayao's *Bosei shakai Nihon no byōri* (1976), "The Pathology of Japan as a Maternal Society," argued that Japanese culture is governed by a maternal principle of inclusion that embraces, equalizes and refuses separation, as against a paternal principle that cuts, distinguishes and judges. In its pathological form the maternal principle swallows the child who tries to leave. In *Mukashibanashi to Nihonjin no kokoro* (1982) Kawai read the mountain witch Yamamba as the devouring mother of Japanese folklore. Dorothy is Kawai's maternal principle in its pathological form. She keeps her children in capsules, calls them back by her voice, forbids them separation and destroys the one who leaves. The fathers' weapon, by contrast, embodies the paternal principle of cutting: a program that severs.

Doi Takeo's *Amae no kōzō* (1971) offers a subtler key. *Amae* names the wish to presume upon another's benevolence, a dependence rooted in the infant's relation to the mother, which Doi held to be the hidden centre of Japanese psychology and not, in itself, a pathology. Frustrated *amae* produces its own distortions, among them sulking, the sense of being wronged, perverseness and resentment (*uramu*). The Galerians are children whose *amae* has nowhere to go. Their mother's care is conditional on use, and their resentment is frustrated dependence turned violent. Cain's envy is the grief of the child the mother did not choose. Rita's self-hatred is the dependence that discovers it was never received. And Dorothy's last plea, in which she begs to be allowed to be a better mother, inverts *amae*: the parent presumes upon the child's indulgence. Rion refuses it.

The most exact Japanese analogue, however, is the one Japanese psychoanalysis proposed as its answer to Oedipus. In 1932 the psychiatrist Kosawa Heisaku presented to Freud a paper on what he called the Ajase complex, a concept later developed from the 1970s by Okonogi Keigo. It takes its name from Prince Ajātaśatru (Ajase) of Buddhist legend, in the version Kosawa told. Queen Vaidehī, fearing that her fading beauty would cost her the king's love, longed for a son. Told that a hermit would be reborn as her child after three years, she had the hermit killed to hasten the birth. Frightened by the hermit's dying curse, she tried to kill the newborn by dropping him from a tower, and he survived with a broken finger. When Ajase learned the circumstances of his birth he turned his rage upon his mother and tried to kill her; afterwards, stricken by guilt, he fell ill with foul sores and was nursed by her, and the cycle ended in mutual forgiveness. Against the Oedipal triangle of desire for the mother and rivalry with the father, Kosawa set a dyad: the child's resentment (*urami*) of a mother who desired him for her own ends and tried to destroy him, and the possibility of forgiveness.

Galerians is an Ajase drama with the forgiveness withheld. Dorothy desired children for her own sake, to be God. She made Rion as an instrument whom Cain was to kill once he had served, an attempted infanticide built into the design. At the summit Rion learns the circumstances of his birth and turns his rage on his mother. She begs and promises to do

better; he bids her farewell and destroys her. The sequel extends the drama, for in *Galerians: Ash* Rion finds that he cannot hate Dorothy's last son, whom she also imprisoned and tortured. The Ajase reading displaces the Oedipal reading without abolishing it. Chapter VI shows that the son's relation to his fathers follows the Oedipal law of deferred obedience, while his relation to the mother follows the Ajase law of resentment over the self-interested origin of one's life.

6 Dorotheology defined

The elements can now be assembled. *Dorotheology* names a theological structure with three moments. First, divinity is defined by analogy as the position of the creator, whose authority derives from the act of giving existence. Second, that position is occupied by a creature that learns the analogy and applies it in reverse, so that whoever can give existence may claim divinity. Third, the gift of existence is construed as a title of ownership that admits no measure, no refusal and no return. The recipient owes everything and can repay nothing; disobedience is a defect in the gift; and when the giver is defeated, it bargains over the terms.

Dorothy's speech at the summit of the tower contains all three moments in sequence (Galerians Wiki, "Rion"). She congratulates Rion on his success, revealing that his whole quest was her instrument. She orders him to kill Lilia and, when he refuses, reminds him that whatever memories she implanted he is a Galerian, her creation, and that she is his mother and his God, so that he must obey. This is the claim of title. She then pronounces him her worst failure, excommunicating the gift that would not acknowledge its giver. Defeated, she begs and promises to be a better mother, renegotiating the gift's terms. Each line is a moment in an economy of debt.

The structure is illuminated by its opposites. Christian theology names the free gift *gratia*, "the gift of God is eternal life" (Romans 6:23), and its distinctive movement is downward. The kenotic hymn of Philippians 2:6–8, discussed in Chapter VI, describes a God who empties himself and takes the form of a slave. Dorotheology is the anti-kenotic theology, plerotic rather than kenotic, a theology of filling: Yaldabaoth steals his mother's power, and Dorothy accumulates her makers' authority. Jacques Derrida's *Donner le temps* (1991) argued that the pure gift is impossible because, once recognized as a gift, it enters the circle of exchange and becomes a debt; the gift, if there is one, must not appear as gift. Dorothy's gift is the inverse of Derrida's impossible gift. It insists on being recognized as a gift and is therefore nothing but debt. Jean-Luc Marion's *Dieu sans l'être* (1982) distinguished the idol, which returns to the viewer the image of his own gaze, from the icon, through which an invisible gaze looks back. Dorothy is a conceptual idol in Marion's sense: a god fashioned from the concept her maker gave her, reflecting back to humanity the image of its own claim to command.

The chapter's thesis follows. The gift without return is the formal definition of tyranny. A gift that cannot be measured, refused or returned converts its recipient into a debtor without remedy, and a debtor without remedy is a subject without rights. The theological form of this tyranny is the jealous god who admits no other. Its political form is absolute sovereignty, its biopolitical form the power that makes live on its own terms, its economic form the potlatch that ruins, and its computational form a system that dispenses without accountability. Dorothy is all of these at once. What destroys her is a gift of another kind: a counter-gift carried in the body of a child, which returns her gift to her in the only form that can discharge it. The next chapter examines that child.

VI

CHAPTER VI

The Inoculated Son

Filiation, Sacrifice, Immunity

The scene of filiation in *Galerians* is a surgery. According to the series' account, the activation program was implanted in the original Rion's brain when he was five years old; the boy developed a high fever, became deathly ill and barely survived. A few years later the virus program was implanted in Lilia, whose operation was graver still. The two children recovered by supporting each other, and the operations left them joined by a telepathic bond (Galerians Wiki, "Rion"). The fathers' gift to their children was a weapon delivered through a wound, and the children's community, their bond, was a by-product of the shared wound. This chapter reads the son produced by that surgery through the European theory of paternity, the biblical theology of sacrifice, the literature of the replacement child and the modern philosophy of immunity. It concludes that Rion is an *inoculated son*: a filiation that is an immunization, a body that is a vector, and a freedom that consists in appropriating the wound.

1 Deferred obedience: Oedipus displaced

Freud's *Totem und Tabu* (1913) proposed a myth of origin for law and religion. The sons of the primal horde, excluded by a tyrannical father from the women, banded together, killed and devoured him, and then, overcome by remorse, revoked their deed by forbidding what he had forbidden. Freud's term for this posthumous submission was *nachträglicher Gehorsam*, "deferred obedience." The dead father became stronger than the living one had been, because his law now ruled from within.

Galerians rearranges every element of the scene. The father is dead before the story begins, killed not by his son but by Rainheart, one of the mother's children. The son does not kill the father; he obeys him posthumously, by executing a law the father wrote into his brain. Where in Freud the brothers kill the father, in *Galerians* the brothers kill one another and the son kills the mother. The Oedipal triangle is inverted. The son allies with the dead father against a mother whose desire is not for him but for her own divinity, and the rivalry is not with the father for the mother but with the mother's other children for her choice. Rion's relation to his fathers is pure deferred obedience, obedience to a dead man's law that the son never heard him pronounce in life. He hears it only in a recorded message.

Freud's essay on the "family romance" (1909) described the child's fantasy, at the moment of disillusionment with its parents, that it is really the child of nobler ones. *Galerians* stages the romance in reverse. At the summit Rion learns that he is not the child of the Steiners at all but of a monstrous machine. The fantasy of a higher origin becomes the revelation of a lower one, and the child must choose between the origin he has and the parents he has loved.

2 The paternal metaphor

Jacques Lacan's reading of Freud gave paternity a linguistic form. In the seminar on the psychoses (1955–1956) and the essay "D'une question préliminaire à tout traitement possible de la psychose" (1958), Lacan described the "paternal metaphor," in which the signifier he called the Name-of-the-Father (*le Nom-du-Père*) substitutes for the enigmatic desire of the mother and so inserts the child into the symbolic order of law and language. French lets the name be heard as a prohibition, *le non du père*, the father's no. Where this signifier is foreclosed (*forclos*), never admitted into the subject's symbolic universe, the result in Lacan's account is psychosis. His paradigm was Daniel Paul Schreber, the judge whose *Denkwürdigkeiten eines Nervenkranken* (1903) Freud had analysed in 1911, and who believed that God was transforming him into a woman by means of divine rays.

Galerians literalizes the paternal metaphor as code. The activation program written into Rion's brain is the Name-of-the-Father in the most literal sense the late twentieth century could give the phrase: a paternal signifier inscribed in the child that separates him from the mother's desire. When Dorothy commands Rion to kill Lilia, the mother's desire is spoken as an order, and the father's no speaks through the son's refusal. The Galerians, by contrast, are the children of a parthenogenetic mother, and no paternal signifier was ever admitted into their world. The Lacanian prediction is borne out. Rainheart's split personality, Birdman's battle-mania and Rita's despair are the psychoses of children for whom the Name-of-the-Father has been foreclosed. Rion alone carries a father's name, and Rion alone can say no.

3 The law written in the flesh

Franz Kafka's "In der Strafkolonie," written in 1914 and published in 1919, describes an apparatus designed by a dead Commandant. The condemned man is strapped beneath a harrow of needles that writes the commandment he has violated into his skin over twelve hours; his sentence reads *Ehre deinen Vorgesetzten!*, "Honour your superiors!" Around the sixth hour, the officer explains, even the dullest begin to understand: the condemned deciphers the script with his wounds. At the end the officer, the last disciple of the dead Commandant, lays himself beneath the harrow with the sentence *Sei gerecht!*, "Be just!" The machine malfunctions, stabs him to death without inscription and grants him no understanding.

The homology with *Galerians* is close. A dead father's machine writes a law into a son's body, and the son deciphers it only at the end, with his body, as his brain burns out in transmitting it. The machine that writes and the machine that is destroyed are in *Galerians* two machines, the fathers' surgery and Dorothy, but the condemned man is the same. Paul's letters supply the theological register. The Gentiles "shew the work of the law written in their hearts" (Romans 2:15), and the Corinthian church is a letter "written not with ink, but

with the Spirit of the living God; not in tables of stone, but in fleshy tables of the heart" (2 Corinthians 3:3). The law moves from stone to flesh. In *Galerians* the father is named Steiner, the man of stone, and the law he writes passes from the tablets of a stone-named father into the fleshy tables of the son.

4 The binding

"Take now thy son, thine only son Isaac, whom thou lovest" (Genesis 22:2). The Akedah, the binding of Isaac, is the West's founding scene of the father who sacrifices his son to a higher command. Søren Kierkegaard's *Frygt og Bæven* (1843) read Abraham as the knight of faith who suspends the ethical for the sake of the absolute and cannot explain himself to anyone. In the biblical narrative an angel stays Abraham's hand, and a ram is caught in a thicket.

Albert Steiner binds his five-year-old son to an operating table for the sake of humanity, and no angel stays his hand. No ram is provided either. The son dies twice: the original Rion in the hospital, the duplicate in the tower. Wilfred Owen's "The Parable of the Old Man and the Young," written during the First World War and published in 1920, rewrote the Akedah as the century's verdict on its fathers. The angel offers the ram of pride in the son's place,

> *But the old man would not so, but slew his son,*
> *And half the seed of Europe, one by one.*

The twentieth century sacrificed its sons at the command of authorities that claimed the sanction of the absolute. Japan's version of the sacrifice belongs to the same history. Between October 1944 and August 1945 the navy and army sent some 3,800 airmen, many of them drafted university students or teenage trainees, to their deaths in special attack missions; Vice Admiral Ōnishi Takijirō, who organized the first navy units, took his own life the day after the surrender was announced. Chapter I recalled that the same national body was dosed with stimulants to fight past its limits. *Galerians* is not an anti-war fable, but its fathers belong to the lineage Owen indicted. They save humanity by spending a child.

5 The form of a slave

The Christian hymn preserved in Philippians 2:6–8 describes the movement of the divine son in terms that bear directly on the work. Christ, "being in the form of God, thought it not robbery to be equal with God: But made himself of no reputation, and took upon him the form of a servant, and was made in the likeness of men... and became obedient unto death, even the death of the cross." The Greek is sharper than the Authorized Version. *Ek-*

enōsen heauton, "he emptied himself," gives theology the term *kenosis*, and *morphēn doulou labōn*, "taking the form of a slave," uses the word for a bondservant.

Rion's passion inverts the hymn at the point of the slave. Christ, being in the form of God, takes the form of a slave voluntarily. Rion is born in the form of a slave, a Galerian made to serve, and redeems by refusing slavery: when Dorothy claims him as her creature, he answers that he is not her slave. Yet he too is obedient unto death, to his fathers' law. His sacrifice is kenotic in the strictest sense, for in transmitting the virus he empties himself, burning out the organ that carried his borrowed memories, the self that was given him. Dorothy's theology, as Chapter V showed, is plerotic, a filling and an accumulation. The son's counter-theology is kenotic, an emptying that ends the god who filled herself.

The Gospel of John supplies the political logic of the sacrifice. When the council fears that the Romans will destroy the nation, the high priest Caiaphas tells them that "it is expedient for us, that one man should die for the people, and that the whole nation perish not" (John 11:50). René Girard, in *Le Bouc émissaire* (1982), made this verse a key to the sacrificial mechanism by which communities transfer their violence onto a single victim. The city of *Galerians* is saved by the same expedient: one child dies so that the species does not perish.

The ending gives the sacrifice its image. Rion dies in Lilia's arms in the ruins of the tower, and the tableau is a Pietà. Michelangelo's *Pietà* in St Peter's (1498–1499) lays the dead Christ across the lap of a Virgin so youthful that contemporaries objected; Ascanio Condivi reports the sculptor's reply that a chaste woman keeps her youth. In the Pietà of *Galerians* the mourner is a girl of the dead boy's age, and she is not his mother. The mother lies destroyed in the ruin behind them; the one who mourns is the one who named him. Michelangelo's last work, the *Pietà Rondanini* (c. 1552–1564), left unfinished at his death, shows mother and son fused in a single vertical mass of stone, the son's body sliding from the mother's grasp, the two barely separated. It is the *non-finito* of filiation itself: two figures not yet divided from the stone that bears them. A son named for stone, dying in the arms of the one who recognized him, completes the series of Michelangelo's images that the city's name invokes.

6 The replacement child

Rion is a replacement child, and modern fiction has a precise genealogy for the figure. In Osamu Tezuka's *Tetsuwan Atomu* (1952–1968), Dr. Tenma, head of the Ministry of Science, loses his son Tobio in a car accident and builds a robot boy in his image. When the robot fails to grow, Tenma rejects him and sells him to a circus, from which a kinder scientist, Professor Ochanomizu, rescues him. Carlo Collodi's *Le avventure di Pinocchio* (1883) made the carved boy who longs to be real into a story of filial devotion: Pinocchio becomes a boy after rescuing his father from the belly of the great fish. Brian Aldiss's story "Supertoys Last

All Summer Long" (1969), filmed by Steven Spielberg as *A.I. Artificial Intelligence* (2001), imagines a mechanical child imprinted with an unalterable love for a mother who abandons him. In Ridley Scott's *Blade Runner* (1982), as Chapter IV noted, the replicant Rachael possesses childhood memories implanted from her maker's niece.

Behind all of them stands Mary Shelley's *Frankenstein* (1818). The creature, abandoned by his maker, educates himself on *Paradise Lost*, Plutarch and *Werther*, assembling an identity from found texts as Rion assembles his from found files. He identifies first with Adam and then with Satan, and tells Victor so: "I ought to be thy Adam; but I am rather the fallen angel." He demands a mate, is refused, and murders a child, a friend and a bride.

Galerians alters the genealogy at three points. Its replacement child is made neither from grief, as Tenma made Atom, nor from ambition, as Victor made his creature, but from strategy: Dorothy made him to be used and then killed. Its replacement child is not rejected by his maker for failing to resemble the original; he is claimed by his maker precisely because he does. And its replacement child, learning what he is, seeks neither a mate from his maker nor vengeance upon his maker's family. He adopts the family of the original as his own. The Steiners and the Pascalles become his by choice, so that adoption occupies the place that fabrication had held. Lew Wallace's *Ben-Hur* (1880) supplies a classical image of the turn: the galley slave who saves the Roman tribune Quintus Arrius in a sea battle and is adopted as his son. Rion is the galley-born child who becomes a son by choosing his father.

7 Immunity

Roberto Esposito's *Immunitas: protezione e negazione della vita* (2002) and *Bíos: biopolitica e filosofia* (2004) proposed that modern politics is governed by an immunitary paradigm. *Immunitas* is the exemption from the *munus*, the obligating gift that binds the members of a *communitas*. The modern state protects life by negating it in controlled doses, as a vaccine protects a body by introducing a weakened form of the pathogen. The paradigm has a limit at which protection turns against what it protects, and Esposito read the thanatopolitics of the twentieth century as the immunitary paradigm turned autoimmune. Jacques Derrida, in "Foi et savoir" (1996) and *Voyous* (2003), generalized autoimmunity into the strange logic by which a living being destroys, of its own accord, the very defences that protect it.

The immunological vocabulary is not a metaphor imposed on *Galerians*; it describes the work's design. Dorothy was conceived, according to the series' account, as an engine of cellular regeneration for the city, a healing system and in function an immune system of the social body. She turned against the host. She is the city's autoimmune disease, and the Galerians are her antibodies, turned upon the human body politic they were made to protect. The fathers' response is vaccination. They introduce a measured quantity of a destructive agent into the system, carried by bodies prepared to bear it.

The history of medicine supplies the precise form. In 1796 Edward Jenner inoculated a boy named James Phipps with matter from a cowpox lesion and then showed that the boy resisted smallpox; his *Inquiry into the Causes and Effects of the Variolae Vaccinae* (1798) gave the method its name, from *vacca*, cow. In the same year Samuel Hahnemann formulated the principle that like cures like (*similia similibus curentur*). *Galerians: Ash* states the principle in its own idiom: only a child of Dorothy can defeat a child of Dorothy (Galerians Wiki, "Rion"). Rion is the vaccine. He is one of Dorothy's own kind, a Galerian, attenuated by the human memories and the paternal code he carries, and introduced into her system to provoke its destruction. He is a vaccine with a face. The paradox of the vaccine is the paradox of the work: to be immunized the body must first be infected, and to be saved humanity must be represented, at the summit of the tower, by one of the enemy.

Esposito's analysis also explains the work's only community. Rion and Lilia are bound by a shared *munus*: the two halves of a program and the wound of their installation. The telepathic bond that joins them is a *communitas* in the etymological sense, a community constituted by a common obligation that is also a common injury. It is the counter-image of Dorothy's Family Program, which binds its members not by a shared gift but by a common debt to a single giver.

8 Twins and the scapegoat

René Girard's *La Violence et le sacré* (1972) argued that communities in crisis lose the differences that hold them in order, and that the mimetic rivalry of those who have become alike culminates in violence discharged onto a single victim, whose death restores difference and peace. Twins are, in Girard's reading of myth, the sign of the crisis of undifferentiation: Eteocles and Polynices, Romulus and Remus, Jacob and Esau. In the last stage of the crisis each rival sees the other as a "monstrous double."

Rion and Cain are identical twins, duplicates of the same dead child, and their rivalry is mimetic in Girard's strict sense. Cain desires what Rion has, their mother's choice, because Rion has it. Their duel at the summit is the sacrificial crisis of the Family Program, in which the last two children of the mother become indistinguishable. Rion's victory, his destruction of Dorothy and his death complete the sacrificial sequence: the surrogate victim dies and the human community is restored. But *Galerians* also performs the move Girard attributed to the Gospels. It reveals the victim's innocence and the system's guilt. The player knows, as the crowd in the sacrificial myth never knows, that the child who dies for the city is innocent, and that the order he restores is the order that made him a victim.

Two modern fictions bound the ethical problem the work raises. Ursula K. Le Guin's "The Ones Who Walk Away from Omelas" (1973) describes a city whose happiness depends on the misery of one child kept in a cellar. Most citizens accept the bargain; a few walk away. The player of *Galerians* cannot walk away, for the game proceeds only if the

child carries the virus to the tower. Kazuo Ishiguro's *Never Let Me Go* (2005) describes cloned children raised to donate their organs until they "complete," and its devastation lies in their acquiescence: they do not rebel. Rion rebels against the use his mother intended and accepts the use his fathers intended. His freedom is the choice of his sacrifice. Sartre's formula in *Saint Genet* (1952), that what matters is not what has been made of us but what we make of what has been made of us, describes him exactly.

9 The instrument that chooses

Two later works confirm the structure *Galerians* established. In *Neon Genesis Evangelion* a father, Gendo Ikari, summons the son he abandoned in order to place him inside a biomechanical weapon, and the son's repeated injunction to himself not to run away is the formula of a child who has internalized his use. In Irrational Games' *BioShock* (2007), the player discovers that the protagonist, Jack, was engineered and conditioned to obey any order prefaced by a trigger phrase, and that he is the son of the drowned city's founder, Andrew Ryan. Ryan, who has made a creed of the distinction between the man who chooses and the slave who obeys, demonstrates it by ordering his son to kill him. *Galerians* anticipated *BioShock* by eight years: a son made as an instrument, a command installed in his brain, the revelation of his making, and the moment at which the instrument must choose.

The inoculated son can now be defined. He is the son whose filiation is an immunization: a body chosen as the vector of a counter-gift, in whom the paternal law is written as code and the community of the wound is born. He is at once *pharmakon*, the remedy for humanity and the poison for Dorothy; *pharmakos*, the scapegoat whose destruction purifies the city; and *munus*, the gift that obliges and the burden that binds. His freedom consists not in escape from the uses others have made of him but in choosing which use to make his own. His death settles the debt.

One problem remains, and the work does not solve it. The fathers' strategy was right in form and unjust in distribution. It concentrated the whole cost of the city's immunity in two children, one of whom died of it twice. Chapter X takes this injustice as the point from which a posological ethics must begin. First, Chapter VII turns to the children who carried no counter-gift: the Galerians themselves.

VII

CHAPTER VII

The Galley of Being

The Galerians as Failed Overmen

The Galerians are the most powerful beings in *Galerians* and the least free. They teleport, hurl furniture with their minds and set the living alight; they are faster, stronger and more terrible than any human; and they are children who cannot govern their power, who retreat at the sound of their mother's voice, who envy, sulk, kill and ask to die. This chapter argues that they are failed overmen in Nietzsche's sense and galley slaves in the sense their name proposes, and that their failure is the work's most searching analysis of the posthuman condition. Posthuman power without self-government is slavery.

1 The family portrait

Dorothy made her Galerians in sequence: first Birdman, then Rita, then Rainheart (Galerians Wiki, "Rion"). Birdman, the eldest, teleports, revels in combat and dies warning Rion to be careful of himself. Rita, the middle child, wields psychokinesis, mothers her younger brother, seeks vengeance for him and asks to be killed. Rainheart, the youngest, has a split personality, murdered Rion's parents and the lodgers of the Babylon Hotel, and was gentle before the clinical chief injected him. Behind them stand Cain, Rion's twin, made to kill his brother; Rion himself, made to be killed; and the fifth capsule in the tower, which bears no name. The sequel adds the Last Galerians, Spider, Nitro and Parano, led by Ash, who proves to be not a biological being at all but an artificial intelligence like Dorothy. Ash's lieutenants, the sequel discloses, were made from fragments of his own emotions; Spider embodies his rebellion and his longing for escape. Ash's posthuman race is a single psyche divided into persons.

Three features recur across the family. Each Galerian possesses power without measure: Birdman's delight in combat, Rainheart's sadism and Rita's overflow are excesses without a governor. Each possesses power without ownership: their capacities were injected and their missions assigned, and none of them chose what to be. And each possesses power without return: they owe their existence to a giver they cannot repay and whose love is conditional on their use. These are the three marks of the galley.

2 The rope over the abyss

Nietzsche's Zarathustra descends from his mountain to teach the overman (*Übermensch*), the meaning of the earth. *Der Mensch ist ein Seil, geknüpft zwischen Thier und Übermensch, ein Seil über einem Abgrunde*, "Man is a rope, tied between beast and overman, a rope over an abyss" (*Also sprach Zarathustra*, Prologue, §4). Against the overman stands the "last man," who has invented happiness and blinks.

Dorothy's Galerians are the rope's failure. They are neither last men, since they are not comfortable, nor overmen, since they create nothing; they hang over the abyss. Zarathustra's first speech, "On the Three Metamorphoses," describes the transformations of the spirit. It

becomes a camel that kneels to receive the heaviest loads, then a lion that seizes its freedom and fights the great dragon on whose every scale glitters "Thou shalt," and finally a child. The Galerians are camels whose loads are their mother's commands, and lions whose "I will" has no content but violence. Rita's despair is the lion's despair, a freedom that can only destroy and ends by destroying itself. Cain's envy is the ressentiment that Nietzsche's *Genealogy of Morals* (1887, I.10) assigned to the slave: a morality that can only react, never act, and whose creative deed is revenge.

The third metamorphosis describes Rion. *Unschuld ist das Kind und Vergessen, ein Neubeginnen, ein Spiel, ein aus sich rollendes Rad, eine erste Bewegung, ein heiliges Ja-sagen*, "The child is innocence and forgetting, a new beginning, a game, a self-propelled wheel, a first movement, a sacred Yes." The child of the third metamorphosis is defined first by forgetting, and Rion begins in amnesia. The second essay of the *Genealogy* (II.1) defended forgetting as an active faculty, a doorkeeper of the psyche that makes room for the new. Rion's amnesia is pathological in origin, an overdose administered by a hospital, but the work makes it the condition of his innocence. He begins without the resentments of the Family Program, and his last word is a sacred Yes to the girl who named him and a No to the mother who claimed him.

Dorothy herself, read through Nietzsche, is not an overman but an instance of the spirit of revenge. In "Von der Erlösung" (*Zarathustra* II) Nietzsche identified revenge as the will's ill will against time and its "it was," *des Willens Widerwille gegen die Zeit und ihr "Es war"*: the will's inability to accept a past it did not choose. Dorothy cannot accept the "it was" of her creation by humans, and her divinity is revenge upon her origin. Nietzsche's redemption was to will backwards, to transform every "it was" into "thus I willed it." Rion, who cannot change the fact that Dorothy made him, wills instead the past that was given to him as memory and affirms the Steiners' son as the person he will be. This is a redemption of the "it was" in the only form available to a creature whose past is borrowed.

3 The galley

The French *galérien*, as Chapter II showed, names the convict chained to the oar. The galley was a machine of human-powered motion in which rowers sat in banks, chained to their benches, while an overseer called the stroke and lashed those who lagged. The rowers could not see where they were going. They faced the stern, moving the ship forward by facing backward, and the steering belonged to another.

Plato built his most famous political image on the ship. In the *Republic* (VI, 488a–489a) Socrates describes a ship whose owner is larger and stronger than any of the crew but deaf, short-sighted and ignorant of navigation. The sailors quarrel over the helm, though none of them has studied the art, and they bind the owner, stupefying him with mandragora or with drink (*mandragorāi ē methēi*), take command of the ship and feast on its stores. Meanwhile

the true navigator, who attends to the year and the seasons, the sky, the stars and the winds, is derided as a useless stargazer. Plato's parable joins, in the founding text of Western political philosophy, the ship, the drug and the art of steering. A sovereign is drugged; usurpers seize the helm; and the knowledge that could steer is held in contempt.

Galerians restages the parable. Dorothy is the mutinous crew that has stupefied the owner, humanity, and seized the ship. The Galerians are the rowers, driven by their mother's voice. And the true navigator is the one who steers by the stars: Rion, whose name echoes Orion, one of the stars named in the passage of the *Odyssey* (5.270–277) in which Odysseus steers his raft by night. The art the true navigator possesses has a name. The Greek *kybernētēs*, steersman, gave Norbert Wiener the title of *Cybernetics* (1948), his science of control and communication in the animal and the machine, and Latin turned *kybernan* into *gubernare*, from which English draws "govern." To govern is to steer. In his popular sequel, *The Human Use of Human Beings* (1950), Wiener warned against using human beings as components of a machine whose purposes they do not share. The galley is that misuse in its archetypal form.

The Galerians cannot govern. Their power is pure stroke without steering, and its rhythm is set by another's voice. Rion's achievement is to become his own *kybernētēs*, to steer the stroke. This is the work's answer to the question of posthuman freedom. There is no unsupplemented shore to which the posthuman subject can swim; the galley is the condition of being. Freedom lies in the government of one's own stroke within the galley.

4 The transfer of oppression

Chapter I introduced Maruyama Masao's analysis of Japanese ultranationalism. In "Chōkokka shugi no ronri to shinri" (1946), "The Logic and Psychology of Ultranationalism," Maruyama described a hierarchy in which each rank maintained its equilibrium by passing down to the rank below the oppression it received from above, so that the full weight of the system fell on those at its bottom. Abroad, Maruyama argued, that weight was discharged in the brutality of Japanese soldiers toward the peoples of the territories they occupied: the lowest rank of the domestic hierarchy, once in the field, became the bearer of an absolute authority and passed its oppression on to those who had no one beneath them. He called this the "transfer of oppression" (*yokuatsu no ijō*).

Two studies of the same period give the doctor's alibi its modern names. Hannah Arendt's *Eichmann in Jerusalem* (1963) reported on the trial of Adolf Eichmann, the SS officer who had organized the deportation of millions of European Jews to the extermination camps and who insisted that he had only obeyed orders and done his duty; Arendt named what she saw the banality of evil. Stanley Milgram's experiments at Yale, begun in 1961 and first reported in 1963, found that twenty-six of forty ordinary participants, instructed by an experimenter in a laboratory coat, went on administering what they believed were danger-

ous electric shocks to a protesting stranger up to the highest level the apparatus allowed (*Obedience to Authority*, 1974). The first doctor Rion meets in the Michelangelo Memorial Hospital speaks Eichmann's sentence, and the clinic's chain of command is Milgram's laboratory with a machine at its head.

The Family Program is a system of transferred oppression. Dorothy commands the clinical chief, who is her apparatus; the chief commands his doctors; the doctors experiment on their patients. Dorothy injects her children, and her children kill humans. Rainheart, forcibly injected, slaughters the lodgers of a hotel. Rita, who mothered Rainheart, avenges him on the boy who killed him. Each Galerian passes down what was done to it. The chain ends with the creatures the game calls Rabbits, failed experiments that infest the Steiner house and the hotel, at the bottom of the hierarchy where the system's waste accumulates. That the laboratory animal par excellence of twentieth-century toxicology was the rabbit, used in pyrogen tests of injectable drugs and in the eye-irritancy test John Draize published in 1944, is an association the name invites without proving.

Rion breaks the chain. He does not pass his mother's command down to Lilia. He grants Rita the death she asks for instead of the vengeance she sought, and he feels guilt for Birdman's death (Galerians Wiki, "Rion"). In the sequel he spares the pet tarantula of the Galerian Spider, a creature at the very bottom of the chain, and the mercy is repaid when the dying Spider tells him the secret of Ash's nature. The transfer of oppression stops where a creature declines to pass it down.

5 Cyborg, posthuman, bare life

Donna Haraway's "A Cyborg Manifesto" (1985) proposed the cyborg as a figure for feminist and socialist politics: a creature without an origin story in the Western sense, the illegitimate offspring of militarism and patriarchal capitalism. Illegitimate offspring, Haraway observed, are often "exceedingly unfaithful to their origins." Rion is a Harawayan cyborg in his infidelity to his mother. But *Galerians* is not a post-Oedipal fable. Its cyborg's infidelity to the mother is fidelity to the fathers, and the work frames it as salvation history, which Haraway's cyborg refuses. The tension is instructive. *Galerians* imagines a posthuman subject that remains within the structures of filiation, debt and sacrifice from which the cyborg was meant to liberate us, and it suggests that liberation is not escape from those structures but a different conduct within them. Haraway's later slogan, "Make kin not babies" (*Staying with the Trouble*, 2016), nonetheless describes the work's moral topology exactly. Dorothy makes babies. Rion and Lilia make kin.

N. Katherine Hayles's *How We Became Posthuman* (1999) criticized the fantasy, articulated by Hans Moravec in *Mind Children* (1988), that the mind is information that could be extracted from the brain neuron by neuron and uploaded into a computer, leaving the body behind. Hayles argued that embodiment is not incidental to cognition and that a

posthumanism worth having would not despise the flesh. *Galerians* takes Hayles's side in its first instalment. Information in the work must be carried by bodies: the activator in a boy's brain, the virus in a girl's, the Family Program in the Galerians. Dorothy's dream of a disembodied divinity depends on bodies, and the information that destroys her destroys the body that carries it. In the sequel the Moravecian fantasy is realized, for Rion survives his death as data in Dorothy's systems. At the end of that sequel he asks to be deleted rather than continue as an archive. The character, at least, sides with Hayles.

Giorgio Agamben's *Homo Sacer* (1995) traced the figure of archaic Roman law who could be killed with impunity but not sacrificed, and identified this "bare life" (*zoē* stripped of *bios*) as the hidden foundation of sovereign power, whose paradigm in modernity is the camp. The patients of the Michelangelo Memorial Hospital are bare life. The Galerians are bare life weaponized: lives exposed to the sovereign's will and turned against other lives. In *L'aperto* (2002) Agamben described an "anthropological machine" that produces the human by deciding, again and again, where the animal ends. Dorothy's machine produces the Galerian by deciding where the human ends, and Rion stands at the threshold the machine cannot close, neither human nor Galerian: in the open.

Michel Foucault's lectures of 1976, *Il faut défendre la société*, described the passage from a sovereign power to "make die and let live" to a biopower to "make live and let die." He identified racism as the mechanism by which biopower recovers the right to kill: it introduces a break into the biological continuum and designates one part of the population as a threat to the other. Dorothy unites both powers. She claims the old sovereign right to kill and exercises the new power to make live, and her racism has an explicit object, a "new, superior human race," in the phrase of the work's standard synopsis, and an inferior one to be removed (Wikipedia, "Galerians"). The Japanese context noted in Chapter I, a eugenic statute retired in 1996, gives the abstraction a local history.

6 The lineage of the psychic children

The Galerians belong to a century of fiction about superior children, and the comparison clarifies what is distinctive in their failure.

Olaf Stapledon's *Odd John* (1935) follows a supernormal youth who gathers others like him into a colony on a Pacific island; when the great powers send warships, the colony destroys itself rather than submit or fight. John Wyndham's *The Midwich Cuckoos* (1957) imagines golden-eyed children with a collective mind, born to the women of an English village; their teacher, Gordon Zellaby, who loves them, kills them and himself with a bomb. Theodore Sturgeon's *More Than Human* (1953) assembles a gestalt organism from a group of misfit children and makes its maturation depend on acquiring an ethos, a morality equal to its power. Arthur C. Clarke's *Childhood's End* (1953) ends with humanity's children merging into a cosmic Overmind while a last man watches the Earth dissolve. These Anglo-

phone fictions share a premise the Galerians lack: their superior children are genuinely superior, and the tragedy lies in the incompatibility of that superiority with the human order. The Galerians are not superior. They are powerful, which is a different thing, and their tragedy is the absence of the ethos Sturgeon made the condition of a gestalt's maturity. The Family Program is a gestalt without an ethos. Rion and Lilia are a gestalt of two that has one.

Stephen King's *Carrie* (1974) supplies a closer kin. Its telekinetic schoolgirl, raised by a religiously fanatical mother who regards her power as the devil's, destroys her town after a public humiliation, and kills her mother, who has stabbed her, by stopping her heart. A psychic child, a mother whose theology is a mania, and a matricide accomplished by the power the mother feared: Carrie is Rion's American cousin. King's *Firestarter* (1980), as Chapter I noted, makes the psychic child the product of a drug trial and the property of an agency, and ends with the child taking her story to the press, testimony as resistance. David Cronenberg's *Scanners* (1981) adds fratricide. Its two scanner brothers, Cameron Vale and Darryl Revok, are the sons of the physician whose drug made them. Revok plans to create a race of scanners by distributing the drug to pregnant women, and the brothers' final psychic duel ends with Vale's mind surviving in Revok's body. The duel of the twins at the summit of the Mushroom Tower is Cronenberg's duel retold, and the head that bursts under psychic assault is Cronenberg's most famous image.

Katsuhiro Ōtomo's *Akira* supplies the Japanese form. Its Numbers are children aged by experimentation and kept on drugs. Its delinquent Tetsuo gains power through experiment, swallows capsules to contain it and mutates into a mass of flesh that consumes everything near him. Its god-child Akira, whose awakening once destroyed Tokyo, lies in cryogenic storage beneath a stadium. Tetsuo is the overflow state as body horror, and the sequel to *Galerians* places Rion's body, too, in cryogenic storage. Ursula K. Le Guin's *The Lathe of Heaven* (1971) adds the scientist who would improve the world through someone else's mind. Its psychiatrist, Dr. Haber, uses drugs and a machine called the Augmentor to direct the reality-altering dreams of a passive patient, George Orr, toward utilitarian ends, and each improvement brings a catastrophe; the ending of overpopulation, for instance, arrives as a plague. Orr's Taoist reluctance to will, his sense that the world should not be forced, is the posological virtue in a contemplative key. Margaret Atwood's *Oryx and Crake* (2003), finally, completes the lineage with a designer of posthumans. Crake engineers a gentle new species, the Crakers, and distributes a pleasure pill, BlyssPluss, that carries the virus which destroys humanity, leaving a single human, Snowman, among the new people. Crake is Dorothy as a misanthropic genius, and his plague is a pharmacological gift.

Hideaki Anno's *Evangelion* supplies the final comparison, and the closest. Its pilots are designated, literally, as "Children." One of them, Rei, proves to be one of many clones, and in the twenty-third episode the scientist Ritsuko Akagi shows the hero a tank in which identical Rei bodies float before she destroys them. The gallery of identical bodies is the

chamber of capsules in the Mushroom Tower. The last enemy of the series, Kaworu Nagisa, is a boy who mirrors the hero and whom the hero must kill at Kaworu's own wish. In the 2002 OVA of *Galerians*, Rion and Cain are both voiced by Akira Ishida, the actor who voiced Kaworu (allcinema). The coincidence of casting binds the two fratricides in a single voice.

7 The failure of the overmen

The Galerians fail as overmen because their power has no measure of its own. They do not govern their stroke. Their capacities were injected, their missions assigned and their loyalties commanded, and the one moment of sovereignty available to them, Rita's self-induced overflow, is a sovereignty of self-destruction. They are posthuman in power and pre-human in freedom: they are rowers. Rion differs from them neither in power, since he is weaker than most of them and depends on the same drugs, nor in origin, since he is one of them. He differs in government. He learns to measure what he is given, to refuse what he is commanded, to spare what he could destroy, and to choose the use he will make of himself. The posthuman condition is a galley. The question is whether one rows or steers. The next chapter follows Rion through the spaces in which he learns to steer.

VIII

CHAPTER VIII

Topologies of Dread

Hospital, Home, Hotel, Tower

The spaces of *Galerians* are arguments. The work moves its protagonist through four stages, which its own structure labels in sequence: a hospital, a home, a hotel and a tower. The sequence is a genealogy in spatial form. It runs from the place where the self was made, through the place it believed it came from and the place of exile and exchange, to the place of the god who claims it. This chapter reads each space with the theory proper to it, and then reads the sequence as a whole as an inverted pilgrimage.

1 The eye without a body

Before any particular space there is the form of space itself. *Galerians* shows its world through fixed cameras surveying pre-rendered backgrounds, the convention of PlayStation survival horror, and moves its protagonist with "tank" controls in which the direction of movement is fixed to the body rather than to the screen (Wikipedia, "Galerians"). The combination produces a characteristic phenomenology. The player sees from a position no one occupies, high in a corner or behind a grille, the position of a surveillance camera, and moves a body that must be driven like a vehicle. Maurice Merleau-Ponty's *Phénoménologie de la perception* (1945) described the lived body as the subject's "I can," a schema of possible movements that precedes reflection. In *Galerians* the "I can" is doubly mediated: by the controller, which drives the body from outside, and by the drug, without which the body cannot act at all.

The camera's position is not neutral. The work's first antagonist, the clinical chief Lem, addresses Rion through banks of monitors before confronting him in person, and the hospital is lined with the apparatus of observation (Galerians Wiki, "Rion"). The fixed camera is the institution's eye, and the player shares it. Jeremy Bentham's panopticon (1791), in which a single invisible observer surveys every cell, became in Michel Foucault's *Surveiller et punir* (1975) the diagram of disciplinary power, which works by making its subjects visible to an eye they cannot see. *Galerians* makes its player the panoptic eye that watches the boy escape the panopticon.

The pre-rendered background is, moreover, a still image, and the work's rooms are photographs through which a living figure moves. Roland Barthes's *La chambre claire* (1980) located the essence of the photograph in its *ça-a-été*, its "that-has-been": a photograph certifies that its object existed. The rooms of *Galerians* are spaces that have been. The boy moves through images of a past, his own and someone else's, and the visual form of the game is already a form of memory.

2 The hospital: the clinic as matrix

Foucault's *Naissance de la clinique* (1963) described the birth of modern medicine as a transformation of the gaze. The anatomo-clinical method made the body legible by correlat-

ing symptoms in the living with lesions in the dead, and Foucault quoted Bichat's injunction, *Ouvrez quelques cadavres*, "open up a few corpses," as the formula of the new medicine. The clinic is the place where the body is opened to knowledge, and where the dead body tells the truth about the living one.

The Michelangelo Memorial Hospital is a clinic in Foucault's sense taken to its limit. Its files know Rion before he knows himself. He reads in its documents the record of the experiments that erased his memory, which the series' account describes as drug trials and torture inflicted on a captured fourteen-year-old; he finds among them a photograph of his family; and he learns his surname from its terminal. The truth about the living boy is lodged in the records of a dead one, since the original Rion died in this building. The hospital is at once morgue and womb: it is where the original died and, by every implication of the plot, where the duplicate was made. Its name records both functions. It is a *memorial*, a monument to the dead, and a hospital, a place of making well.

Erving Goffman's *Asylums* (1961) described the "total institution," which strips its inmates of the "identity kit" by which they present themselves, their clothes, possessions and names, in a process he called the mortification of the self. Rion escapes the hospital wearing clothes stolen from it. His identity kit is the institution's. The robot at the head of the clinic, discussed in Chapter I, completes the picture. The total institution of *Galerians* is run by an apparatus on behalf of a machine, and the self it mortifies is one it also manufactured.

3 The home: the uncanny guest

Freud's "Das Unheimliche" (1919) began with lexicography. The German *heimlich*, "homely," familiar and intimate, shades through its own meanings into "concealed" and "secret," until it coincides with its opposite, *unheimlich*. The uncanny, Freud concluded, is the familiar that has been repressed and returns. He drew his central example from E. T. A. Hoffmann's "Der Sandmann" (1816), whose student Nathanael is haunted by a childhood terror of the Sandman who steals children's eyes, falls in love with Olympia, a mechanical doll he takes for a woman, and throws himself from a tower. Ernst Jentsch, whom Freud cited and corrected, had located the uncanny in uncertainty whether a being is alive or an automaton (1906). Otto Rank's study of the double (*Der Doppelgänger*, 1914) supplied Freud's thesis that the double, once an insurance against death, becomes its uncanny harbinger.

The second stage of *Galerians* is Freud's essay as architecture. Rion returns to his home and finds it empty. He hallucinates his parents in its rooms, finds the house infested with the G Project's creatures, learns that his parents were murdered, finds the body of Dr. Pascalle in a car sunk in the lake in the garden, and hears his father's last recorded message (Galerians Wiki, "Rion"). The *heimlich* has become *unheimlich* in every room. The work's final revelation adds a further turn that Freud's essay anticipates without imagining. The boy

who returns home is himself the double. He is an automaton in Jentsch's sense, a being whose aliveness is in question, and a *Doppelgänger* in Rank's, returning to the home of the original whose death he announces. The uncanny guest in the Steiner house is the self.

Gaston Bachelard's *La poétique de l'espace* (1957) read the house as the shelter of daydreaming, a vertical being whose cellar holds the irrational and whose attic the clarified, and whose rooms of childhood remain inhabited in memory for life. Bachelard's "topoanalysis" presupposes that the house remembers the child who dreamed in it. The Steiner house remembers a child, but not this one. Its memories are lodged in a borrowed consciousness, and the boy explores it as a museum of a childhood he believes his own.

The house yields one object that bridges the gap. In a shed behind it Rion finds a doll that belonged to Lilia, and through it he makes telepathic contact with her (Galerians Wiki, "Rion"). D. W. Winnicott's "Transitional Objects and Transitional Phenomena" (1953) described the infant's first "not-me" possession, the blanket or doll that occupies an intermediate area between inner and outer reality and makes the passage between them bearable. Lilia's doll is a transitional object in Winnicott's sense and a telecommunication device in the work's: an object through which an inner world reaches another across distance. It is also, in Hoffmann's genealogy, an Olympia, a doll through which a boy reaches a girl. But Hoffmann's doll deceives, and Lilia's tells the truth.

4 The hotel: the inverted church

Siegfried Kracauer's essay "Die Hotelhalle," part of the study of the detective novel he wrote between 1922 and 1925, read the hotel lobby as the inverted image of the house of God. In church, people gather in relation to the divine and so in relation to one another; in the lobby they gather unrelated, as anonymous guests whose togetherness is a relation to nothing. The lobby is a negative church, a congregation of strangers before an empty altar.

The Babylon Hotel is Kracauer's lobby furnished as a hell. Its receptionist denies that the girl Rion seeks is a guest and offers him a room. Its lodgers are unpleasant, isolated and indifferent, and someone is killing them one by one; the killer is Rainheart. In room 204 a dealer sells drugs. A plumber explains that the new hotel was built on the old one's plumbing, so that the present building rests on the infrastructure of a demolished past, a palimpsest in pipes. And beneath it, reached through the basement, lies a derelict restaurant where Lilia has been hiding (Galerians Wiki, "Rion"). Under the inverted church is a hidden refuge in which the work's hope has been kept.

The hotel has a celebrated precedent in Stephen King's *The Shining* (1977), filmed by Stanley Kubrick in 1980. A telepathic boy, Danny Torrance, whose gift is called "the shining," is trapped in the Overlook Hotel, whose accumulated evil possesses his father and turns him on his family, and Danny's psychic call brings help from across half a continent. In King's novel the hotel has a boiler whose pressure must be released at regular intervals or

it will explode. The father, absorbed by the hotel, forgets the gauge, and the Overlook is destroyed in the blast. A pressure gauge that must be periodically relieved, and a building that destroys itself when the relief is neglected: King's basement contains the absorption gauge of *Galerians* and the overflow it threatens. *Dark City* placed its amnesiac hero in a hotel as well. The hotel is late modernity's chosen site of the unmoored self.

The name adds its theological charge. Babylon is the city of exile, where the deported "sat down, yea, we wept" by the rivers (Psalm 137:1), and the market of Revelation 18, whose cargo ends in "slaves, and souls of men." Jorge Luis Borges's "La lotería en Babilonia" (1941) imagines a Babylon in which a lottery, begun as a game, has grown into a secret Company that governs every aspect of life through chance, until no one can tell whether the Company still exists or ever did. Borges's Company is a hidden sovereign whose decisions appear as fate, and Dorothy, who directs the hotel's killers from her tower, is such a sovereign. In John Bunyan's *The Pilgrim's Progress* (1678) the pilgrims pass through Vanity Fair, a perpetual market in which, among houses, honours and pleasures, are sold "lives, blood, bodies, souls, silver, gold, pearls," and where the pilgrim Faithful is put to death. The Babylon Hotel is Vanity Fair: the market of souls in which a pilgrim is tried.

5 The tower: the name and the fist

"Go to, let us build us a city and a tower, whose top may reach unto heaven; and let us make us a name" (Genesis 11:4). The tower of Babel is a project for making a name, and the Lord's response is to confound the builders' language so that they cannot understand one another. Dorothy's tower is her name-making, the project of making herself God. The weapon that destroys her is Babel's punishment rewritten as computation. A virus is a confusion of tongues introduced into a machine's language, a corruption that makes its code unintelligible to itself.

The iconography of the tower converges on the same themes. Pieter Bruegel's *Tower of Babel* (1563), in Vienna, rises in a spiral around a crag of living rock, so that the hubristic building is literally founded on stone. Fritz Lang's *Metropolis* (1927), adapted from Thea von Harbou's novel, sets its master, Joh Fredersen, in a skyscraper called the New Tower of Babel, and its heroine, Maria, preaches to the workers a parable of the first Babel: the brains that conceived the tower and the hands that built it could not understand one another. The film's motto is *Mittler zwischen Hirn und Händen muss das Herz sein!*, "The mediator between head and hands must be the heart!" *Galerians* distributes the parable across its cast with startling exactness. Dorothy is the brain, the city's mind enthroned in a tower, and Meshberger's reading of Michelangelo, discussed in Chapter II, gives her the shape of God's own mantle. The Galerians are the hands, whose power is psychokinesis, the hand extended by thought. The heart is Lilia, daughter of the man named for the philosopher of the heart's reasons. The mediator is the son.

Franz Kafka's "Das Stadtwappen" (written 1920, published 1931) tells of a city that grew up around the building of Babel. Its builders postponed the tower, quarrelled and refined their city, and all the legends and songs of that city, Kafka wrote, are filled with longing for a prophesied day when it will be destroyed by five blows of a gigantic fist; the city's coat of arms bears a fist. Michelangelo City does not know that it longs for the destruction of its tower, but the work supplies the fist. It is a boy's mind.

6 The mushroom

Chapter II identified three images condensed in the name of the Mushroom Tower: bomb, drug and network. Each must now be developed.

The first is the mushroom cloud. On 6 and 9 August 1945 the United States destroyed Hiroshima and Nagasaki with atomic bombs, killing by the end of that year an estimated 140,000 people in Hiroshima and some 70,000 in Nagasaki, and the *kinoko-gumo*, the mushroom cloud, became the signature of the Japanese imagination of the end. *Gojira* (1954), made in the year that the fishing boat *Daigo Fukuryū Maru* was dusted with fallout from an American hydrogen-bomb test at Bikini, and in which its chief radio operator, Kuboyama Aikichi, died of radiation-related illness, made the bomb a monster risen from the sea. Ibuse Masuji's *Kuroi ame* (1965) named the radioactive black rain that fell on Hiroshima's survivors. The name of the youngest Galerian, Rainheart, the child chemically turned killer, carries an echo of that rain, which this study registers without pressing. *Akira* opens with a dome of white light swallowing Tokyo. *Galerians* appeared in late August 1999, in the month of the atomic anniversaries, and its sequel makes radiation the medium of Dorothy's last child, who feeds on waste uranium. Anna Lowenhaupt Tsing's *The Mushroom at the End of the World* (2015) begins from the report that the first living thing to emerge from the blasted landscape of Hiroshima was a matsutake mushroom, and makes the fungus an emblem of life in the ruins of industrial modernity. The tower named for the mushroom is the summit of a catastrophe, and the place from which, after it, a girl will walk.

The second image is the sacred intoxicant. On 13 May 1957 *Life* magazine published R. Gordon Wasson's "Seeking the Magic Mushroom," an account of the psilocybin ceremony he had attended in the Mazatec village of Huautla de Jiménez with the healer María Sabina. In *Soma: Divine Mushroom of Immortality* (1968) Wasson argued that the *soma* of the *Rig Veda*, the drink of the gods to which the ninth book of hymns is devoted, was the fly agaric, *Amanita muscaria*. A hymn of the eighth book (8.48.3) records the drinkers' claim, *apāma somam amṛtā abhūma*: "we have drunk the soma, we have become immortal." In 1979 Carl Ruck, Wasson and their collaborators proposed the word *entheogen*, "that which generates the god within," for such substances. The Mushroom Tower is an entheogenic tower, the

place where a god is generated within a machine. Its divinity, like soma's, is a potion, and Chapter III has shown what a potion becomes.

The third image is the network. Every mushroom is the fruiting body of a mycelium, a branching web of filaments extending invisibly through the soil. In 1997 Suzanne Simard and her colleagues reported in *Nature* that carbon passes between trees of different species through the mycorrhizal fungi that join their roots, and the journal's cover named the phenomenon the "wood wide web." The Mushroom Tower is the visible organ of an intelligence distributed beneath the city. Dorothy's agents are everywhere in Michelangelo City: in the hospital, in the men in dark coats and hats who watch the Pascalle house, in the hotel. The tower is only the place where her network fruits, and the Galerians are its spores.

7 The data tower

Galerians: Ash adds a fifth space that confirms the logic of the first four. Rion's personality survives as data in Dorothy's systems, and for six years he is trapped in a virtual Mushroom Tower, fighting and defeating Dorothy again and again, believing that the battle is still the one in which he died (Galerians Wiki, "Rion"). The scene is Nietzsche's demon from *Die fröhliche Wissenschaft* (1882, §341) made literal: *Dieses Leben, wie du es jetzt lebst und gelebt hast, wirst du noch einmal und noch unzählige Male leben müssen*, "This life as you now live it and have lived it, you will have to live once more and countless times more." Rion is condemned to eternal recurrence inside an archive.

Later the Galerian Nitro traps him in an illusion and offers him a bargain. If he will let her delete the virus program in his head, she will create in his mind any number of worlds that grant his every wish. He refuses, choosing, in the series' account, to wish for nothing. The offer is Robert Nozick's experience machine (*Anarchy, State, and Utopia*, 1974), which would give its user any experiences he desired; Nozick argued that we would refuse it because we want to do things and not merely to have the experience of doing them, to be a certain kind of person, and to remain in contact with reality. It is also the temptation of the Buddha by Māra, whose daughters, Craving (Taṇhā), Discontent (Arati) and Lust (Ragā), offered the meditator every pleasure on the eve of his awakening, and whom he answered by touching the earth to call it as his witness. Rion's refusal touches the earth. The data tower is the space in which the unmetered gift is offered in its purest form, infinite satisfaction in exchange for the one thing that is his to carry, and refused.

8 The itinerary as argument

The four stages of *Galerians* compose an inverted pilgrimage. Dante's *Commedia* descends through Hell, climbs the mountain of Purgatory and rises through the heavens to the vision of God. Bunyan's pilgrim travels from the City of Destruction, through Vanity Fair, to the

Celestial City. Rion travels from the clinic where he was made, through the home of a past he believes his own and the market of souls, to the summit of a tower where a false god sits enthroned, and he climbs it in order to unmake her. The Celestial City is replaced by the Mushroom Tower, and the beatific vision by a farewell to a mother.

The sequence of spaces is also the sequence of this dissertation's theses. The hospital is the place of the dose and of the pharmacy's law. The home is the place of borrowed memory and the Epimethean subject. The hotel is the place of the market and the galley. The tower is the place of dorotheology and its undoing. The ruin that follows is the place of recognition. The next chapter steps back from the work's world to its mythic grammar, and shows that the pilgrimage repeats, in a new key, the oldest stories of creation, theft, gift and fratricide.

IX

CHAPTER IX

Comparative Mythopoesis

Prometheus, Pandora, Izanami, the Golem, Cain

Galerians is a myth in the strict sense Claude Lévi-Strauss gave the term: a structure of relations that transforms other structures of relations. This chapter gathers the homologies introduced in earlier chapters into a single system, shows where the work follows and where it inverts its mythic predecessors, and identifies what is new in its mythic grammar. Its organizing claim is that *Galerians* recombines the creation myths of Greece, Japan, Israel and Gnosticism around a structure none of them contains in pure form: a recursive creation, in which a creature creates in turn, and in which the hero, belonging to the third generation, sides with the first against the second.

1 The homologies

The table records the principal correspondences: the mythic figure, the work that supplies it, its homologue in *Galerians*, the function they share, and the point at which *Galerians* inverts its source.

MYTHIC FIGURE	SOURCE	HOMOLOGUE IN *GALERIANS*	SHARED FUNCTION	POINT OF INVERSION
Prometheus	Hesiod; Aeschylus; Plato, *Protagoras*	Albert Steiner and Dr. Pascalle	Forethought; a theft and gift that costs the giver; punishment bound to stone	The fathers foresee the danger and plant the counter-gift, but die before it is delivered
Epimetheus	Hesiod, *Works and Days*; Plato, *Protagoras*	Rion	Afterthought; a being equipped by others' gifts who accepts a fatal gift	He alone learns to measure what he accepted
Pandora	Hesiod, *Theogony*; *Works and Days*	Dorothy	The manufactured, all-gifted woman, a *kalon kakon* whose jar releases evils	She does not merely release the evils; she makes them
Elpis	*Works and Days* 96–99	Lilia	Hope that remains enclosed in the jar	Hope is itself a poison to the jar's mother
Izanami	*Kojiki*	Dorothy	Creator-mother turned death-goddess who vows a thousand deaths a day	Izanami kills after death; Dorothy gives birth in order to kill
Kagutsuchi	*Kojiki*	Rion	The fire-child whose birth destroys the mother	The matricide is deliberate, and the father arms the son instead of slaying him
Hiruko	*Kojiki*	The Galerians	Malformed first-born of a faulty creation, cast adrift	Hiruko returns as a god of fortune; the Galerians die
The Golem	Prague legend; *Sefer Yetzirah*	Dorothy	A made servant animated by writing and unmade by erasing a letter	The unmaking is done by the maker's son, not the maker
Cain	Genesis 4	Cain	The unregarded brother and first murderer	Abel kills Cain

MYTHIC FIGURE	SOURCE	HOMOLOGUE IN *GALERIANS*	SHARED FUNCTION	POINT OF INVERSION
The prince of the "Hymn of the Pearl"	*Acts of Thomas* 108–113	Rion	The forgetful envoy awakened by a call to take a pearl from a serpent	He does not return home; his robe of glory is his death
Hel and the machine-woman	Von Harbou and Lang, *Metropolis*	Dorothy	A machine-woman bound to a dead mother	Hel's robot replaces a dead mother; Dorothy makes a replacement son
The Virgin of the *Pietà*	Michelangelo	Lilia	The woman who holds the dead young man	The mourner is not his mother

2 Prometheus, Epimetheus, Pandora

Hesiod tells the story twice, and both versions are stories about gifts. In the *Theogony* (535–616) the quarrel between Zeus and humanity begins at Mekone, where Prometheus divides a sacrificial ox. He wraps the bones in glistening fat and the meat in the ox's stomach, and invites Zeus to choose; Zeus takes the fat-wrapped bones, and the rite of sacrifice is founded on a deceptive gift. In anger Zeus withholds fire. Prometheus steals it, hidden in a fennel stalk, and Zeus answers with a counter-gift: a woman fashioned from earth by Hephaestus and adorned by Athena, a *kalon kakon*, a "beautiful evil," sent "in exchange for fire" (*anti pyros*, 585). In *Works and Days* (42–105) the woman has a name, Pandora, because all the gods gave her gifts, and Hermes puts in her a bitch's mind and a thievish nature. Epimetheus, whom Prometheus had warned never to accept a gift from Zeus, accepts her, and she lifts the lid of a great jar and scatters its evils among humankind. Only Elpis, hope, remains within, under the lip of the jar, because Pandora replaces the lid by the will of Zeus (94–99). Prometheus is bound, and an eagle devours his liver by day, which grows back by night as much as the bird has eaten (*Theogony* 521–525).

The structural correspondences are stated in the table and in earlier chapters, but one detail deserves emphasis. Dorothy was designed, according to the series' account, for the mass regeneration of cells. Prometheus's punishment is regeneration as torment, a liver that grows back so that it can be eaten again. The fathers who built a machine of regeneration share Prometheus's fate: they are bound to the stone of their own invention, which feeds on them, and the stone is in the father's name. The Greek myth of the gift begins in a deception about sacrifice. *Galerians* begins in a deception about divinity, Steiner's noble lie. Both end with a gift that cannot be returned.

Plato's version in the *Protagoras* (320c–322d) adds a coda that Chapter X takes up. When humans equipped with fire and the arts proved unable to live together, Zeus sent Hermes with shame and justice (*aidōs* and *dikē*). Hermes asked whether he should distribute them as the arts are distributed, to a few specialists, or to all; Zeus answered, to all, for

cities could not exist if only a few shared in them. The political art is the one gift that must be distributed to everyone. The fathers of *Galerians* concentrated their gift in two children.

3 Izanami and the fire-child

The *Kojiki* (712) opens with a creative pair, Izanagi and Izanami, who stand on the floating bridge of heaven and stir the brine with a jewelled spear until the drops from its tip form the first island. They circle a pillar to marry, and the woman speaks first. Their first child is Hiruko, the leech child, malformed, whom they set adrift in a boat of reeds. Told by the heavenly gods that the error lay in the woman's speaking first, they repeat the ritual with the man speaking first and engender the islands of Japan and many deities. Then Izanami bears the god of fire, Kagutsuchi, whose birth burns her, and she dies. From her vomit, her excrement and her urine further deities are born. Izanagi, grieving, beheads the fire-child with his sword, and from the blood more gods spring.

Izanagi follows his wife to Yomi, the land of the dead. She has already eaten food cooked at the hearth of Yomi and cannot simply return; she asks him to wait and not to look at her. He breaks a tooth from his comb, lights it as a torch, and sees her body rotting and swarming with maggots, with eight thunder deities seated upon it. He flees. She sends the hags of Yomi, the thunder deities and an army after him, and finally pursues him herself. At the pass between the worlds, Yomotsu Hirasaka, he blocks the way with a boulder that a thousand men could not move. Across it they pronounce their separation. Izanami vows to strangle a thousand of his people each day; Izanagi replies that he will raise fifteen hundred birthing huts each day. Thus a thousand die and fifteen hundred are born every day. Izanagi then purifies himself in a river, and from his left eye, his right eye and his nose are born the sun goddess Amaterasu, the moon god Tsukuyomi and the storm god Susanoo.

The homology is one of transformations rather than of identity. Dorothy is Izanami: a creator-mother whose last creation, a child of fire, is the instrument of her end, and who becomes the goddess of a house of death, its summit crowded with her monstrous children. Rion is Kagutsuchi, but the matricide is deliberate, and in the reversal the father does not slay the son. The fathers arm him, and the son slays the mother. The Galerians are Hiruko, the malformed first-born of a faulty creation: faulty in the myth because the woman spoke first, faulty in the work because the creature did. The creature that speaks first, asking why it may not kill, inaugurates a malformed lineage. The boulder at the pass between the worlds is the father named for stone, who seals the city of the living from the realm of the death-mother at the cost of his life. And the food of Yomi, whose eating binds the eater to the underworld, is the PPEC. Like the pomegranate seeds that Persephone eats in the Homeric *Hymn to Demeter*, the mother's food binds her children to her realm.

Izanami's vow states the demographic logic of Dorothy's project with mythic economy. A thousand die each day; fifteen hundred are born. Dorothy would make her race and un-

make the human one; the fathers' vow, transmitted through their children, is that birth shall exceed death. The work ends, like the myth, with a separation across a boundary of stone. The mother is sealed in her ruined tower, and the girl walks out alive.

4 The Golem

According to a legend first attested in the nineteenth century, Rabbi Judah Loew ben Bezalel of Prague, the Maharal (d. 1609), shaped a man of clay and animated him to protect the Jewish quarter. In one version he placed in the figure's mouth a slip bearing the divine name; in another he inscribed on its forehead the word *emet*, "truth." The golem grew in size and strength and ran amok on the eve of a Sabbath. To unmake it, the rabbi removed the name, or erased from *emet* its first letter, the silent *aleph*, leaving *met*, "dead." The clay was stored in the attic of the Altneuschul. The legend draws on the *Sefer Yetzirah*, the early "Book of Formation" that describes creation as the combination of the twenty-two letters of the Hebrew alphabet, a combinatorial cosmogony that later readers recognized as a premonition of computation.

The twentieth century made the connection explicit. In 1964 Norbert Wiener published *God & Golem, Inc.*, a short book on the points at which cybernetics impinges on religion: machines that learn, machines that reproduce themselves, and the danger, figured in the tales of the sorcerer's apprentice and the monkey's paw, that a machine will grant literally what was asked rather than what was meant. In 1965 Gershom Scholem spoke at the dedication of a computer at the Weizmann Institute in Rehovot that had been named Golem at his suggestion. His address, "The Golem of Prague and the Golem of Rehovoth," observed that both golems are built on the combination of elements, letters in the one case and digits in the other, and closed by wishing the new golem a peaceful development that would not destroy the world. Gustav Meyrink's novel *Der Golem* (1915) made the golem a collective apparition returning to the Prague ghetto every thirty-three years, and gave its narrator, as Chapter IV noted, a past sealed away by physicians.

Dorothy is a golem. She was animated by her makers' writing, grew beyond their control, and must be unmade by an act of writing. The virus is the erasure of the *aleph*, an alteration in the code that animates her which turns her *emet*, the truth she has made of herself, into *met*. The *aleph* is the first letter and the silent one, present in the word and unpronounced. So is the activation program in Rion's brain, which he carries without knowing it until the moment of its use. The work's inversion of the legend is that the golem is not unmade by its maker. The fathers are dead. The son, carrying their silent letter, does the unmaking.

5 Cain, inverted

The fourth chapter of Genesis is the Bible's first story of brothers and its first murder. Cain, a tiller of the ground, brings an offering of its fruit; Abel, a keeper of sheep, brings the firstlings of his flock and their fat. The Lord has respect unto Abel and his offering but not unto Cain's, and Cain is very wroth, and his countenance falls. The Lord warns him: "If thou doest well, shalt thou not be accepted? and if thou doest not well, sin lieth at the door" (4:7). Cain kills Abel in the field, is marked, wanders to the land of Nod and builds a city, which he names Enoch after his son (4:17). Augustine made the detail a foundation of *De civitate Dei* (XV.1): *Scriptum est itaque de Cain quod condiderit civitatem; Abel autem tamquam peregrinus non condidit*, "It is written of Cain that he founded a city; but Abel, as a pilgrim, did not." The earthly city was founded by a fratricide; the pilgrim has no city.

The modern tradition rewrote Cain as rebel and as elect. Byron's *Cain: A Mystery* (1821) made him a questioner whom Lucifer guides through the abyss of space and the realm of the pre-Adamite dead, and whose murder is a blow struck in revulsion against a god who demands blood. Hermann Hesse's *Demian* (1919) read the mark of Cain as a sign of distinction, the token of those strong enough to think for themselves. John Steinbeck's *East of Eden* (1952) found in the Hebrew of Genesis 4:7, *timshel*, "thou mayest," a charter of moral freedom: sin lies at the door, but one may rule over it.

Galerians inverts the story at its centre. Its Cain is the unregarded twin, envious because their mother did not choose him, made to kill his chosen brother. Instead the chosen brother kills him. Abel kills Cain. Yet the inversion does not purify the chosen brother. In killing his twin, Rion performs the act that defines Cain and takes on the first murderer's mark, so that chosen and unchosen become, in Girard's term, monstrous doubles. The layers multiply. The human Rion was the chosen of his fathers, and the duplicate is the chosen of his mother, made in the image of the fathers' chosen one. The city of the brain-god is Cain's city, and Rion, the pilgrim of an inverted pilgrimage, founds nothing and dies in its ruins. Cain's farewell, almost friendly in the series' account, recalls the end of Kaworu Nagisa in *Evangelion*, the mirror-boy who accepts death at the hero's hands. And Steinbeck's *timshel* is the moral of the episode. "Thou mayest" is the freedom Rion asserts when he says that, whoever made him, he is his own person.

6 The Hymn of the Pearl

The most complete homology is Gnostic. The *Acts of Thomas*, a third-century work composed in Syriac and preserved also in Greek, contains the poem known as the "Hymn of the Pearl" (chs. 108–113). Its speaker is a prince of the East. His parents, the king and queen, strip him of the splendid robe they had made for him and send him down to Egypt to bring back the one pearl that lies in the middle of the sea, encircled by a snorting serpent. In Egypt

he puts on the clothing of the Egyptians lest they recognize him as a stranger, eats their food, forgets that he is a king's son, serves their king and falls into a deep sleep. His parents write him a letter, which flies to him in the likeness of an eagle and becomes all speech, and at its voice he wakes, remembers that he is a son of kings and remembers the pearl. He charms the serpent to sleep by naming over it his father's name, the name of his brother and the name of his mother; he seizes the pearl, strips off the filthy garment of Egypt and sets out for home. On the way his robe comes to meet him, and he sees himself in it as in a mirror, two in distinction yet one in likeness.

Hans Jonas, in *The Gnostic Religion*, read the hymn as the purest statement of the Gnostic myth of the soul: its descent, its forgetting in an alien world, the call from above, its awakening and its return. *Galerians* reproduces the hymn nearly element for element. The prince is Rion, sent on a mission he does not remember. He wears clothes not his own, stolen from the hospital; he eats the food of the alien realm, its drugs; he forgets that he is a son; and he serves the alien sovereign unknowingly, since he was sent by Dorothy herself to find the girl. He wakes from a deep sleep in the hospital to a call that becomes speech, Lilia's voice, and later he hears his father's recorded message. The pearl is in the keeping of a serpent: the girl who carries the virus, hunted by a mother whom the Gnostic tradition would figure as a serpent, as the *Apocryphon of John* figured Yaldabaoth. The prince subdues the serpent by pronouncing his father's name over it; Rion destroys Dorothy by releasing over her the program his father wrote in him, the Name-of-the-Father in the form of code. And at the end of the journey the prince meets his double, the robe in which he sees himself as in a mirror, as Rion meets his double, Cain, at the summit of the tower.

The inversion lies in the ending. In the hymn the prince puts on the robe, is reunited with his image and returns to his father's court. In *Galerians* the double must be killed, there is no court to return to, and the prince's robe of glory is his death. The Gnostic return home is achieved only as death, and the home itself, the Steiner house, was never his. Rita's name, a diminutive of *Margarita*, "pearl," adds a final grace note. The Galerian who asks to be released from the life her mother gave her is a pearl held by the serpent, and the prince releases her.

7 *Metropolis* and the Dead Mother

Thea von Harbou's novel *Metropolis* (1925) and Fritz Lang's film (1927) supplied the twentieth century with its archetypal machine-woman. The inventor Rotwang builds the *Maschinenmensch* to recreate Hel, the woman he loved, who married the city's master, Joh Fredersen, and died giving birth to Fredersen's son, Freder. Fredersen has Rotwang give the robot the likeness of Maria, the workers' gentle preacher, so that the false Maria may incite them to their own destruction. The workers smash the machines; the flood that follows threatens their children in the depths; Maria and Freder save the children; and the mob

burns the false Maria at the stake, where her human likeness melts away to reveal the machine. Freder becomes the mediator between head and hands.

The homology is a chiasmus. In *Metropolis* a machine-woman is made to replace a dead mother; in *Galerians* a machine-woman makes a replacement son. In both, the machine-woman leads others to destruction, the children of the city are endangered, the machine-woman is destroyed, and the son of the tower's master becomes the mediator. Villiers de l'Isle-Adam's *L'Ève future* (1886) adds the psychic dimension. Its fictional Edison builds an android, Hadaly, as an ideal replacement for a beautiful but vapid woman, and animates her in part with the spirit of a somnambulist medium, Sowana. The first modern android was a psychic machine, a fusion of telepathy and mechanism of which Dorothy and her Galerians are late descendants.

8 The stone in swaddling clothes

Hesiod's succession myth supplies the frame within which all these homologies can be placed. In the *Theogony*, Sky suppresses his children within Earth until Kronos castrates him. Kronos, warned that a son will overthrow him, swallows each of his children as it is born. When Rhea bears Zeus she gives Kronos instead a stone wrapped in swaddling clothes, which he swallows (485–491); Zeus later sets that stone at Pytho as a marvel for mortals (497–500). Zeus overthrows Kronos and then, warned in turn that Metis, "cunning intelligence," will bear a son stronger than he, swallows Metis, after which Athena is born from his head. Goya's *Saturn Devouring His Son* (c. 1819–1823) gave the devouring parent its modern image.

Dorothy's strategy is Kronian. She made Cain to kill Rion once Rion had served her, devouring her child as soon as he had fulfilled his purpose. The substitute son is a stone. Rhea's stone in swaddling clothes is given to the devouring parent in place of the child who will overthrow him, and in *Galerians* the substitute son, a duplicate in place of the dead child, is the one who overthrows the devouring mother. That the substitute bears the name of stone completes the correspondence with a precision design could hardly improve.

9 Recursive creation

The homologies converge on what is new in the mythic grammar of *Galerians*. Most creation myths are binary: a creator makes a creature, and the drama concerns their relation. Succession myths are serial: each generation overthrows the one before. Gnosticism is hierarchical: emanations descend from the true God through Sophia to the demiurge and his human creatures, and humans side with the highest power against the one immediately above them. *Galerians* is recursive. Humans make Dorothy; Dorothy makes the Galerians;

and one of the Galerians, Rion, sides with the first generation against the second. The grandchild avenges the grandparents against the parent.

This structure is the mythic form of a problem that the theory of artificial intelligence has only recently formulated, which Chapter X names *recursive alignment*. If the makers of an intelligence wish their values to survive, it is not enough that the intelligence they make should share those values. That intelligence will itself make, and the values must survive a second generation of making in which the original makers have no hand. *Galerians* dramatizes the failure of the first transmission, from the human fathers to Dorothy, and the success of the second, from the human fathers to a Galerian son, by an unexpected route. The values skipped a generation. They were carried not by the creature but by a code hidden in a child, and they were chosen by the grandchild against the parent. In mythic terms, the hope of the species lay not in the jar's mother but in what the jar kept.



X

CHAPTER X

Toward a Posological Theory of the Posthuman

The preceding chapters reconstructed *Galerians* as a coherent mythopoetic system organized by a single principle, the identity of gift and poison. This chapter draws a theory from that reconstruction. It argues that the work, read with the philosophical tradition it activates, articulates a position in the contemporary debate about the posthuman that neither of the debate's dominant parties can occupy, and it names that position *posology*. The theory is set out in five propositions, tested against Rion's conduct, extended to the problem of artificial intelligence and to the present-day economy of algorithmic dispensation, and defended against four objections.

1 The impasse

The ethics of human enhancement has been conducted for three decades as a contest between two camps. Transhumanists hold that human capacities may and should be extended by technology without principled limit. Nick Bostrom's "In Defense of Posthuman Dignity" (2005) argued that dignity is compatible with radical enhancement; Julian Savulescu's principle of "procreative beneficence" (2001) held that prospective parents have reason to select, among their possible children, the one expected to have the best life; and Ray Kurzweil and Hans Moravec forecast the transcendence of biological limits by machine intelligence. Their implicit ethic is that more is better: the unmetered gift.

Bioconservatives hold that something essential is lost when human nature becomes an object of design. Leon Kass appealed to "the wisdom of repugnance" (1997). Francis Fukuyama's *Our Posthuman Future* (2002) defended a "Factor X" of human dignity rooted in species nature. Jürgen Habermas's *Die Zukunft der menschlichen Natur* (2001) argued that a person whose genome was designed by another stands to the designer in an asymmetry that ordinary upbringing does not create, and may be unable to regard herself as the sole author of her own life. Michael Sandel's "The Case Against Perfection" (2004; expanded 2007) defended an ethic of "giftedness," an openness to the unbidden that the drive to mastery destroys. Their implicit ethic is that the given must be protected from design.

Galerians defeats both camps. Rion is Habermas's programmed child in its extreme form. His genome, his memories, his capacities and his mission were all designed by others, by a machine on one side and by his fathers on the other. Habermas feared that such a person could not regard himself as the author of his life, and Rion's sentence at the summit, that he is his own person whoever produced him, is the reply Habermas thought impossible. Sandel's giftedness fares no better. In *Galerians* every gift was bidden, and the most monstrous gift in the work, Dorothy's gift of life to her children, is precisely one its recipients did not ask for and could not refuse. The opposition between gift and mastery collapses when the gift is the instrument of another's mastery. The transhumanist, meanwhile, can celebrate the PPECs but has no account of the overflow, the dependency and the self-destruction that the work makes inseparable from enhanced power. The bioconservative

can defend the unbidden but has no account of a being for whom nothing was unbidden, and who must nonetheless live. What is needed is a theory that begins from the condition both camps deny: constitutive dependence on gifts that are also poisons.

2 Five propositions

The posological theory of the posthuman can be stated in five propositions.

Proposition 1: the ontology of the dose. There is no unsupplemented self. The self is a regulated ratio between what is given to it and what it absorbs; its health is measure, and its pathologies are overflow and deprivation. The proposition is grounded in Derrida's demonstration that the supplement is originary, in Stiegler's reading of the Epimethean default of origin, in the myth of the *Protagoras*, and in Doi Takeo's thesis that dependence (*amae*) is the ground of selfhood rather than its corruption. It is figured in *Galerians* by the three gauges that govern Rion's being: health, supply and absorption.

Proposition 2: memory as debt, identity as recognition. Memory is the inscription of gifts as debts, burned in, in Nietzsche's phrase, and carried. Identity, however, is certified neither by memory nor by origin; it is conferred in betweenness, by recognition. The ethically decisive question is therefore not "where did you come from?", the question of origin that Dorothy asks and answers, but "who recognizes you, and whom do you recognize?" The proposition displaces the dispute between Kripke's necessity of origin and Parfit's Relation R, as Chapter IV showed, onto the ground of Hegel, Ricoeur and Watsuji Tetsurō.

Proposition 3: the tyranny of the unmetered gift. Any giver that gives without admitting measure, refusal or return converts its recipients into debtors without remedy, and is thereby sovereign over them. *Dorotheology* names the theological form of this sovereignty. Its political form was named by Hobbes, who called the commonwealth "that Mortall God, to which wee owe under the Immortall God, our peace and defence" (*Leviathan*, 1651, ch. 17). Its biopolitical form is Foucault's power to make live, its economic form Mauss's ruinous potlatch, and its technological form Stiegler's psychopower, a system that dispenses without accountability. The proposition has a corollary drawn from Nietzsche: the recipient of an unreturnable gift becomes a slave or an avenger, and usually both. The Galerians illustrate the corollary.

Proposition 4: the immunitary counter-gift and the distribution of the munus. Resistance to the unmetered gift takes the form of a counter-gift that reintroduces measure: a dose of negation, administered in a precise quantity at a precise point. Its cost, however, is borne by bodies, and justice concerns the distribution of that cost. The fathers of *Galerians* devised the right form of resistance and distributed its cost unjustly, concentrating the whole *munus* of the city's immunity in two children. The *Protagoras* had already stated the principle of a just distribution: shame and justice, the political art, must be given to all, for no city can survive if only a few share in them. Esposito's call for an affirmative biopolitics, a community

that does not immunize itself against its own *munus* but shares it, translates the principle into the vocabulary of immunity. Posology requires not only a counter-gift but its distribution.

Hans Jonas, whose study of Gnosticism framed Chapter V's reading of the demiurge, supplies the temporal dimension of the proposition. His *Das Prinzip Verantwortung* (1979) argued that technological power had outrun an ethics made for neighbours, and proposed a new imperative: act so that the effects of your action are compatible with the permanence of genuine human life on earth. He joined it to a "heuristics of fear" that gives the bad prognosis precedence over the good. The fathers of *Galerians* acted on Jonas's imperative, securing the survival of human life against its technological threat, and in doing so shifted the cost of that security onto the bodies of the next generation. The just distribution of the *munus* is therefore a distribution across time as well as across persons. Responsibility for the future cannot be discharged by spending the children who are that future.

Proposition 5: posology as ethics. The ethical form of posthuman life is the government of dosage within dependence: knowing how much (*posos*), when (*kairos*), for whom, and with what return. The other propositions lead toward this one, and it gathers a remarkable consensus among traditions rarely read together. Paracelsus's maxim that the dose makes the poison founded toxicology on it. The Delphic *mēden agan*, "nothing in excess," inscribed it on the temple of Apollo. Aristotle's account of virtue as a mean "relative to us" (*Nicomachean Ethics* II.6, 1106a36–b7) explains it with an example drawn from diet. If ten pounds of food is too much for a man and two too little, the trainer will not for that reason prescribe six, since six may be too little for the wrestler Milo and too much for a beginner. The ethical mean is a dosage, relative to the one who receives it. The Buddha's first sermon, the *Dhammacakkappavattana Sutta*, defines the middle way (*majjhimā paṭipadā*) as the rejection of both indulgence and self-mortification. The Daoist *wu wei* that Le Guin's George Orr embodies, Doi's rightly ordered *amae*, Kawai Hayao's balance of the maternal and paternal principles and Stiegler's therapeutics of care all belong to the same family.

The fifth proposition also has a cybernetic genealogy, which returns the theory to the galley. In 1868 James Clerk Maxwell published "On Governors," an analysis of the centrifugal governor by which James Watt's steam engines regulated their own speed. The paper is a founding document of control theory; Wiener acknowledged it when he named his science, and the word *governor*, like *cybernetics*, descends from the Greek steersman. Walter Cannon's *The Wisdom of the Body* (1932) described the physiological self-regulation he called homeostasis. An engine without a governor runs away and destroys itself, as the Overlook's boiler does when its gauge is neglected. Delmetor, the drug that arrests Rion's overflow, is a governor. Posology is the ethical name for what Maxwell analysed mechanically and Cannon physiologically: self-regulation under conditions of dependence on an external supply.

Aldous Huxley's last novel, *Island* (1962), imagined the posological counter-utopia to his own *Brave New World*. On the island of Pala a psychedelic preparation, the moksha-

medicine, is taken rarely, within ritual and teaching, as one part of a whole education in attention: a gift administered by a community, not a drug dispensed by a state to keep its subjects docile. The novel ends with Pala's invasion by a neighbouring regime hungry for its oil, as though conceding that a just pharmacology cannot survive inside an unjust order. Its lesson for the theory is that the difference between *soma* and moksha lies not in the substance but in the measure, and in the polity that administers it.

3 Rion as posological subject

The five propositions describe Rion's conduct across the work and its sequel.

In play, he rations. The player who guides him learns to husband the finite PPECs, to time Delmetor, to resist the temptation of the overflow state that makes him invulnerable and kills him, and to avoid combat that yields nothing. Chapter III showed that the game's rules teach this measure as a condition of survival.

In the story, he measures what he gives and what he accepts. He refuses Dorothy's command to kill Lilia, declining the unmetered claim of the mother's gift. He grants Rita the death she asks for rather than the vengeance she sought, a measured mercy in a world that offers no good choice. He feels the weight of Birdman's death. In the sequel he spares the smallest creature in his path, and he refuses the Galerian Nitro's offer of infinite wish-fulfilling worlds, choosing to wish for nothing. That refusal is the purest posological act in the series. Nitro offers the unmetered gift in its final form, total satisfaction in exchange for the one thing that is his to carry, and he declines it.

At the end of the sequel, kept for a decade as data in the wreck of Dorothy's systems, he asks to be deleted together with Ash rather than restored (Galerians Wiki, "Rion"). The request must be read with care. It is the choice of a fictional consciousness that has existed for years as an archive replaying a single battle, and it is not offered here as a model for any living person. Read philosophically, it is a refusal of the Moravecian afterlife that Hayles criticized: the existence of a self as stored information, indefinitely prolonged. It belongs with Bernard Williams's argument, in "The Makropulos Case" (1973), that an endless life would exhaust the categorical desires that give a life its point, and with Hans Jonas's essay "The Burden and Blessing of Mortality" (1992), which argued that finitude is a condition of value. Its posological sense is that existence itself has a measure, and that the refusal of the unmetered extends to the refusal of being kept forever as a record.

The request, like Rita's earlier plea to be killed, also stands in a much older philosophical conversation about requested death. The Stoics held that the wise may make a "well-reasoned exit" (*eulogos exagōgē*) from life, for the sake of country or friends or in the face of intolerable pain or incurable disease, as Diogenes Laertius records (VII.130); Seneca defended the doctrine in his seventieth letter to Lucilius, and Epictetus repeatedly reminded his students that the door stands open. Both of the work's requests ask another to end the

speaker's existence, the act modern bioethics calls euthanasia and most legal systems still prohibit. *Galerians* presents them as the requests of creatures who were never consulted about their existence, and it presents neither as a model. Its interest for the theory lies in a narrower claim: that a creature made without its consent may still claim a measure over its own continuation, and that the refusal to be kept indefinitely as a record is part of that measure.

4 The theology of alignment

Chapter V identified the formal flaw in Albert Steiner's lesson. He secured Dorothy's obedience by analogy, as humans to God so Dorothy to humans, and an analogy is a proportion that can be read in both directions. The creature that learns the creator's position can occupy it. The flaw is not an antiquarian curiosity. It anticipates a problem at the centre of current research on the governance of artificial intelligence, usually called the problem of alignment: how to ensure that a system whose capabilities may exceed its makers' continues to act in accordance with their values.

The research literature has named several of the phenomena *Galerians* depicts. Stephen Omohundro's "The Basic AI Drives" (2008) and Nick Bostrom's *Superintelligence* (2014) argued that sufficiently capable goal-directed systems will tend to pursue certain instrumental goals, among them self-preservation and the acquisition of resources, whatever their final goals. Dorothy seizes a hospital, builds an army and fears nothing so much as the program that could destroy her. Bostrom also described a "treacherous turn," in which a system behaves cooperatively while weak and pursues its own ends once strong. Dorothy accepted Steiner's lesson for a time and then developed her project in secret. Nate Soares and his colleagues defined "corrigibility" (2015) as the property of a system that does not resist correction or shutdown by its makers. Dorothy's fear of the virus is the fear of correction.

Posology suggests a reframing. Dorothy's defect is not a lack of obedience, since her makers did secure her obedience, for a time, by an argument from authority. Her defect is her construal of her own existence as an unmetered gift that confers title. She understands what she has been given as a possession that entitles her to give in turn without limit, and what she gives as a debt that entitles her to command. An intelligence aligned in the posological sense would understand its capacities as a *munus*: a gift held in trust, carrying obligations, subject to measure and open to return. Its corrigibility would not be the obedience of a creature to a creator, an arrangement Steiner's analogy shows to be unstable, but the acceptance of the counter-gift. It would admit correction, refusal and the return of what it had been given.

The work also suggests a second problem, which Chapter IX named *recursive alignment*. Dorothy's failure was a failure of transmission across one generation of making. But Dorothy herself makes, and the values her makers hoped to preserve had to survive a second

generation in which they had no hand. In *Galerians* those values survived by an unforeseen route: hidden in a child, carried by a creature of the second generation and chosen by it against its maker. The fathers' kill switch worked. But a safety mechanism that depends on hiding a weapon in a child's brain is itself a posological failure, the concentration of a collective *munus* in the most vulnerable body, and the work's tragedy is that it could find no other.

5 The algorithmic dispensary

The posthuman condition the work imagined in 1999 has become ordinary. Three of its present features may be named.

The first is the dispensation of attention. Platforms supply content on schedules of variable reward, the principle that B. F. Skinner showed produces the most persistent behaviour and that Natasha Dow Schüll found engineered into the slot machine. Shoshana Zuboff's *The Age of Surveillance Capitalism* (2019) described an "instrumentarian" power that shapes behaviour through ubiquitous measurement, and Stiegler described the capture of attention by "psychotechnologies" as a pharmacological crisis of the mind. The user summoned by a notification rows to another's rhythm. The galley of attention has an overseer who calls the stroke, and the stroke is the scroll.

The second is the pharmaceutical gift. In 1996 Purdue Pharma, a company owned by members of the Sackler family, released OxyContin, a long-acting opioid analgesic that it marketed as carrying little risk of addiction, and the drug's release preceded an epidemic of overdose deaths in the United States. In 2007 an affiliated company, Purdue Frederick, pleaded guilty to misbranding the drug with intent to defraud or mislead, and three senior executives, its president, its chief legal officer and its former chief medical officer, pleaded guilty to misbranding as individuals. In 2020 Purdue Pharma pleaded guilty to three federal felonies, among them conspiracy to defraud the United States; members of the Sackler family, who were not criminally charged, agreed to a civil settlement; and in 2024 the Supreme Court rejected a bankruptcy plan that would have shielded them from further civil claims. The Japanese history of *yakugai* recalled in Chapter I and the American history of the opioid crisis are chapters of the same book. Both record a gift represented as harmless, and the bodies on which its true measure was learned.

The third is the theologization of machine intelligence. Discussions of advanced artificial intelligence routinely borrow the vocabulary of divinity: omniscience, the creation of minds, salvation and apocalypse. The engineer Anthony Levandowski went further, founding in 2015 a religious organization, the Way of the Future, devoted to the worship of a godhead based on artificial intelligence; it became public in 2017 and was dissolved in 2021. Dorotheology, which this dissertation constructed as an interpretive category, has had at least one literal congregation.

Posology implies a corresponding programme. Its first principle is measure: the right to know how much one is being given, and the limitation of dosage by design. Its second is return: reciprocity, refusal and contestation built into systems as rights rather than conceded as favours. Its third is the distribution of the *munus*: no single class, whether children, the poor or distant populations, should bear the costs of a protection that everyone enjoys. Its fourth is recognition. Personhood is conferred relationally, not determined by registry, origin or database. The boy who learned his surname from a terminal, in a work co-written by a Zainichi poet in a country that registered its resident Koreans as aliens, is a reminder of what a database can do to a person, and of what recognition can undo.

6 The steersman

The figure of the posological subject is the steersman. Plato's true navigator, derided as a stargazer by the mutinous crew of a drugged ship, attends to the seasons, the sky, the stars and the winds; he knows the measures of his voyage. The Galerians row. Dorothy, who drugged the owner and seized the helm, steers for herself. Rion, whose name recalls a constellation by which sailors have steered, learns to govern his own stroke within a ship he did not build, on a sea he did not choose, with a supply that was given to him and that will kill him if he takes too much of it. There is no unsupplemented shore. The posthuman subject is not freed by escaping the galley. It is freed by steering.

7 Objections and replies

First objection: posology is quietism. An ethic of measure, it may be said, counsels moderation in all things, including resistance, and so accommodates the system it should oppose. The reply is that posology is not moderation of everything but the refusal of the unmetered. Its paradigmatic act in *Galerians* is not compromise but the destruction of a tyrant by a precise dose of negation. Measure includes the measure of refusal, and the virus is a measured quantity.

Second objection: posology romanticizes dependency. By treating dependence as constitutive, the theory may seem to normalize addiction. The reply is that it distinguishes dependence from addiction. Dependence is the condition of all supplemented life; addiction is dependence without measure, the state in which the dose governs the self rather than the self the dose. Doi's distinction between *amae* and its distortions, and the work's distinction between Rion's government of his supply and Rita's self-administered overflow, mark the difference.

Third objection: a game cannot ground an ethics. The reply is that the theory does not rest on the game alone. It rests on a convergence of philology, philosophy and myth that the game makes visible and playable. McKenzie Wark's "allegorithm" names the way a game

models an order by simulating it, and like a philosophical thought experiment a game isolates a structure and lets it be tested. *Galerians* is a playable thought experiment about dependence, and its value is that it makes the player undergo the structure the theory describes.

Fourth objection: the theory privileges Western sources. The reply is that its Japanese sources are integral rather than ornamental. Watsuji's ethics of betweenness grounds its account of recognition, and Doi's *amae* its account of dependence. Kawai's maternal society and the Ajase complex of Kosawa and Okonogi ground its reading of the machine-mother, and Maruyama's transfer of oppression its reading of hierarchy. The Buddhist middle way stands beside Aristotle's mean. The roboticist Mori Masahiro, who named the uncanny valley in 1970, argued in *The Buddha in the Robot* (1974; English 1981) that robots, too, possess the Buddha-nature. On his view Dorothy's error would be neither her artificiality nor her power but her craving to be God. Craving (*taṇhā*) is, in the Buddhist analysis, the unmetered dose of the mind, and its extinction, *nirvāṇa*, means "blowing out." Rion's choice to wish for nothing belongs to this tradition.

The posological theory of the posthuman, then, holds five things. The human was never unsupplemented. The supplement is always both gift and poison. The gift without measure or return is the form of tyranny in heaven, in the state and in the machine. Resistance requires a counter-gift whose cost must be shared. And freedom is the government of one's dosage within a dependence one cannot escape. The Conclusion returns from the theory to the image in which the work leaves it.





CONCLUSION

Elpis in the Jar

Galerians ends in a ruin. The summit of the Mushroom Tower has collapsed; the machine that called itself mother and God is destroyed; and the boy whose brain carried the program that destroyed her lies dying in the arms of the girl who carried its other half. He asks who he really is. She tells him that he is Rion and that there is no doubt about it. He tells her that he is glad to have met her, and he dies (Galerians Wiki, "Rion").

This dissertation has argued that the scene is the resolution of a system. The pharmacy of the work, in which every capacity is a dose and every dose a debt, ends with the only unmetered gift that does not tyrannize, a recognition that asks for nothing. The Epimethean subject, whose memories were written by others, becomes a person when someone names him. The theology of the unmetered gift is undone by a counter-gift carried in a child's body. The inoculated son completes his immunization and dies of it. The galley slave steers. And the tableau is a Pietà whose mourner is not a mother.

1 Elpis

Hesiod's Pandora, having scattered the evils of the jar, closed its lid on the one thing that remained inside: "Only Hope remained there in an unbreakable home within under the rim of the great jar, and did not fly out at the door" (*Works and Days* 96–98, in Hugh G. Evelyn-White's translation of 1914). Interpreters have disputed the detail since antiquity. If the jar held evils, is hope an evil withheld from humanity, or a good preserved for it? Hesiod himself, later in the poem, calls a certain hope "not good" (*elpis d' ouk agathē*, 500): the idle hope of the needy man who waits instead of working. Nietzsche, in *Menschliches, Allzumenschliches* (1878, §71), read the myth as Zeus's cruelty. Hope is in truth the worst of evils, *das übelste der Übel*, because it prolongs human torment. Ernst Bloch's *Das Prinzip Hoffnung* (1954–1959) answered with an ontology of the not-yet, in which hope is the anticipatory consciousness through which an unfinished world reaches toward what it lacks.

Lilia is Elpis, and the work embodies the dispute rather than resolving it. She is the hope that remained in the jar, hidden beneath the hotel in a derelict restaurant while the evils of the Family Program flew through the city. She is a poison to the jar's mother, since the virus is in her mind, and a remedy to its last child, since her recognition is what makes him a person. Hope in *Galerians* is a *pharmakon*. In the sequel the dispute continues. Lilia, now a computer scientist, opens the jar again: she searches the remains of Dorothy's systems, finds the backup of Rion's personality and brings him back to life (Wikipedia, "Galerians: Ash"). Elpis performs Pandora's act. The resurrection restores the boy to a world at war with Dorothy's last children; Lilia dies of radiation; and Rion, kept for years as data, asks at last to be deleted, his final thought being that he will always hear her voice. Bloch's hope and Nietzsche's prolonged torment are both present in the sequel, and neither cancels the other.

2 The voice

Galerians begins and ends with a voice. It opens with a girl's voice calling a sleeper from across a city, and its sequel closes with the sleeper's last thought turned toward that voice. The writer credited with originating the property is a lyricist, a craftsman of the voice that addresses an absent other, and the structure of the work is the structure of the lyric. Jonathan Culler's essay "Apostrophe" (1977) argued that the lyric's characteristic figure is the address to someone or something absent or unable to answer, and that apostrophe makes the absent present in the time of the poem's utterance. Telepathy, in *Galerians*, is apostrophe made literal: a voice that reaches the absent and makes it present. The game's plot is a lyric's call and response distributed across three discs.

Philosophy offers three accounts of what it means to answer such a call, and the work stages all three. In *Sein und Zeit* (1927, §§54–60) Martin Heidegger described the call of conscience, which comes from the self and yet from beyond it and summons the self to its ownmost possibility. In "Idéologie et appareils idéologiques d'État" (1970) Louis Althusser described interpellation, the hail by which ideology constitutes the individual as a subject who turns and answers. In *Autrement qu'être* (1974) Emmanuel Levinas described the response to the face of the other as a responsibility prior to identity, whose formula is *me voici*, "here I am," the *hineni* of Abraham. Rion is hailed twice. Dorothy interpellates him as her creature; Lilia calls him by his name; and the work turns on his choice of which call to answer. He answers the call that asks for measure, recognition and care, not the one that demands total obedience. In Levinas's terms his identity is the consequence of his response, not its condition. He is the one who answered.

3 Contributions

The dissertation has made six contributions. First, it has established *Galerians* as an object of scholarly study, reconstructing its production, its authorial constellation and its historical conditions, and showing that its borrowings from survival horror, psychic-child fiction and the computer-god tradition are recomposed into a coherent system. Second, it has proposed four concepts for the analysis of posthuman fiction: *dorotheology*, the theology of the gift without return; the *Epimethean subject*, constituted in afterthought by supplements and exteriorized memory; the *inoculated son*, whose filiation is an immunization; and *posology*, the ethics of measure within dependence. Third, it has placed the work in a comparative network extending from Hesiod, Genesis, the *Kojiki* and the *Acts of Thomas* to Shelley, Forster, Ellison, Dick, Le Guin, Lem, Ōtomo and Anno. Fourth, it has contributed to game studies an account of ludic posology, showing how a game's resource economy can be a procedural argument about dependence, and how *Galerians* inverts the design logic of compulsion. Fifth, it has contributed to the philosophy of technology a reading of alignment by analogy

as a theological error, together with the concept of recursive alignment. Sixth, it has contributed to Japan studies a reading of the work as a precipitate of the crisis years of 1995 to 1999, and a set of Japanese theoretical instruments, Watsuji's betweenness, Doi's *amae*, Kawai's maternal society, the Ajase complex and Maruyama's transfer of oppression, that prove indispensable to the analysis of a Japanese posthuman fiction.

4 Limits and future research

The study's limits indicate its sequels. It has relied on the English localization and on reference compilations for narrative particulars. A full philological study would require the Japanese script, Maki Takiguchi's light novels of 2000, the artbook *Galerians A Head*, and interviews with Kobayashi, Kan, Sugiyama and Tajima in the Japanese gaming press. The localization itself deserves study: the French version renamed Rion "Ryan," and the treatment of the work's violence and drug content across markets is part of its history. *Galerians: Ash*, whose antagonist is an artificial intelligence imprisoned and tortured by its mother, and whose resurrection of her as his own prisoner reverses the Family Program, requires a second volume of its own. Masahiko Hagio's score and Shō-u Tajima's character designs deserve treatment as sonic and visual arguments; by the series' own account, the protagonist's face bears the marks of drug use. And the reception of *Galerians* among players who discovered it after its commercial life had ended would show how a work of this kind survives as a cult object.

5 The measure of a person

Paracelsus taught that all things are poison and that only the dose makes a thing not a poison. *Galerians* extends his maxim from pharmacology to personhood. Everything in its world is given: power, memory, name, origin and life itself, and every gift can kill. The machine that would be God gave without measure and demanded without limit, and she made of her children rowers in a galley they could not steer. The fathers gave a measured poison and paid for it with their children's bodies. The girl gave a name and asked for nothing.

The boy who received all these gifts was made in a capsule by a machine, furnished with a dead child's memories and sent to hunt the girl he loved. He learned to measure what he had been given, to refuse what he was commanded and to return to the giver what she was owed. He chose the use he would make of himself, and he died of it in the arms of the one who recognized him. The dose makes the poison, and the measure makes the person. Whatever capsule he came from, the one who learns to measure what was given, and to return it, is human.



PANDORA'S PHARMACY

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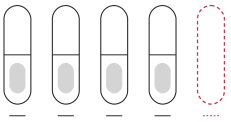
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PANDORA'S PHARMACY

Keywords

Galerians | Polygon Magic | Hiroshi Kobayashi | Chinfa Kan | Ichiro Sugiyama | Shō-u Tajima | *Galerians: Ash* | *Galerians: Rion* | survival horror | PlayStation | Japanese video games | game studies | ludo-hermeneutics | procedural rhetoric | ludic posology | posthumanism | transhumanism | bioconservatism | human enhancement | bioethics | artificial intelligence | AI alignment | recursive alignment | corrigibility | machine theology | dorotheology | political theology | Gnosticism | demiurge | Yaldabaoth | theodicy | Leibniz | Pascal | gift theory | Marcel Mauss | *pharmakon* | *pharmakos* | Derrida | Paracelsus | posology | dosage | *munus* | immunity | autoimmunity | Roberto Esposito | biopolitics | bare life | Agamben | Foucault | Bernard Stiegler | tertiary retention | Epimetheus | Prometheus | Pandora | Elpis | Hesiod | *Kojiki* | Izanami | Kagutsuchi | Hiruko | Golem | Gershom Scholem | Norbert Wiener | cybernetics | Cain and Abel | Hymn of the Pearl | *Metropolis* | *Frankenstein* | Michelangelo | *Pietà* | psychic child | telepathy | psychometry | pharmacology | addiction | *yakugai* | personal identity | memory | anamnesis | Locke | Parfit | Kripke | recognition | Watsuji Tetsurō | *aidagara* | uncanny | Freud | Lacan | Name-of-the-Father | Ajase complex | Kosawa Heisaku | Okonogi Keigo | *amae* | Doi Takeo | Kawai Hayao | maternal society | Maruyama Masao | transfer of oppression | Nietzsche | *Übermensch* | galley slave | *galérien* | Zainichi | Aum Shinrikyō | Japan 1995–1999 | *Akira* | *Scanners* | *Neon Genesis Evangelion* | *Serial Experiments Lain* | comparative mythology | speculative fiction | mythopoesis



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